

How Do Housing Markets Price Public Capital Depreciation? Evidence from Local Infrastructure Tax Cuts*

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May 29, 2026

Abstract

Do housing markets efficiently price the physical depreciation of public capital? We investigate this question using a quasi-experimental design centered on narrow-margin referendums to renew local road-maintenance property taxes in Ohio. Combining 3,184 close-margin elections (1991–2021) with a novel ConvNeXt-V2 computer-vision model fine-tuned to measure physical road condition from satellite imagery, we provide evidence on the capitalization of infrastructure disinvestment into property values. A failed renewal cuts road-maintenance funding by 11 percent at the moment of the vote. Physical road quality declines by 23 percent over the subsequent ten years, with statistical significance emerging in the four-year-ahead window. House prices, however, show no immediate response: the price gap between narrowly-failing and narrowly-passing jurisdictions opens only in the fourth year, the same year visible road deterioration becomes statistically detectable. The cumulative ten-year price decline is 9 percent (\$15,350 on average), against an average per-household tax saving whose perpetuity present value is approximately \$1,900 — a capitalization gap of roughly eight to one. Under rational expectations, the full future amenity decline should have been capitalized at the moment of the vote; the data decisively reject this benchmark.

We rationalize the delayed-capitalization pattern using a dynamic general-equilibrium

*Acknowledgements: We are grateful for comments from CK Tang, Guoyang Yang, Dan Boles, Eunjee Kwon, Gary Painter, Olivier Parent, Jeffrey Mills, Nayoung Lee, Matias Cattaneo, and Chris Bollinger, to CoreLogic® for leasing us the housing dataset, to Albert Saiz, Lu Han, Siqi Zheng, David Albouy, Chao Liu, Tony Yezer and other attendees at the AREUEA-ASSA Conference, AREUEA National Conference and UEA North American Meeting for their insightful comments. We thank John Schroeder, Public Service Director of Beavercreek township, Ohio, for his insights on funding and maintenance of local roads in Ohio. No external funding was received for this project.

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model in which households Bayesian-update on a noisy signal of the public-capital stock, recovering the rational-expectations benchmark as a knife-edge restriction. The calibrated model matches both the dynamic price path and the cross-sectional heterogeneity by urban density and price quantile. The implied marginal value of public funds is -1.4 , with the experience-based counterpart in $[-2.1, -1.6]$, identifying the road-maintenance tax cut as a welfare-destroying policy whose cost is concealed from voters by the visibility lag of physical decay.

JEL Codes: R42, H71, R10, G14

1 Introduction

A central question in public finance and asset pricing is whether markets efficiently internalize the long-term consequences of infrastructure disinvestment. While a vast literature has established the positive capitalization of new public investment (Oates, 1969; Cellini, Ferreira and Rothstein, 2010), we know surprisingly little about how and when markets price the physical depreciation of the existing public commons. In theory, rational expectations should trigger an immediate adjustment in property values the moment a fiscal decision to reduce maintenance funding is made. In practice, if information frictions or fiscal illusions persist, the economic cost of disinvestment may remain hidden until physical decay becomes tangible. We investigate this disconnect through a quasi-experimental setting in which the underlying fiscal shock — a local road-maintenance tax cut — is cleanly identified by close-margin renewal referendums, and in which the underlying public-capital depreciation is directly measured from satellite imagery at sub-county resolution over thirty years.

The empirical setting is well-suited to the question because the deterioration of roads is a *visible* phenomenon. Voters and prospective home-buyers can, in principle, observe the consequences of reduced maintenance directly. Whether they do so in a manner that reconciles with the rational-expectations capitalization benchmark is an empirical question that this paper answers in the negative.

Setting and identification. We assemble a panel of 3,184 renewal-tax referendums on local road maintenance in Ohio from 1991 through 2021, covering 617 local governments. The timing of each referendum is determined by the predetermined expiration cycle of the prior levy, an average of five years, making the *timing* of the vote orthogonal to local economic conditions. The *outcome* of close-margin votes is, by construction, unforecastable within a narrow window around the 50-percent threshold. We exploit this discontinuity in a sharp regression-discontinuity design that delivers two empirical objects: the dynamic event-study path of (i) house-price RD coefficients in event time $\tau \in \{-3, \dots, +10\}$ and (ii) road-quality

RD coefficients in rolling three-year post-election windows.

The first contribution of the paper is the measurement of G_t — the underlying physical road-capital stock — at a resolution and frequency that existing engineering and smartphone-based instruments cannot deliver outside of dense metropolitan settings. We fine-tune a ConvNeXt-V2 vision model on 53,677 labeled satellite road images from [Brewer et al. \(2021\)](#) and apply it to the National Agriculture Imagery Program (NAIP) and the Ohio Statewide Imagery Program (OSIP3) image panels. The resulting Road Quality Score, defined on a 0–100 continuous scale, is available year-by-year at sub-county resolution over the full sample window. This is, to our knowledge, the first measurement of dynamic changes in the physical road-capital stock at this resolution over this horizon.

Main empirical findings. Three empirical patterns emerge from the analysis. First, jurisdictions that narrowly fail to renew their road-maintenance levy experience an immediate 11-percent decline in maintenance spending — the direct fiscal margin. Second, road quality declines progressively in the years following the vote, with the rolling-window RD coefficient on the Road Quality Score becoming statistically significant in the $\tau \in [2, 4]$ window (a 16-percent decline) and reaching a 23-percent decline at the 10-year horizon. Third, and most importantly, the response of housing prices does not follow the path predicted by rational-expectations capitalization. The pre-period coefficients are economically small and statistically indistinguishable from zero — which rules out an anticipation story driven by autocorrelated political characteristics. But the price response immediately *after* the vote, in $\tau \in [0, 3]$, is also indistinguishable from zero. The price gap opens only in $\tau \in [4, 9]$, peaks at roughly nine percent of the mean sale price, and partly closes by $\tau = 10$. The total cumulative ten-year discount is approximately \$15,350, or nine percent of the mean median sale price (\$166,000).

The puzzle. Under rational expectations, the entire present value of the future amenity decline should have been capitalized at $\tau = 0$, the moment the close-election outcome was

realized. The pre-period null effectively rejects an anticipation explanation. The contemporaneous null effectively rejects rational expectations. The price gap opens not when the fiscal shock occurs, but when the *physical signal* — the visible degradation of roads, captured by our ConvNeXt-based measure — becomes statistically detectable. Approximately one-twentieth of the long-run capitalization arrives in the first three post-election years; nineteen-twentieths arrives only after physical decay is observable.

The synchronization between the visibility of decay and the price response is the central piece of evidence in this paper. It is decisive in distinguishing rational-expectations capitalization from limited-attention and signal-extraction alternatives in which buyers update their belief about the public-capital stock only when an observable signal arrives. We formalize this distinction in Section 6 and develop a structural counterpart in Section 7.

Structural test and welfare. We embed the empirical wedge in a dynamic general-equilibrium model with a Rioja-style road-depreciation channel. Under perfect foresight — the rational-expectations benchmark — the model overshoots the data in the short run, generating an immediate price drop that is decisively rejected by the empirical event study. We then introduce a Bayesian signal-extraction microfoundation in which households observe the public-capital stock with noise that shrinks endogenously with the magnitude of physical decay. The microfoundation nests the rational-expectations benchmark as a knife-edge restriction ($\bar{\sigma} = 0$) and identifies the friction parameter $\bar{\sigma}$ from a single transparent moment in the data: the ratio of the cumulative discounted price gap accumulated through $\tau = 3$ to the long-run capitalization at $\tau \geq 9$. The calibrated belief-updating model lies within the empirical 95-percent confidence band throughout the event-time window and reproduces the cross-sectional heterogeneity by urban density and by housing-price quantile without additional free parameters.

The implied welfare loss is substantial. Capitalizing the average per-household tax savings (\$76 per year) as a perpetuity at the user-cost rate of 0.04 yields a present value of

approximately \$1,900. The corresponding capitalized wealth loss is \$15,350. The ratio — approximately eight to one — is the *capitalization gap* we report in the body of the paper. Translated into the marginal-value-of-public-funds (MVPF) framework of [Hendren and Sprung-Keyser \(2020\)](#), the policy delivers an MVPF of -1.4 : every dollar of revenue forgone by the government generates approximately \$1.40 of net welfare loss to households. Under the belief-updating microfoundation, this number is best interpreted as a conservative *choice-based* measure; the *experience-based* counterpart, evaluated under perfect foresight, lies in the range $[-2.1, -1.6]$.

Contributions. The paper makes three contributions. First, it provides the first credibly-identified estimates of the capitalization of public-capital *depreciation* into property values — a literature in which positive shocks (new bonds, new highways) have dominated and the negative-shock case has been substantially understudied despite its first-order policy importance for an aging U.S. infrastructure stock ([American Society of Civil Engineers, 2025](#); [Ramey, 2021](#)). Second, the paper introduces a novel computer-vision-based measurement of physical road quality at sub-county resolution over thirty years, complementing the smartphone-acceleration approach of [Currier, Glaeser and Kreindler \(2023\)](#) and the engineering-survey benchmarks (IRI, PCI) which are sparse outside large metropolitan areas. Third, the paper develops a formal test and rejection of rational-expectations capitalization in a salient local asset market and provides a tractable structural alternative — a belief-updating microfoundation — that quantitatively rationalizes the empirical pattern and delivers welfare-relevant policy implications, including pre-vote disclosure of road-condition assessments (in the spirit of [Brasington and Parent, 2024](#); [Chetty, Looney and Kroft, 2009](#)).

Roadmap. The remainder of the paper is organized as follows. Section 2 positions the contribution relative to the literatures on public-capital capitalization, information frictions in housing markets, and computer-vision-based economic measurement. Section 3 describes the data and the fine-tuned vision model. Section 4 develops the close-election regression-

discontinuity design and formalizes the anticipation test. Section 5 presents the empirical results. Section 6 formalizes the rational-expectations benchmark, documents the empirical wedge, and identifies the candidate mechanisms. Section 7 develops the structural counterpart, including the belief-updating microfoundation. Section 8 reports the welfare analysis. Section 9 concludes. Technical derivations, additional robustness checks, and the vision-model fine-tuning details are deferred to the online appendix.

2 Related Literature

This paper sits at the intersection of three literatures: the long-running tradition on the capitalization of public capital into property values, a more recent body of work on information frictions and slow capitalization in asset markets, and a fast-growing literature on the use of computer-vision methods to construct economic measurements from high-resolution imagery. We discuss each in turn and locate our contribution.

Public Capital and Capitalization

Following the cross-jurisdictional regressions of [Oates \(1969\)](#) and the Oates Decentralization Theorem ([Oates, 1972](#)), the subsequent literature has been overwhelmingly concerned with the capitalization of *positive* shocks to the public-capital stock. [Cellini, Ferreira and Rothstein \(2010\)](#) provide the modern reference, studying close-margin school-bond referendums in California with a dynamic regression-discontinuity design and reporting substantial capitalization of new capital projects into housing prices. [Biasi, Lafortune and Schönholzer \(2025\)](#) sharpen the dynamic-RD methodology and isolate the share of capitalization attributable to teacher-quality investments. [Edel and Sclar \(1974\)](#) is among the few early studies to attempt to estimate the role of *maintenance* spending using cross-jurisdictional Boston data; the cross-sectional design recovers null effects, and the dimension of public-capital depreciation goes largely unexamined in the subsequent literature.

The transportation-infrastructure literature has likewise focused on new construction. [Gonzalez-Navarro and Quintana-Domeque \(2016\)](#) documents that paving previously-unpaved roads in Mexico produces a 28-percent appraisal-based increase in property values. [Theisen and Emblem \(2021\)](#) estimates a 13-percent house-price increase associated with a new highway in Norway. [Asher and Novosad \(2020\)](#) documents that close-margin rural roads in India have no detectable effect on village incomes or assets, attributing prior estimates to selection on growth potential. [Gertler et al. \(2024\)](#) estimates effects of road maintenance in Indonesia, with mixed evidence on long-run local economic development. The classical theoretical reference for road-maintenance dynamics is [Rioja \(2003\)](#), whose dynamic general-equilibrium calibration provides the law of motion for public-capital depreciation that we adopt in Section 7. [Ramey \(2021\)](#) surveys the macroeconomic literature on the consequences of infrastructure investment, emphasizing that the multiplier is largest when starting capital is well below the optimal level — a finding that resonates with our welfare estimates for the case of disinvestment from existing capital.

Two relatively recent papers approach the public-services capitalization question from angles directly relevant to our setting. [Brinkman and Lin \(2024\)](#) use historical highway-revolt episodes to identify capitalization of *both* positive and negative shocks to neighborhood transportation infrastructure, broadening the literature beyond positive-shock identification. [Brasington and Parent \(2024\)](#) use a close-election RD to study capitalization of fire-protection services and explicitly document a *limited-awareness* mechanism in which capitalization is muted because residents are slow to register changes in public-service quality. Their evidence is the closest existing analog to the mechanism we develop, and we view our findings as complementary: in their setting, the public-service shock is binary (fire-station closure) and observable; in ours, the underlying public-capital depreciation is continuous and only gradually observable, generating a delayed-capitalization pattern absent from their setting.

Information Frictions and Slow Capitalization

A separate strand of literature establishes that markets capitalize information about asset fundamentals with substantial delay when the information itself is non-salient, costly to acquire, or arriving in noisy increments. [Chetty, Looney and Kroft \(2009\)](#) and [Finkelstein \(2009\)](#) provide the canonical evidence from the tax-salience literature: agents respond more weakly to non-salient tax changes than to mechanically-equivalent salient ones, and the wedge is large enough to overturn welfare conclusions derived from fully-rational benchmarks. [Bakkensen and Barrage \(2022\)](#) extend this logic to housing markets, showing that undercapitalization of long-horizon flood risk is correlated with belief heterogeneity in the relevant local population. [Ouazad and Rancière \(2024\)](#) provide the most directly relevant theoretical apparatus, deriving conditions under which amenity capitalization is time-varying in the presence of arbitrage costs and information frictions. [Bishop and Murphy \(2019\)](#) provide methodological guidance on when a dynamic hedonic specification is necessary in the presence of time-varying attributes.

Our paper contributes to this literature in three respects. First, our quasi-experimental design generates exogenous variation in a fiscal decision whose long-run consequence (physical decay of the public-capital stock) is observable to all parties only with a substantial visibility lag. This is a particularly clean laboratory for testing rational-expectations asset pricing against limited-attention or signal-extraction alternatives. Second, our novel measurement of the underlying physical state of the public-capital stock — via computer vision on aerial imagery (Section 3) — allows us to time the observable signal precisely and demonstrate empirically that capitalization tracks the signal rather than the underlying fiscal shock. Third, the belief-updating microfoundation in Section 7.12 nests the rational-expectations benchmark as a knife-edge case and identifies the friction parameter from a single transparent moment in the data, providing a structural counterpart to the reduced-form rejection of RE.

Computer Vision in Economics

Our paper contributes to a fast-growing literature on the use of computer-vision methods to construct economic measurements from high-resolution imagery. [Naik et al. \(2017\)](#) use street-view imagery and a convolutional architecture to predict physical urban change, demonstrating that physical-appearance features carry forward-looking economic content. [Glaeser, Kincaid and Naik \(2018\)](#) apply convolutional networks to real-estate imagery and document that visual features predict property values beyond conventional hedonic controls. [Yeh et al. \(2020\)](#) use Landsat imagery and deep learning to estimate economic well-being in Africa, validating the methodology against ground-truth survey data and demonstrating its scalability across geographies. [Currier, Glaeser and Kreindler \(2023\)](#) construct a Road Roughness Index from vertical-acceleration signals collected by an Uber smartphone app, providing the closest existing analog to our outcome variable but limited to areas with active Uber service.

Two prior measurement traditions exist for road quality. The first relies on engineering benchmarks such as the International Roughness Index (IRI) and the Pavement Condition Index (PCI), both of which require costly on-site inspection (profilometer vans, trained surveyor visits) and are updated irregularly with substantial geographic gaps. The second relies on smartphone-sourced acceleration data ([Currier, Glaeser and Kreindler, 2023](#)), which has broad coverage in metropolitan areas but is sparse in suburban and rural settings precisely where many of the renewal-tax referendums in our sample occur. Our contribution is to apply a fine-tuned ConvNeXt-V2 vision model to the aerial-imagery panel from the National Agriculture Imagery Program (NAIP) and the Ohio Statewide Imagery Program (OSIP3), fine-tuned on the labeled road-surface dataset of [Brewer et al. \(2021\)](#), and generating consistent road-quality measurements at sub-county resolution across thirty years of imagery. To our knowledge this is the first measurement of *dynamic changes* in road quality at this resolution and over this horizon.

3 Background & Data

3.1 How are roads funded in Ohio?

Roads in Ohio are funded through a combination of federal, state, and local sources. A significant portion of road funding provided by federal and state governments comes from gas taxes, which are currently set at \$0.18 per gallon for federal tax and \$0.38 per gallon for Ohio state tax. Additional sources for these two levels of government include vehicle registration fees, license plate fees, tolls, and driver’s license fees. Funding for local governments largely comes from property taxes. Below, we provide an overview of road funding in Ohio from national, state, and local sources.

Federal Funding. U.S. Department of Transportation is responsible for federal road infrastructure funding and allocates funds through the Federal Highway Administration (FHWA). The FHWA provides funding for the construction, maintenance, and operation of highways, bridges, and tunnels through the Highway Trust Fund (HTF), which is funded by the federal gas tax. The HTF is divided into two accounts: the Highway Account and the Mass Transit Account. The Highway Account is used to fund highway construction and maintenance, while the Mass Transit Account is used to fund public transportation projects. Under the Bipartisan Infrastructure Law (BIL) ([U.S. Department of Transportation, 2022](#)), the HTF funds key programs like the National Highway Performance Program (NHPP), which targets the National Highway System (NHS), and the Surface Transportation Block Grant (STBG) program, which provides flexibility for a broader range of federal-aid roads. However, federal funding for local road infrastructure is largely restricted to major roads, leaving most neighborhood streets dependent on local revenue. As of 2022, about 4/5th of all road funding came from state and local governments ([Peter G. Peterson Foundation, 2024](#)).

State Funding. The Ohio Department of Transportation (ODOT) is responsible for maintaining the Interstate system, the U.S. and State Routes. In Ohio, gas taxes, licensing fees and user fees account for 69% of the state’s road funding ([Boesen, 2021](#)), which can

be used for both maintenance and new construction. However, about 70% of funds goes to highway construction¹, 2% is given to local governments as grants and only 4% of state funds are directed towards roadway maintenance ([Ohio Department of Transportation, 2023](#)). The remaining funds are part of payroll and operating expenses and other miscellaneous expenses. Hence, most of the road maintenance funding for local, neighborhood-specific roads in Ohio comes from local governments.

Local Funding. Local governments in Ohio fund roads mainly through property taxes, although the extent of this funding varies across different localities. These municipalities have the authority to levy property taxes specifically for road maintenance, providing a crucial source of funding for the upkeep of local infrastructure. For instance, as per our correspondence with Beavercreek Township, 61% of its road funds come from property taxes, and only 8% from gas taxes. Moreover, 77% of funding for roads in Beavercreek Township is provided by the local government and 11% of the overall local government budget is allocated to roadway maintenance ([Public Service Department of Beavercreek Township, 2025](#)). We can see that local roads are primarily funded by local governments in Ohio and road maintenance funding is a significant part of the local government’s budget.

3.2 Local Taxation in Ohio

Background. Ohio consists of 88 counties, each covering about 464 miles² (1,200 kilometers²). Each county was historically divided into about 15 equally-sized townships, which do not cross county lines. Citizens can petition to incorporate as a village, which has a different type of government structure than a township and the ability to levy both income and property taxes, whereas townships may only levy property taxes. When a village exceeds 5,000 population in Ohio, it is reclassified as a city. Villages and cities may cross township and county lines, dissolve, or annex parts of contiguous townships. Villages, cities and townships, which we call “cities” for brevity, are the most local governmental unit in Ohio. Each local

¹According to ([Ohio Department of Transportation, 2023](#)), most of this category is used for preexisting infrastructure rather than new capacity.

government covers about 18.2 miles² (47.1 km²) on average. The Ohio Revised Code lets local governments collect a small amount of tax without a vote. Beyond this limited amount local governments put tax levies on the ballot to ask for additional money from voters. Our data on renewal road tax levies include 3,184 referendums and cover 617 local governments in Ohio.

The type of tax levy we study is for the renewal of road tax levies. Most of the renewal taxes we consider have stated purposes of “road and bridges repair”, “road repair”, “street fund”, and “street improvements”, although there are less common stated purposes like “repair and maintenance of streets and sewers system” and “resurfacing and rehabilitation of city streets.” The construction of new roads and bridges, in contrast, would be funded with a tax levy for additional money, not a renewal tax; and it would likely be funded by a bond levy lasting 20 or 30 years. We eliminate from our dataset stated purposes that might suggest new road construction like a 30-year 1.9-mill new tax in Moscow Village for “permanent improvements” and 0.5-mill new tax for 20 years in Shawnee Hills Village for “general construction and road and bridges repair.” Our dataset includes tax levies such as a 2-mill, 5-year renewal in Adams Township (Champaign County) in 1995; a 3-mill, 5-year renewal in Lore City Village in 2016; and a 2.5-mill, 5-year renewal in Pataskala City in 2007.

Levies that had originally passed typically expire, and the most common duration to collect a levy is five years, representing about 90% of the road tax levies in our sample. If a tax levy is renewed, taxes and funding continue. If 50% or fewer votes approve the levy, it fails. When a tax levy for additional funding fails, there is no increase in funding, but existing funding from other tax levies continues as normal. When a renewal tax levy fails, funding from that tax levy stops. 99% of the road tax levies in our sample are property taxes and 1% are income taxes.

Renewal levies. Most RD studies that use voting data to look at the impact of funding changes examine new tax levies for additional funding. [Cellini, Ferreira and Rothstein \(2010\)](#)

observes that votes for additional tax money may not be statistically independent; a vote may be proposed until it passes. We minimize this source of endogeneity of new votes by only considering renewal votes ([Brasington, 2017](#)). While a government may choose when to put a vote for additional funding on the ballot and keep proposing the new tax until it is passed, when a vote passes, it has an expiration date. So if a road tax levy for additional funding passes in 2007 to last five years, in 2012 voters will have the chance to renew or reject the tax. The timing of the vote in 2012 is not endogenous, having been set in 2007. If voters renew it in 2012, it will be up for renewal again in 2017. The exogeneity of the timing of a renewal tax levy contrasts with the timing of new taxes for additional money as explained by [Cellini, Ferreira and Rothstein \(2010\)](#) because local governments can endogenously choose the timing of a new tax levy to coincide with positive economic conditions or positive sentiment toward government action, making an election more likely to pass.

Spending impact of failing to renew a levy. When a renewal tax levy fails, the local government loses the funding from that tax levy. The local government may still have tax levies for other purposes in effect, but the road tax levy that failed to renew is no longer collected. We determine the dollar amount in consideration when a household is making a decision on whether to vote for or against a renewal road tax levy, and we call it Average Road Tax per household, which is the tax amount that a household will pay if the levy is successfully renewed. Average Road Tax per household is computed following Equation 1:

$$\text{Average Road Tax per household}_{it} = \text{millage rate}_{it} \times \text{Average Assessed value}_{it} \quad (1)$$

where i is the city, t is the year, and millage-rate ² is the predetermined millage amount set for the road tax levy. The Average Assessed value is 35% of the average appraisal value³, which for our study equals the average sale price of houses in the city for that year. From

²Millage-rate is property tax rate expressed in mills (tax per \$1,000 of assessed value).

³Assessment ratio of 35%, as set by [Ohio Department of Taxation \(2020\)](#).

this, we can also compute the average road tax per city by multiplying the average road tax per household with the number of households in the city.

Although we acknowledge a large variation in the appraisal value across cities and in millage rates across referendums, we find that the average road tax per household is \$76, and the average road tax per city is \$167,011 as shown in Table 1. We do not observe any significant difference in the average road tax per household between cities that renew and cut road tax levies, which suggests the levies up for renewal are not systematically different between areas that renew and areas that fail to renew them. Whenever a renewal tax levy fails, the local government loses the funding from that tax levy, which is the average road tax per city. This loss in funding directly impacts local government’s spending budget needed to maintain local roads in Ohio and accounts for 11% of a local government’s budget for road maintenance ⁴.

⁴For townships within our effective bandwidth, we use the average expenditure by the public works department as our base, which is \$1,528,404 as per their audited financial reports. All local roads in such municipalities are funded via this department. Average loss in maintenance spending for areas that cut their renewal levy is \$163,547, thus giving us $11\% = \frac{163,547}{1,528,404} \times 100$

Table 1: Spending impact of failing to renew a Road Tax Levy

	Aggregate	Renewed	Cut
Panel A: road tax per household			
Mean	76	75	79
	(55)	(53)	(62)
Panel B: road tax per city			
Mean	167,011	167,648	163,547
	(340,268)	(340,628)	(338,671)

Notes: This table presents descriptive statistics for two measures of the money collected via road tax levies. **Panel A** reports the mean and standard deviation (SD) of the road tax per household per year. **Panel B** reports the mean and SD of the total road tax collected per city. “Aggregate” denotes the full sample, while “Renewed” and “Cut” refer to levies that were renewed and failed to renew respectively. All monetary values are in constant 2010 U.S. dollars and rounded to the nearest integer.

3.3 Running Variable

The running variable plays a critical role in RD, which in this study represents the proportion of votes against the renewal of a road tax levy. A vote share of 50% or more against the renewal road tax levy means the levy fails and the tax will no longer be collected, resulting in a stoppage of road funding via that particular tax levy. There are 3,184 referendum results in our sample, 83% of which renew the tax, and 17% of which cut taxes and road maintenance. The Great Financial Crisis falls in the middle of our dataset, so readers might wonder if voting behavior was affected, but we find vote share the same to two decimal points during and outside the years 2008-2009. Our key identification assumption is that

the election results are not predetermined and vote share is not precisely manipulated to fall just above or below the cutoff. This assumption allows us to exploit the randomization around the cutoff and provides the variation needed to identify the causal effect of cutting road tax. We test this assumption using a density test detailed below and covariate balance tests (see Appendix C).

Density test. A classic RD assumption states that agents cannot precisely manipulate the running variable to fall just above or below the cutoff. In our context, it means that the election results are not determined prior to when the ballot takes place. In other words, no individuals, organizations, higher levels of government, foreign governments, or the firm that programs the voting machines are dictating the precise vote share for the renewal road tax levy referendums raised by a city. The standard way to test this assumption is to perform a density test like that of [Cattaneo, Jansson and Ma \(2020\)](#), which is based on the idea that manipulation of elections might cause a clustering of votes just to one side of the cutoff, with a pronounced drop-off on the other side of the cutoff. The p -value of this density test is 0.98. A histogram of vote share is shown in Figure 1 that graphically illustrates the lack of abrupt change in density.

Although Table 3 shows covariate balance between sets of cities that pass and fail to renew road tax levies, covariate values could still jump from one side of the cutoff to another. A drop in education levels, for example, could cause a drop in house prices that might coincide with a change in treatment, so that what might look like a treatment effect from cutting taxes and spending might in fact be caused by lower education levels. Graphs that suggest covariate smoothness around the cutoff are found in Appendix C. A formal way to assess covariate smoothness is to use each covariate as an outcome variable in a regression of the running variable and the treatment effect dummy. When we do so, the p -value of the treatment effect dummy varies from 0.234 to 0.981, as shown in Table 15, indicating no statistically significant jump in covariate values.

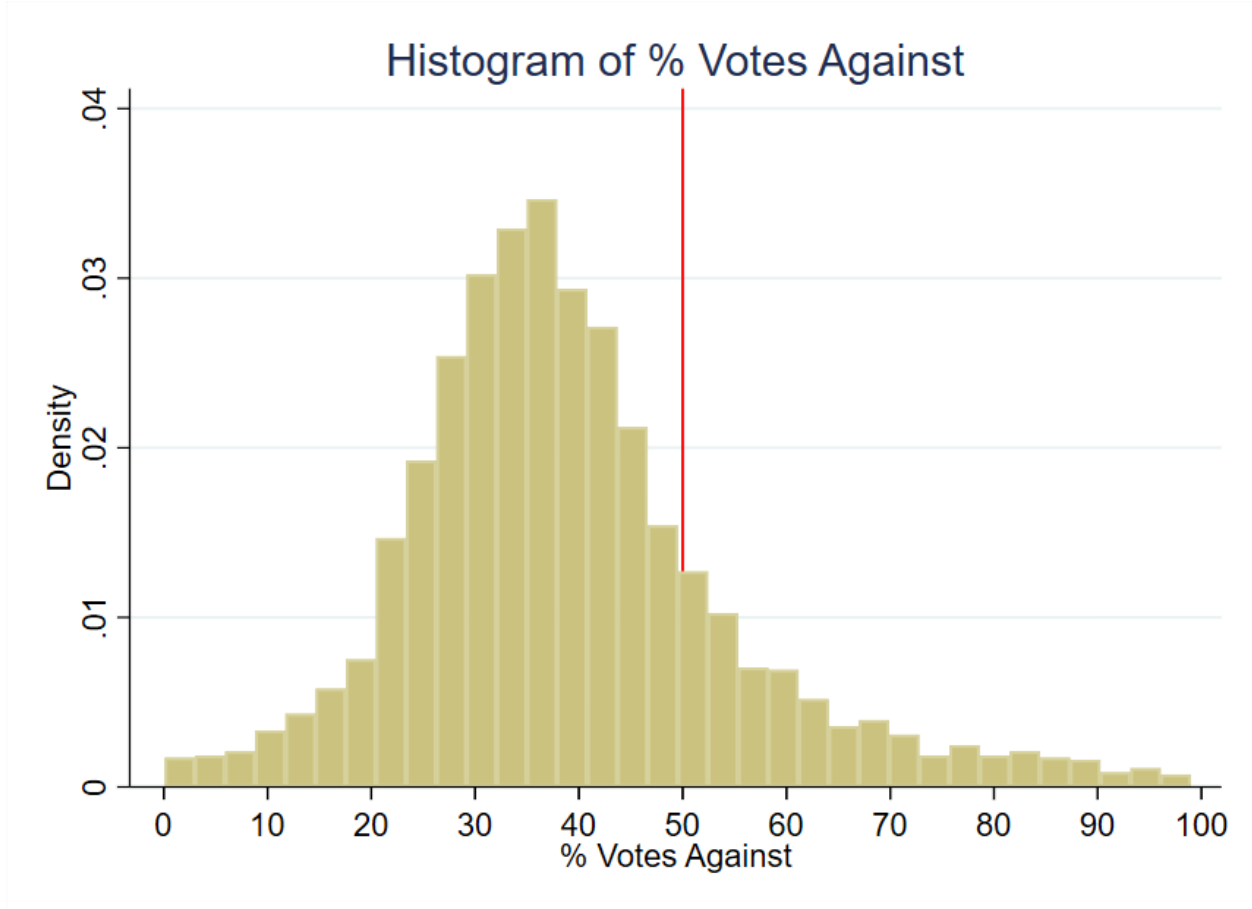


Figure 1: Histogram of Running Variable

3.4 Tax Cuts & Evidence of Road Quality

Public road capital depreciates at an annual rate of approximately three percent according to U.S. Bureau of Economic Analysis fixed-asset accounts, and the cumulative under-investment in U.S. infrastructure is estimated at \$2.6 trillion through 2029 ([American Society of Civil Engineers, 2025](#)). Two features of this depreciation process are central to the framing of this paper. First, depreciation is gradual: physical decay accumulates over multiple years before becoming visually apparent. Second, residents and prospective home-buyers observe physical road decay *directly*, but they observe the underlying maintenance budget only indirectly through ballot questions and intermediate news reports. The output of the fine-tuned vision model introduced below is best understood, accordingly, not merely as a measurement instrument but as a measure of the *observable signal* that drives the belief-updating

mechanism in Section 7.12.

Roads in Ohio last 15-20 years and deteriorate faster with poor drainage, utility cuts to the road, and snowplow damage (City of Hudson, 2020). To understand the effect of cutting local road taxes on road quality, we fine-tune an AI vision model on satellite imagery data from Brewer et al. (2021). Following the seminal work on text-based Transformers in Natural Language Processing (NLP) by Vaswani (2017), vision transformers (ViTs) were introduced by Google Brain’s team and are part of a recent class of deep learning models that have shown to outperform classic Convolutional Neural Networks (CNNs) on image classification tasks (Dosovitskiy, 2020). Nevertheless, a more recent class of vision models learns from ViTs and uses CNNs to achieve high accuracy on image classification tasks (Woo et al., 2023). We use one such model to classify road quality from satellite images⁵. Below, we outline the steps we take to assess road quality.

Satellite Images for fine-tuning. Fine-tuning a vision model involves taking a pre-trained model and adapting it for a specific image-classification task. In order to ensure appropriate fine-tuning and enable our vision model to accurately predict road quality, we need large-scale road-image data. We use the road-image dataset of Brewer et al. (2021) which consists of 53,677 labeled satellite images of roads in different conditions, and a classification representing quality of each road: 0 (poor), 1 (decent) and 2 (high). We divide the data into training, validation and test datasets, and use 70% of the data for training, 15% for validation and 15% for testing. For a detailed analysis of the road-image dataset, see the Appendix, Section J. In section 5.1, we present the results from our fine-tuned model, when applied to roads in areas with close elections within the effective bandwidth.

Ohio Satellite Images. While the fine-tuning of our vision image-classification model uses images from Brewer et al. (2021), our Ohio application uses aerial imagery from the National Agriculture Imagery Program (NAIP) and Ohio Statewide Imagery Program (OSIP3). These sources provide substantially broader coverage than the earlier hand-collected sample,

⁵see the Appendix, Section J for details.

but image resolution and timing still vary across products and years. For that reason, our empirical analysis does not rely on a single before-and-after image pair. Instead, Section 5.1 aggregates predicted road quality within pre-election and post-election windows around each renewal election, excludes election-year imagery, and censors each window at adjacent renewal elections so the same subdivision-year is not reused across levy cycles. For interpretability, the main Table 4 reports NAIP-based Road Quality Rating and Road Quality Score outcomes, while pooled NAIP-OSIP3 standardized estimates are used only in supporting diagnostics. Figure 2 presents two illustrative NAIP road images from Ohio, one classified as poor quality and one classified as high quality.



Figure 2: Road Quality Satellite Images for Ohio

Road Quality Rating. Our fine-tuned vision model classifies each satellite image into one of three categories: 0 (poor), 1 (average), or 2 (high). This rating is based on visual cues such as pavement quality, visible cracks, patchwork, and overall surface condition, as learned by the vision model from the labeled training data. For each image, the model outputs both the predicted class and a confidence score, which reflects the model’s certainty in its prediction. These ratings provide a standardized, objective measure of road quality across locations and time periods, enabling us to compare changes in road conditions before and after tax levy referendums for both treated and control groups.

Road Quality Score. Even though our original model output is a road quality rating⁶, we

⁶This is a direct result of our training data from [Brewer et al. \(2021\)](#) which uses the discrete 0 (low), 1

also convert the discrete model output to a continuous measure of road quality between 1 and 100. This Road Quality Score (RQS) combines the predicted road quality rating with the model’s confidence so that higher values correspond to better predicted pavement condition. We use RQS as a companion outcome to RQR because it preserves the ordinal classification while allowing finer comparisons across images and election windows.

Next, we share a summary of means and standard deviations of the predicted road quality rating and road quality score for the treatment and control groups, before and after the referendum.

Table 2: Predicted Road Quality by Treatment Status and Period

	Treated (Failed Levy)	Control (Renewed Levy)
Panel A: Road Quality Rating (0, 1, 2)		
Pre-Election	1.62	1.56
	(0.490)	(0.499)
Post-Election	1.47	1.56
	(0.505)	(0.499)
Panel B: Road Quality Score (1–100)		
Pre-Election	74	71
	(17.3)	(17.9)
Post-Election	68	70
	(18.0)	(17.2)

Notes: The control group comprises townships within the effective RD bandwidth that successfully renewed their road tax levies; the treated group comprises those that failed to renew. Ratings are derived from our fine-tuned vision model. Scores combine the predicted class and model confidence into a 1–100 measure. Standard deviations are in parentheses.

(medium), 2 (high) road-quality metric.

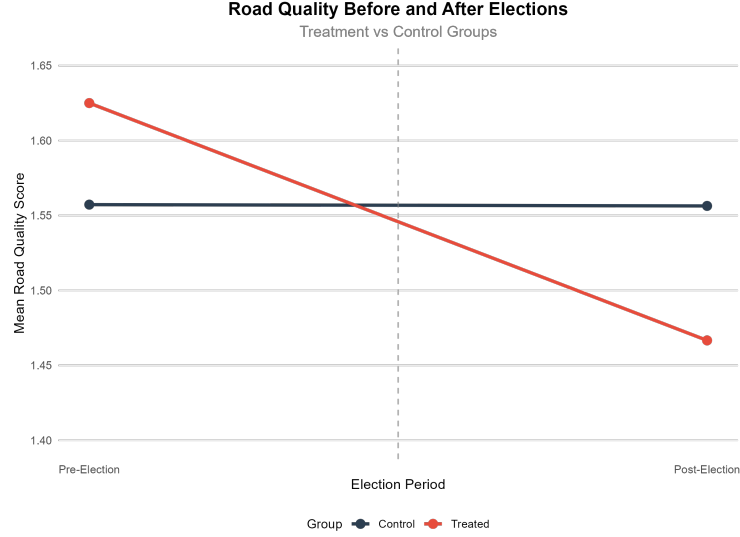
Panel A of Table 2 summarizes the Road Quality Rating (RQR) in a simple descriptive comparison. Treated areas, i.e. jurisdictions that cut their renewal road tax levies, started with a slightly higher average RQR (1.62) than control areas (1.56) before the referendum, but experienced a decline from 1.62 to 1.47 after the referendum, while control areas remained at 1.56.

Panel B of Table 2 presents the continuous Road Quality Score (RQS), where treated areas also began with a higher average RQS (74) compared to control areas (71) before the referendum. After the referendum, RQS fell from 74 to 68 in treated cities, but only slightly from 71 to 70 in control cities. These descriptive differences are suggestive, but the formal evidence in Section 5.1 comes from the stacked-event regression discontinuity design.

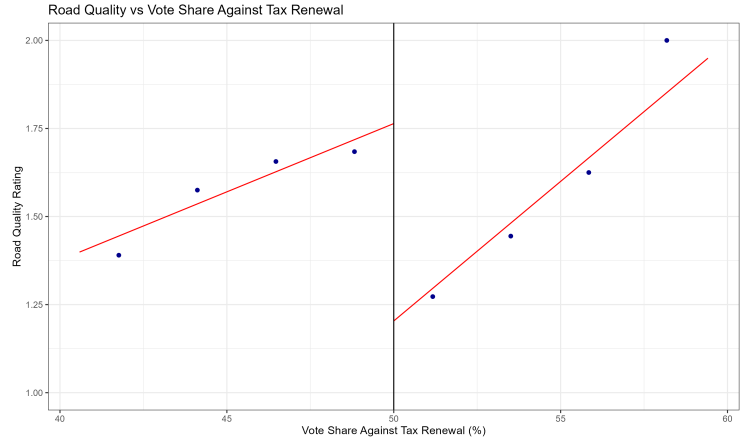
Figure 3 provides descriptive evidence on road quality around the referendum. Panel A illustrates the before-and-after difference in road quality rating for treated and control groups, while Panel B plots post-election road quality against vote share against renewal. Both panels are consistent with lower road quality on the fail-levy side of the cutoff, but Section 5.1 provides the formal stacked-event RD estimates.

3.5 Outcome Variable: Median House Price

Our house price data comes from a CoreLogic® dataset of actual sales transaction prices in Ohio from 1995 through 2021 containing over 7 million observations. The dependent variable, *Median House Price*, reflects the median sale price of houses within a specific city and year. For example, for the houses sold in Delaware Township during the year 2002, the median sale price was \$205,041. We take precaution to only include arm’s-length transactions, and we restrict our attention to single-family residential structures for comparability. The overall sample mean for the 10-year period from the time of votes considered in this study is \$166,082 in constant 2010 dollars with a standard deviation of \$372,135 which suggests the presence of outliers. Although our use of median sale price addresses outliers, one of our robustness checks drops 1% tails and re-estimates the treatment effects (Figure 10).



(A) Before and after referendum



(B) Road Quality vs Vote Share Against Renewal (%)

Figure 3: Difference in Road Quality Rating by Treatment Status and Vote Share

Figure 4 shows house prices from five years after the vote graphed alongside the percent of votes against the tax levy⁷. The points represent the mean house price for the 10 representative bins of vote shares for the average effective bandwidth around the cutoff of 50%. The graph suggests a discontinuity in house prices at the cutoff, which is the basis of our identification strategy, suggesting that the failure to renew a road tax levy has a negative impact on house prices.

⁷after deflating to 2010 U.S. dollars

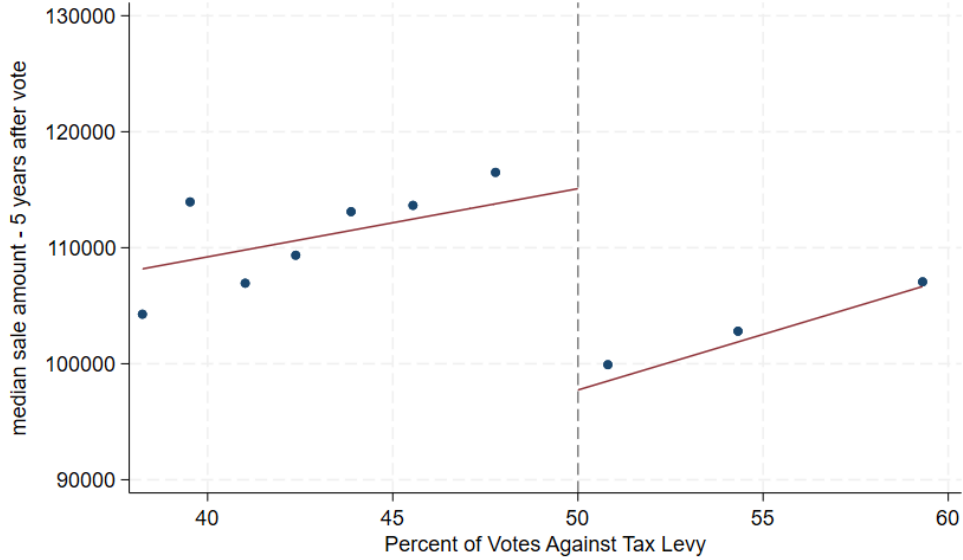


Figure 4: Median Sale Price of Houses: 5 years after vote

3.6 Covariates

Table 3 shows covariate means for both the global sample of all votes in the data set as well as the local sample within a representative effective bandwidth of the decision cutoff. The effective bandwidth displayed in Table 3 is the mean bandwidth for all the housing outcome regressions. The first columns for the global sample show similar values of characteristics between cities that renew and cut road taxes and spending. The second set of columns for the effective bandwidth show even closer similarity between cities that renew and cut road taxes and spending.

The data, captured at the time of the vote and as observed in Table 3, shows minimal differences within the effective bandwidth: mean population differs by only 301, and median family income varies by \$436, measured in 2010 U.S. dollars. Other variables, including poverty rates, married household percentages, educational attainment, age distribution, and racial composition, show differences of two percentage points or less, bolstering the comparability of the treatment and control groups. This similarity suggests that any observed differences in outcomes can more confidently be attributed to the treatment effect rather

than to pre-existing differences⁸.

4 Empirical Strategy

In this section, we describe our empirical strategy to estimate the causal effect of cutting road maintenance spending on house prices. One key feature of our quasi-experimental design is the exogeneity of the timing of the election. The timing is determined by the natural expiration of a road maintenance tax levy, which is typically 5 years, and is not impacted by factors such as the prevailing economic conditions or whether a road tax levy was passed or failed in earlier years.

4.1 Regression Discontinuity in Panel Data Setting

Suppose that local government in area i and year t conducts a referendum to renew an existing road tax levy. Let v_{it} be the vote share against the renewal tax levy and v^* be the threshold determining the result of the referendum (levy fails to renew if $v_{it} > v^*$). Let $F_{it} = 1(v_{it} > v^*)$ be an indicator to represent if the renewal road tax levy fails, and y_{it} be the outcome variable median housing sale price. We can write Equation 2 as follows:

$$y_{it} = \alpha + \theta F_{it} + \epsilon_{it} \tag{2}$$

where α is the intercept, θ is the parameter of interest representing the causal effect of cutting a renewal road tax levy and ϵ_{it} is the error term representing all other determinants of the outcome. Around a narrow enough window around the threshold v^* , we can estimate the causal effect of cutting a renewal road tax levy on the outcome variable y_{it} by comparing the outcome variable for cities that narrowly pass the referendum to those that narrowly fail it.

⁸We also find similar covariate means for cities that have no road tax levies, which we explain in Section 5.4

Table 3: Variable Means by Road Tax Levy Renewal Status

Variable	Global			Effective	
	Full Sample	Renewed	Cut	Renewed (Control)	Cut (Treatment)
Population	5,072 (7,936)	4,733 (7,291)	5,139 (8,058)	4,885 (7,036)	5,186 (8,229)
Poverty Rate	0.11 (0.08)	0.11 (0.07)	0.11 (0.08)	0.10 (0.07)	0.11 (0.08)
% with Kids	0.39 (0.08)	0.40 (0.08)	0.39 (0.08)	0.39 (0.07)	0.39 (0.08)
% Households with Children under 18	0.09 (0.06)	0.09 (0.05)	0.09 (0.06)	0.09 (0.06)	0.08 (0.05)
% Less than High School Education	0.16 (0.11)	0.18 (0.12)	0.15 (0.11)	0.18 (0.12)	0.16 (0.10)
% Some College Education	0.25 (0.06)	0.24 (0.06)	0.25 (0.06)	0.24 (0.07)	0.25 (0.06)
% Renters	0.20 (0.11)	0.20 (0.10)	0.20 (0.11)	0.19 (0.09)	0.20 (0.11)
Unemployment Rate	0.05 (0.04)	0.05 (0.03)	0.05 (0.04)	0.05 (0.03)	0.06 (0.04)
% White	0.96 (0.07)	0.97 (0.07)	0.96 (0.07)	0.97 (0.07)	0.97 (0.08)

Table 3: Variable Means by Road Tax Levy Renewal Status (continued)

Variable	Global			Effective	
	Full Sample	Renewed	Cut	Renewed (Control)	Cut (Treatment)
% Black	0.02 (0.07)	0.02 (0.06)	0.02 (0.07)	0.02 (0.07)	0.02 (0.07)
% Married	0.59 (0.09)	0.60 (0.08)	0.59 (0.09)	0.61 (0.08)	0.60 (0.09)
% Separated	0.01 (0.01)	0.01 (0.01)	0.01 (0.01)	0.01 (0.01)	0.01 (0.01)
Income Heterogeneity Index	0.10 (0.08)	0.09 (0.07)	0.10 (0.08)	0.09 (0.07)	0.09 (0.06)
Median Family Income	61,018 (17,649)	58,761 (13,915)	61,467 (18,270)	59,934 (13,655)	60,370 (15,713)
% Under 5 Years Old	0.06 (0.02)	0.06 (0.02)	0.06 (0.02)	0.06 (0.02)	0.06 (0.02)
% Aged 5 to 17	0.20 (0.05)	0.21 (0.04)	0.20 (0.05)	0.20 (0.04)	0.20 (0.05)
% Aged 18 to 64	0.60 (0.05)	0.60 (0.05)	0.60 (0.05)	0.60 (0.04)	0.60 (0.05)
% Racial Minority	0.04 (0.07)	0.03 (0.07)	0.04 (0.07)	0.03 (0.08)	0.03 (0.07)
Number of Observations	3,184	2,656	528	653	269

4.2 Intent-to-Treat (ITT) Estimator

We follow a model of RD design similar to [Cellini, Ferreira and Rothstein \(2010\)](#) and estimate the **Intent-to-Treat** or ITT estimator. We prefer using the ITT estimator instead of the alternative **Treatment on the Treated** (TOT) estimator because the ITT estimator is more suited to our setting given the independence of the renewal elections. As described in [Cellini, Ferreira and Rothstein \(2010\)](#), when the elections are independent, the ITT estimator equals the TOT estimator.

We operationalize our ITT estimator using Equation 3:

$$Y_{i,t+\tau} = \alpha_\tau + \kappa_t + F_{it}\theta_\tau^{ITT} + P_g(v_{it}, \gamma_\tau) + Z_{it}\beta_\tau + \epsilon_{i,t+\tau} \quad (3)$$

Equation 3 shows a city i that holds an election in year t and we study this city's outcome τ years later. $Y_{i,t+\tau}$ represents the outcome variable for city i at year $t + \tau$. We define treatment as failure of a city, village or township to renew its road maintenance tax levy, which is represented by the indicator F_{it} . θ_τ^{ITT} is the causal effect of failing to renew road tax on the outcome. $P_g(v_{it}, \gamma_\tau)$ is a polynomial function of the running variable v_{it} , which is the percent of votes against the renewal tax levy. α_τ and κ_t represent timing and year-specific fixed effects. Z_{it} is a vector of control variables that include city-level demographics, economic conditions, and other relevant covariates. $\epsilon_{i,t+\tau}$ is the error term.

We use the bandwidth selection method of [Calonico et al. \(2019\)](#) to find the mean optimal bandwidth h and then conduct a local polynomial regression after choosing a weighting scheme k . The bandwidth h determines the size of the neighborhood around the cutoff v^* , defined as $(v^* - h, v^* + h)$. Only observations within this neighborhood are used to compute the bias-corrected treatment effect estimate $\hat{\tau}$. For a sufficiently small neighborhood, the continuity assumption central to the RD estimator is considered valid. We also cluster the standard errors by city to account for any serial correlation between years within each city. The weighting scheme k determines the weights of the observations within the neighborhood

$(v^* - h, v^* + h)$ and is crucial in estimating θ_τ . Common weighting schemes include uniform, triangular, and Epanechnikov. We use the default Mean Squared Error Regression Discontinuity (MSERD) method to compute the effective bandwidth (h) and bias bandwidth (b) for the outcome variable. This method identifies the bandwidth that minimizes the trade-off between bias and variance of the treatment effect estimate. All observations are used to estimate h and b , but only those within the effective bandwidth h are used to identify our treatment effect estimates θ_τ for different τ years.

For road-quality outcomes, we apply the same close-election RD logic in a stacked-event setting. Each renewal election defines an event, and predicted road quality is aggregated over pre-election and post-election windows around that vote. We omit election-year imagery ($t + 0$) and censor each event window at the adjacent renewal elections so that the same subdivision-year is not assigned to multiple levy cycles. The main road-quality table fixes the pre-election window at $t - 3$ to $t - 1$ and reports rolling 3-year post-election windows, controlling for the corresponding pre-election level. Table 4 reports NAIP-based raw Road Quality Rating and Road Quality Score outcomes so the units remain directly interpretable across columns. Supporting outputs report pooled standardized NAIP-OSIP3 outcomes, placebo tests, and alternative event-window specifications.

4.3 Testing for Anticipation under Rational Expectations

A central premise of the present paper is that the close-election design also delivers a direct test of the rational-expectations (RE) capitalization benchmark. Under RE, the marginal buyer at date t should price housing as the present value of the future service-flow from the public-capital stock $G_{t+\tau}$ that the property gives access to. Because each renewal election is scheduled by the predetermined expiration of the prior levy (a 5-year cycle), the timing of the referendum itself is anticipated. The *outcome* of a close election is, by construction, unpredictable in a narrow window around $v^* = 0.5$. RE therefore predicts that the price gap between narrowly-passing and narrowly-failing jurisdictions should be (i) close to zero

in periods strictly before t , and (ii) approximately equal to the present value of the entire future maintenance shortfall at $t + 0$, with no further drift thereafter.

We test the first prediction directly. Equation 3 delivers an estimate of θ_{τ}^{ITT} for $\tau \in \{-3, -2, -1\}$, the three years strictly preceding the vote. Under RE these pre-period coefficients should be statistically indistinguishable from zero (Bishop and Murphy, 2019). Failure to reject this joint null would imply that sale prices do not move in advance of a then-unanticipated tax cut — ruling out a “prices already adjusted before the close election” story that would otherwise contaminate the post-period response.

The second prediction — a one-shot capitalization at $t + 0$ of the entire future shortfall — is examined in conjunction with the post-period coefficient path in Section 5. Section 6 formalizes the wedge between the empirical price path and the RE benchmark, and Section 7 translates that wedge into a structural test using the calibrated dynamic general-equilibrium model.

5 Results

5.1 Road Quality Decline

First, we present the results of our analysis on the impact of cutting road maintenance taxes on road quality. As discussed in Section 3.4, we use a fine-tuned AI vision model to assess road quality based on satellite imagery data. The model gives us a road quality rating of 0 (poor), 1 (medium), or 2 (high), which we also convert to a continuous variable called Road Quality Score (RQS) ranging from 1 to 100. The main estimates in Table 4 use NAIP-based raw outcomes so the reported units are directly interpretable, while pooled NAIP-OSIP3 standardized specifications are used only in supporting diagnostics.

Table 4 fixes the pre-election window at $t - 3$ to $t - 1$ and begins with a placebo RD in Column (1) using average pre-election road quality over those same years. Columns (2) through (6) then report RD estimates for rolling 3-year post-election windows using the Road

Table 4: Road Quality Decline after a Failed Renewal Election

	$t-3$ to $t-1$	$t+1$ to $t+3$	$t+2$ to $t+4$	$t+3$ to $t+5$	$t+4$ to $t+6$	$t+5$ to $t+7$
Panel A: Road Quality Rating						
Change in Road Quality	0.042 (0.089)	0.017 (0.080)	-0.188* (0.099)	-0.238*** (0.070)	-0.250 (0.163)	-0.075 (0.232)
Eff. bandwidth (h)	7.47	10.85	4.53	4.43	5.11	7.06
Bias bandwidth (b)	13.21	18.55	8.73	8.34	7.11	8.96
Eff. observations	170	209	53	46	34	23
Total observations	515	378	279	221	132	46
Panel B: Road Quality Score						
Change in Road Quality	1.144 (3.035)	0.648 (3.088)	-6.914** (3.445)	-8.362*** (2.621)	-9.626 (6.401)	-5.038 (9.094)
Eff. bandwidth (h)	7.38	8.84	4.66	4.68	5.15	7.09
Bias bandwidth (b)	13.01	15.31	8.83	8.73	7.11	8.99
Eff. observations	167	160	55	51	34	24
Total observations	515	378	279	221	132	46

Notes: Entries are bias-corrected sharp RD estimates at the 50% vote-share cutoff. Column (1) is a placebo RD using average pre-election road quality over $t-3$ to $t-1$. Columns (2)–(6) use rolling 3-year post-election windows, exclude $t+0$, and control for population and the corresponding pre-election outcome. Standard errors clustered by ten-digit FIPS code are in parentheses.

Quality Rating (RQR, 0–2) and Road Quality Score (RQS, 1–100). The placebo estimate is small in both outcomes, and the post-election columns exclude $t+0$ and control for the corresponding pre-election level.

The rolling-window pattern suggests that road deterioration emerges with a lag. In Column (2), which uses the post-election window $t+1$ to $t+3$, the estimated effects are close to zero and statistically insignificant. By Column (3), which uses $t+2$ to $t+4$, the estimates become negative and statistically significant: RQR falls by about 0.19 points and RQS by about 6.9 points. Relative to the pass-side local counterfactual, these estimates imply declines of about 16% for RQR and about 14% for RQS. Column (4), which uses $t+3$ to $t+5$, remains strongly negative, with RQR lower by about 0.24 points and RQS lower by about 8.4 points. In Column (5), which uses $t+4$ to $t+6$, the estimates remain negative but are less precise. By Column (6), which uses $t+5$ to $t+7$, the estimates are no longer informative because the available sample becomes thin. Taken together, the rolling 3-year

windows are consistent with gradual infrastructure deterioration rather than an immediate visible break right after the vote.

Supporting diagnostics from the stacked-event analysis include pooled standardized specifications, a pre-election placebo RD, the full pre-specified window grid, bandwidth sensitivity, sample flow, and continuity checks for image availability and source composition. In those supporting results, the pooled standardized estimates are generally negative and the pre-election placebo is small, which is consistent with deterioration after the tax cut rather than a discontinuity in baseline road quality.

The next section presents our main results on the impact of cutting road maintenance taxes on house prices. Before doing so, Figure 5 overlays the dynamic RD coefficients on road quality with those on house prices in event time. The visual comparison is the central piece of reduced-form evidence in this paper: the price gap between narrowly-failing and narrowly-passing jurisdictions opens only after the road-quality gap is statistically detectable. We return to the implications of this synchronization in Section 6.

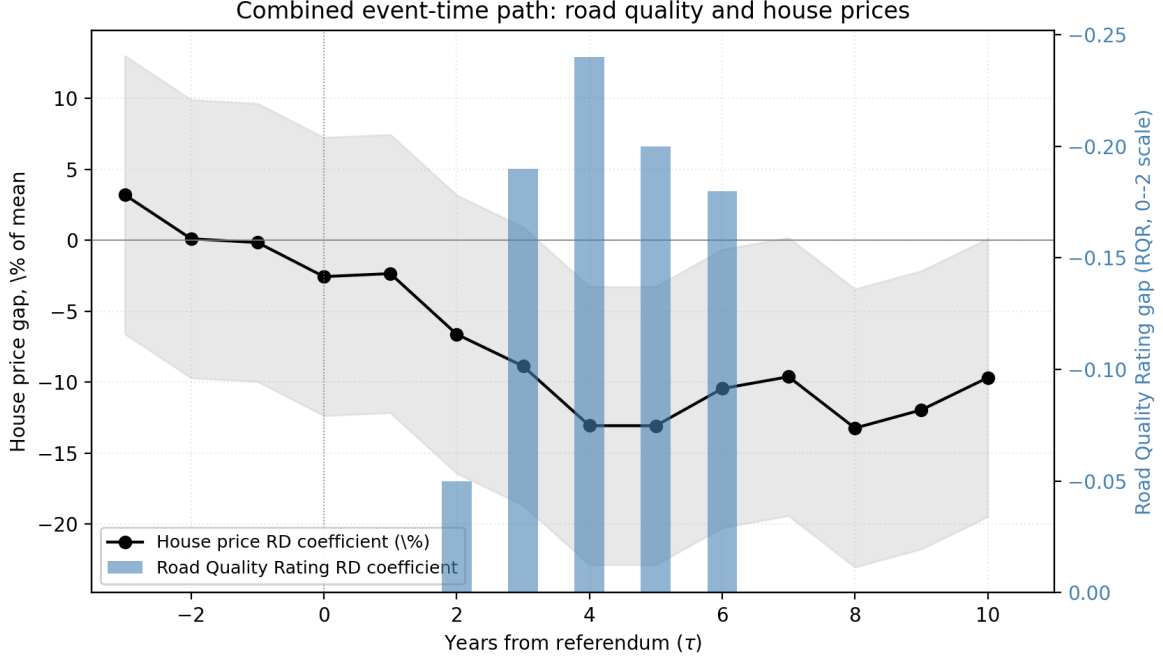


Figure 5: Combined event-time path of road-quality and house-price RD coefficients. House-price coefficients (black, left axis, percent of mean) are flat in the pre-period and through $\tau = +3$, then open sharply between $\tau = +4$ and $\tau = +9$. Road-quality coefficients (blue bars, right axis on inverted scale, Road Quality Rating units) are reported for rolling 3-year post-election windows and become statistically detectable in the $\tau \in [2, 4]$ window before peaking in the $\tau \in [3, 5]$ window. The synchronization between the visibility of physical decay and the timing of the price response is the central piece of reduced-form evidence in this paper.

5.2 House Prices Decline

Table 5 below shows the ITT estimates of failing a road tax levy on housing sale prices.

Each treatment effect estimate represents the discount in median sale price for cities that cut road taxes relative to otherwise similar cities that renew the taxes. We show dynamic effects starting three years before the vote and extending to ten years after the vote. Treatment effect estimates for years 4 through 9 after the vote are statistically significant in Table 5. The estimate for year $t + 10$ has a smaller estimate and is only significant at the 10% level, suggesting that the effect of tax and service cuts on house prices may peter out ten years after the vote. Overall, we find an average reduction of \$15,349 in median house price over

the 10-year period for houses in cities that vote to cut road tax levies, representing 9% of overall house value⁹. The 9% reduction we estimate may be compared to [Gonzalez-Navarro and Quintana-Domeque \(2016\)](#) which observes a change in property values of 17-28% after paving previously unpaved roads and the 13% increase in house prices [Theisen and Emblem \(2021\)](#) finds in towns nearest to a newly-constructed highway¹⁰.

⁹The average treatment effect estimate of \$15,349 was divided by the mean house sale price in the dataset of \$166,000 to get 9%.

¹⁰Note that these studies focus on the effect of developing paved roads whereas we focus on deterioration of existing paved roads.

Table 5: Effect on median house prices of failing to renew a road tax levy

Panel A: Years $t - 3$ through $t + 3$							
Year	$t-3$	$t-2$	$t-1$	t	$t+1$	$t+2$	$t+3$
Treatment effect	5,307	166	-273	-4,261	-3,908	-11,001	-14,733*
Standard error	(7,341)	(6,943)	(7,391)	(7,955)	(8,719)	(9,405)	(7,989)
Effective band- width (h)	10.30	10.06	8.76	7.97	9.78	9.63	11.93
Bias bandwidth (b)	20.22	17.48	14.39	13.03	17.50	18.22	22.69
Effective observa- tions	724	725	650	587	774	778	1,003
Total observations	2,552	2,631	2,696	2,784	2,764	2,699	2,640
Panel B: Years $t + 4$ through $t + 10$							
Year	$t+4$	$t+5$	$t+6$	$t+7$	$t+8$	$t+9$	$t+10$
Treatment effect	-21,701***	-21,706**	-17,365**	-15,975**	-21,984**	-19,857**	-16,090*
Standard error	(7,747)	(8,751)	(8,355)	(7,248)	(9,074)	(7,751)	(9,027)
Effective band- width (h)	8.52	11.20	9.79	13.16	8.25	7.28	6.24
Bias bandwidth (b)	16.25	20.16	17.32	23.42	15.28	14.09	16.27
Effective observa- tions	688	922	764	1,061	591	505	402
Total observations	2,614	2,531	2,438	2,324	2,199	2,115	2,017

Notes: Outcome is median house price in constant 2010 U.S. dollars. Unit of observation is the city-year, so a treatment effect of -\$21,701 in year $t + 4$ means that four years after the vote, cities that fail to renew road taxes have median sale prices \$21,701 lower than similar cities that renew. Covariates include the demographic and socioeconomic controls listed in Table 3.

Figure 6 provides an event-study plot that summarizes the treatment effects for each

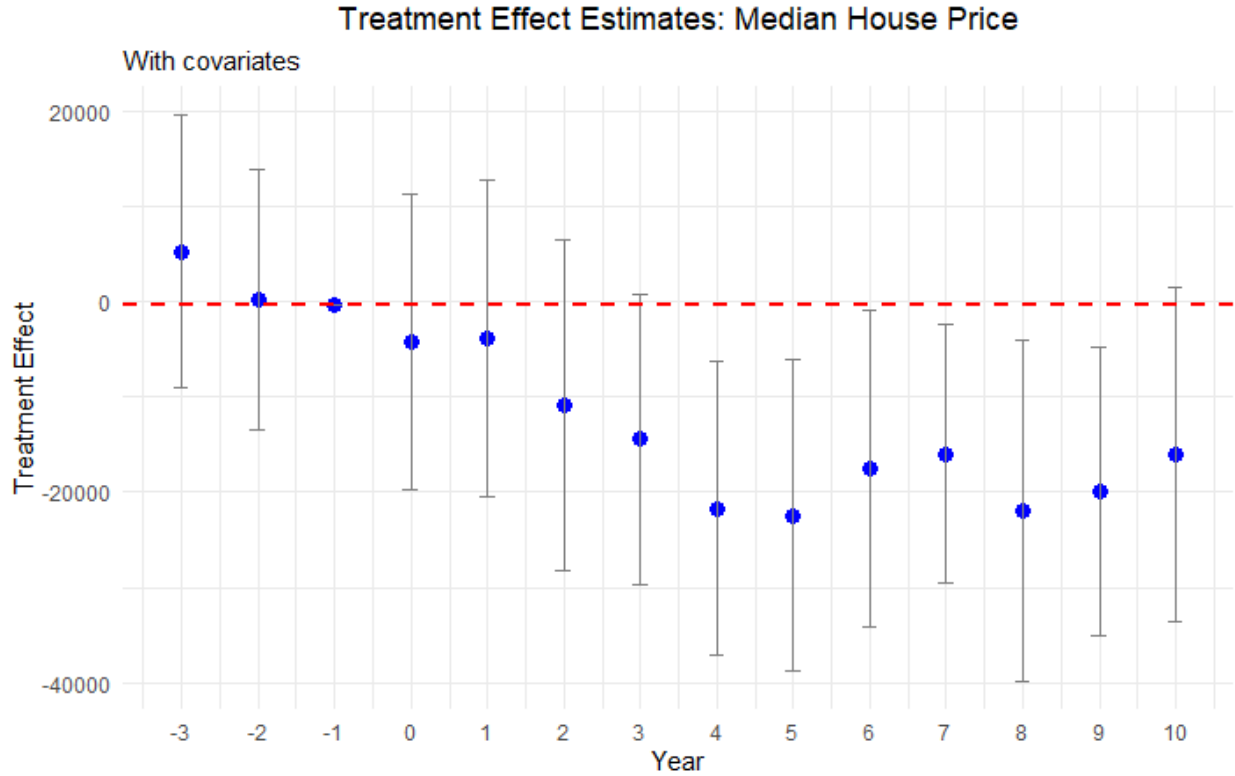


Figure 6: Effect plot for Median Housing Price

time period. In the graph, we include placebo years up to 3 years before the treatment to show that the median housing prices are statistically identical for cities above and below the threshold prior to the treatment. Each dot represents the treatment effect point estimate for that year and the bar around it represents the 95% confidence interval for that estimate. For year 0, which is the year of the vote, we see a slight decrease in the estimate. However, this effect is not statistically significant, as evidenced by the confidence interval containing the null effect. Up to year 3, we observe that the treatment effect estimates are fairly close to zero, and the confidence interval overlaps with zero. As stated previously and shown in Table 5, we start to see a sizable increase in treatment effect from year 4 onwards and continue to observe it through year 9 after the vote.

5.3 Heterogeneity Analysis

We show the results of our heterogeneity analysis, where we explore the differential impact of cutting road maintenance spending on house prices in urban and rural neighborhoods, for tax levies of different sizes, and for different housing price quantiles.

Urban vs Rural neighborhoods: Analyzing regional heterogeneity is important because the impact of public goods like road maintenance on house prices can differ substantially between areas due to factors such as differences in elasticity of housing supply. In this section, we check how treatment effects differ between urban and rural areas¹¹.

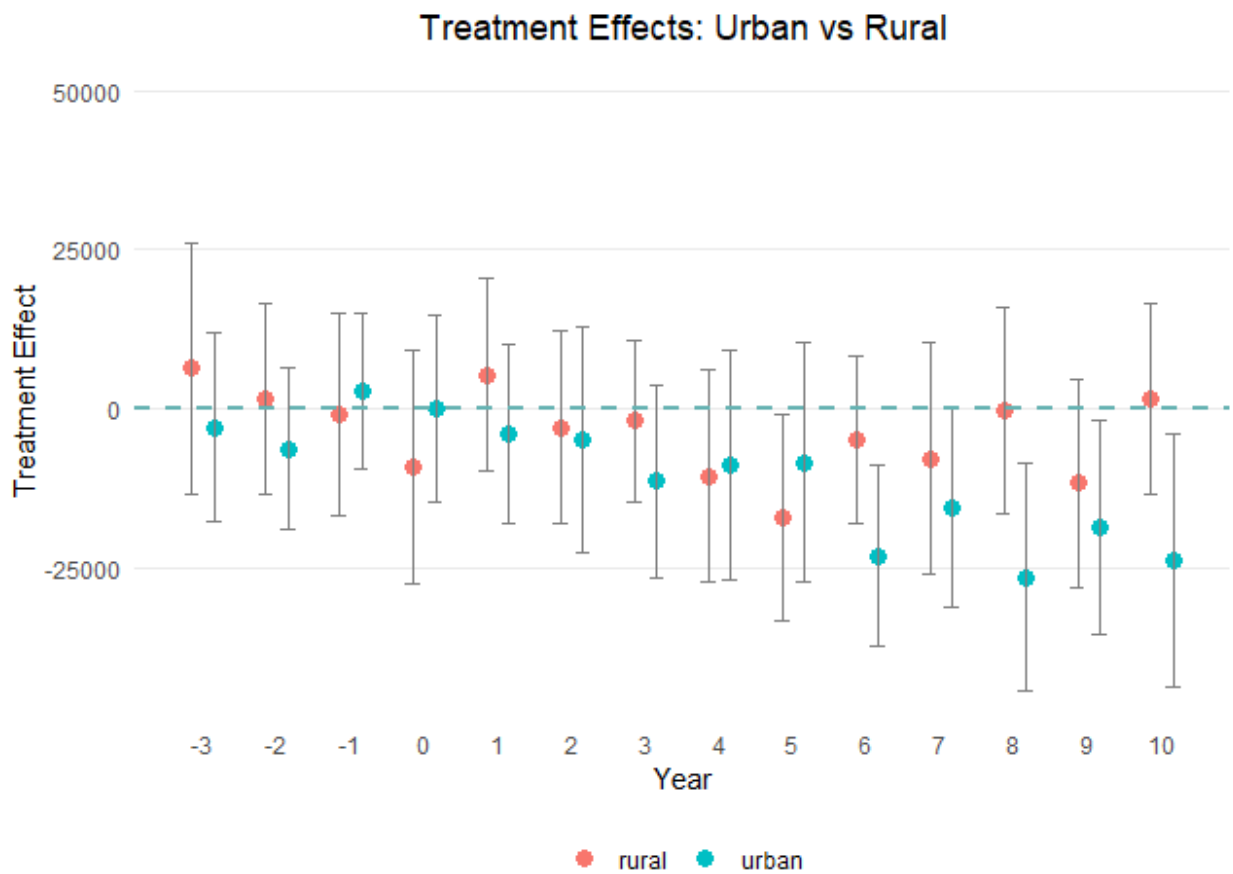


Figure 7: Median Housing Price in Urban and Rural Areas

¹¹Urban areas are identified using data from U.S Census Bureau. We strictly consider urban areas with at least 0.3 allocation factor in 2010 and do not consider urbanized clusters, making our classification of urban areas more restrictive.

As shown by Figure 7, we do not find any significant differences in housing prices in rural areas after a renewal tax levy fails to pass. On the other hand, we do find a statistically significant decline in housing prices in urban areas starting six years after voting. The standard errors are somewhat smaller for the rural estimates due to a larger number of observations. The stronger effects in urban than rural areas may stem from a more inelastic housing supply (Brasington, 2002). Overall, we find that housing prices decrease by \$13,302 on average over the decade¹² after cutting road maintenance tax levies in urban areas.

Tax magnitude: We check for dose-response by splitting the sample by the size of the tax levy. First, we consider splitting the sample based on the median size of the tax levies. The median size of the tax levies in our sample is 1.9 mills, however we find that the tax levies above and below the median size do not show any significant differences in treatment effects. Next, we focus on the top quartile of the tax levies, which in our sample is the set of tax levies ranging from 2.1 to 8 mills. Figure 8 below presents the results for this subsample. The treatment effect estimates for these larger tax cuts are both statistically significant and of greater magnitude, indicating a more pronounced decline in house prices when the reduction in road tax levies is more substantial.

Despite the reduction in sample size leading to wider standard errors in Figure 8, the overall pattern in decline of house price remains, with mean treatment effect for this subsample being approximately \$30,000 which is about double the effect observed in the full sample. This finding supports the presence of an intensive margin effect: the larger the tax cut, the greater the negative impact on housing prices. Such a pattern is consistent with a dose-response relationship, where the magnitude of the fiscal shock translates directly into the size of the decline in property values.

¹²This is equal to $7.6\% = \frac{13,302}{175,217} \times 100$, where the denominator is average sale price of houses in urban areas in our dataset.

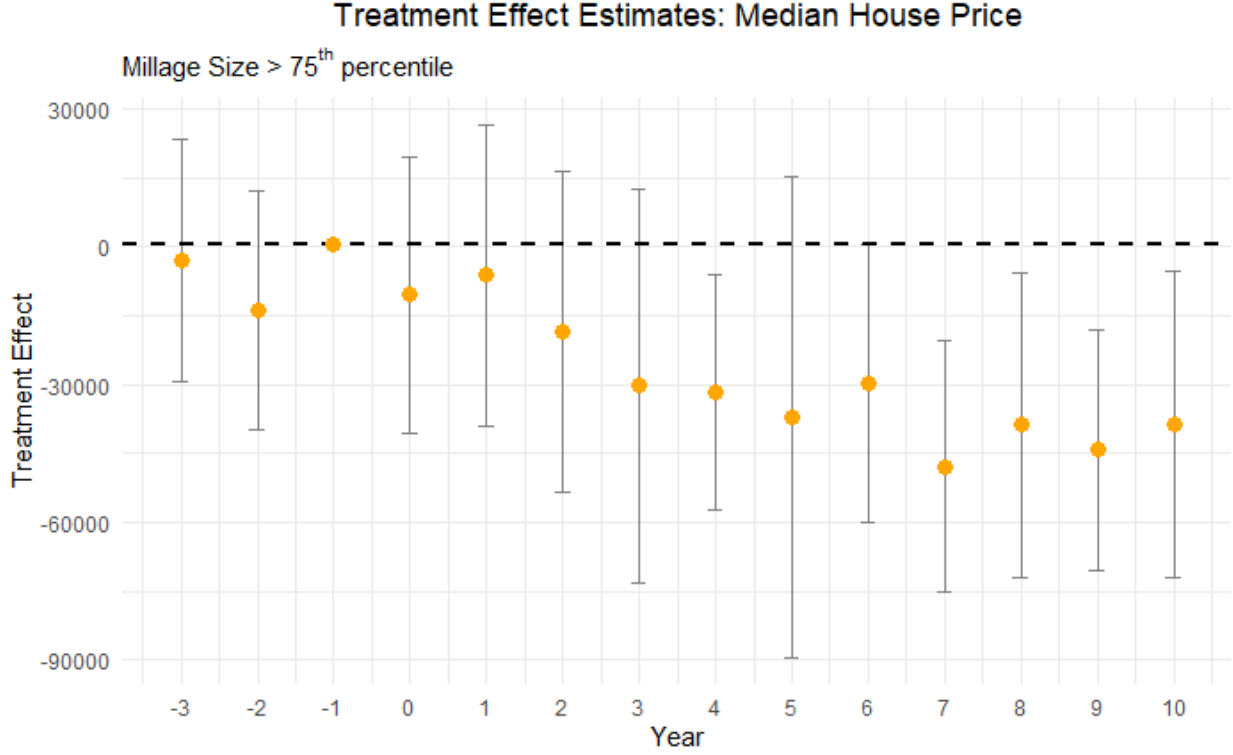


Figure 8: Effect of Large Road Maintenance Tax Cuts (>2 mills) on Median House Price

RD Quantile Estimation: We analyze our results by estimating quantile-level treatment effects, as suggested by [Frandsen, Frölich and Melly \(2012\)](#), to study how the treatment's impact varies at different quantiles of the outcome variable.

Table 6 shows the treatment effect heterogeneity of cutting road spending on high and low quantiles of median house prices. The top percentiles consistently exhibit a statistically significant decline in house sale prices, beginning in year 6 after the reduction in road spending. In contrast, the lower percentiles do not demonstrate a consistent treatment effect. This suggests a differential impact, where higher-valued properties are more sensitive to road disrepair than lower-valued houses. Figure 9 contrasts the treatment effects of the 20th and 80th percentiles of house sale prices in an effect plot to highlight this differential impact of reduction in road maintenance spending.

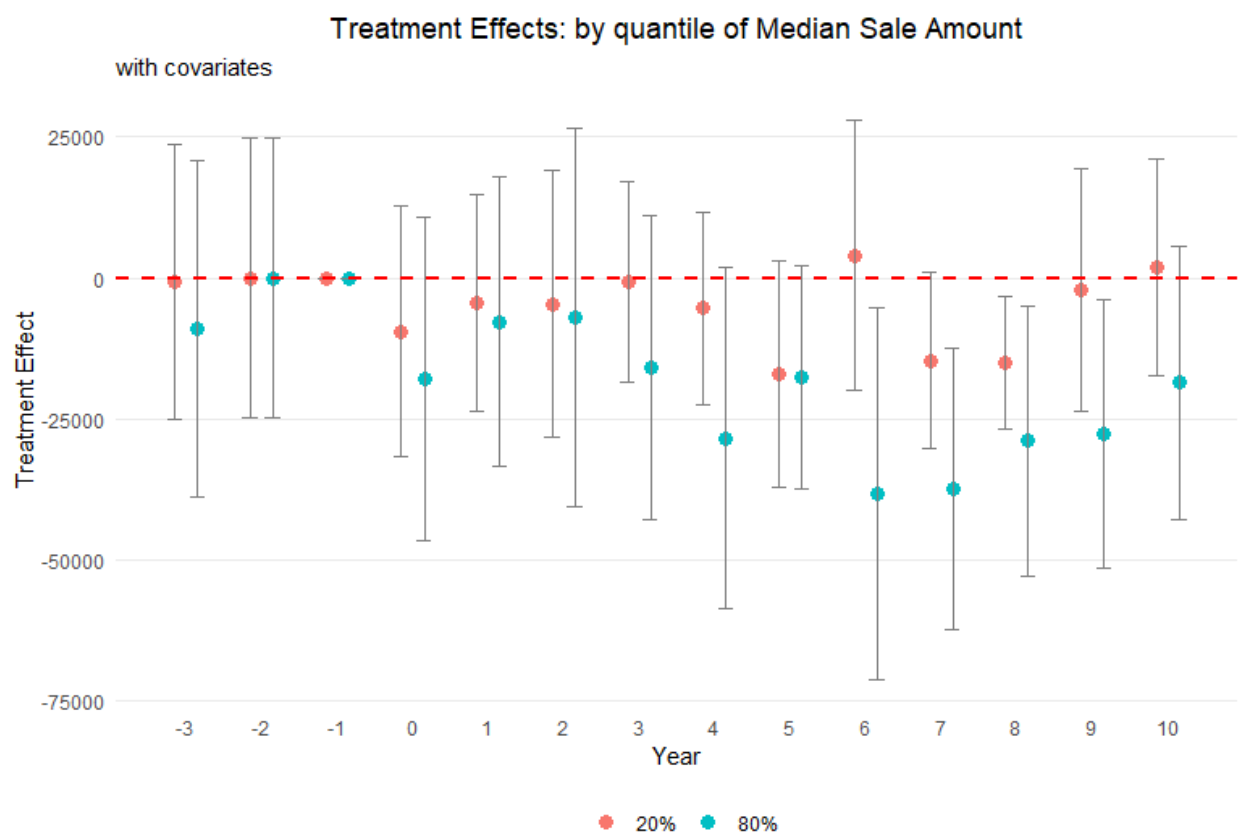


Figure 9: Treatment Effect of Cutting Road Maintenance Taxes for 20th and 80th Percentiles of Median House Price

Table 6: Quantile-level Treatment Effects of Cutting Road Spending on Median House Prices

Percentile	$t + 4$	$t + 5$	$t + 6$	$t + 7$	$t + 8$	$t + 9$	$t + 10$
10%	-6,433 (9,364)	-22,570** (9,065)	-9,602 (9,205)	-12,984 (8,420)	-11,217 (9,136)	-6,569 (10,809)	-1,326 (8,793)
20%	-5,400 (9,983)	-15,070 (9,886)	4,014 (7,443)	-14,682* (8,502)	-15,040* (8,160)	-3,228 (10,435)	624 (8,509)
70%	-21,760* (12,333)	-11,171 (11,806)	-38,082*** (12,835)	-36,685*** (12,163)	-21,356* (12,218)	-25,605* (13,984)	-18,600* (9,872)
80%	-28,478** (13,343)	-16,379 (11,404)	-38,460** (18,623)	-37,470*** (12,169)	-28,950** (12,507)	-27,800** (12,421)	-18,658 (11,808)
90%	-51,470*** (18,409)	-34,604** (15,837)	-38,510* (22,194)	-27,039* (16,308)	-29,010* (16,640)	-49,093*** (14,498)	-36,662* (19,110)

Notes: The outcome is median house price in constant 2010 U.S. dollars. The unit of observation is the city-year, so a treatment effect of $-\$28,478$ means that at the 80th percentile of house prices four years after the vote, cities that fail to renew road taxes and the associated spending have houses that sell for $\$28,478$ less than cities that vote to renew them. Covariates include demographic and socioeconomic characteristics as described in Table 3.

5.4 Robustness Tests

We conduct several robustness tests to ensure the validity of our results. We test the sensitivity of our treatment effect estimates by removing contaminated observations and winsorizing the outcome variable to reduce the influence of outliers. Lastly, we consider other tax levies to alleviate endogeneity concerns.

Removing contaminated observations: In this test, we focus on independence and exogeneity. Estimates may be biased if tax levies for additional money pass after renewal tax levy decisions. To address this concern, we exclude observations from our analysis if a tax levy for additional funding is introduced and passed within a ten-year period following a renewal tax levy vote, as these votes may contaminate the treatment effect of the renewal tax levy vote. For example, consider a scenario where a city votes to fail its renewal tax levy in the year 2000 i.e. funding for road maintenance is cut. If that city subsequently introduces and passes a tax levy for additional road spending in 2004, we exclude all votes for that city from 2005 through 2010. This exclusion ensures that the effect on house prices from the 2000 vote are captured for uncontaminated years but not for years after 2004 when

the effect of additional road taxes may counteract the drop in tax money from the elections in the year 2000.

Table 7: Effect on median house prices of failing to renew a road tax levy – uncontaminated observations only

Panel A: Years $t - 3$ through $t + 3$							
Year relative to vote	$t - 3$	$t - 2$	$t - 1$	t	$t + 1$	$t + 2$	$t + 3$
Treatment effect	5,295	−369	1,207	−2,482	−3,107	−12,236	−15,026*
Standard error	(7,255)	(7,359)	(7,776)	(8,482)	(9,270)	(10,123)	(8,746)
Effective bandwidth (h)	10.25	9.70	8.84	7.48	9.19	8.20	10.09
Bias bandwidth (b)	20.08	16.41	14.45	12.03	16.04	16.33	20.21
Effective observations	719	682	630	510	680	604	763
Total observations	2,552	2,574	2,593	2,640	2,627	2,536	2,449
Panel B: Years $t + 4$ through $t + 10$							
Year relative to vote	$t + 4$	$t + 5$	$t + 6$	$t + 7$	$t + 8$	$t + 9$	$t + 10$
Treatment effect	−21,082**	−16,827	−17,565*	−18,179**	−20,427**	−29,970***	−23,593**
Standard error	(9,145)	(11,228)	(10,128)	(9,202)	(9,829)	(9,059)	(10,838)
Effective bandwidth (h)	6.85	5.64	8.48	9.34	7.41	5.51	6.20
Bias bandwidth (b)	13.19	14.82	14.53	18.06	14.23	11.74	16.54
Effective observations	473	367	573	609	442	303	322
Total observations	2,389	2,274	2,145	2,016	1,890	1,787	1,666

Notes: Sample excludes any potentially contaminated city–year observations. Outcome is median house price in constant 2010 U.S. dollars. Each coefficient is the difference between treated and control cities in the indicated year; standard errors are heteroskedasticity-robust. All regressions control for the demographic and socioeconomic covariates listed in Table 3.

Upon implementing this data filtration, we observe that the treatment effect of the re-

newal levies on housing prices, measured from $t + 1$ to $t + 10$, remains consistent with our initial findings. This consistency in treatment effect, despite the exclusion of potentially confounding data, lends credence to our results. The standard errors increase slightly due to the reduction in sample size caused by the aforementioned data filtration process.

Placebo cutoffs: In our primary analysis, the pivotal threshold for the vote share running variable is 50%, indicating whether a renewal levy passes or fails. Although we find significant treatment effects using this 50% threshold, it could be random jumps in the data rather than cutting road taxes and funding that are responsible for the significant estimates. To this end we conduct a series of placebo tests using alternative cutoffs: 30%, 40%, 60%, and 70%. Table 8 below summarizes the results from the placebo cutoffs analysis.

Table 8: Robust Treatment Effect Estimate for Placebo Cutoffs

Years after vote	30%	40%	60%	70%
$t + 4$	2,578 (8,209)	9,149 (7,284)	9,419 (11,462)	-12,987 (14,365)
$t + 5$	-6,381 (9,086)	-29,077 (20,680)	6,383 (10,786)	41,683** (17,836)
$t + 6$	7,681 (9,616)	5,573 (8,120)	-1,095 (8,612)	-14,226 (15,733)
$t + 7$	1,162 (10,468)	3,982 (8,191)	-12,050 (9,396)	22,261 (19,765)
$t + 8$	4,334 (9,670)	12,881 (8,625)	3,593 (10,061)	31,696*** (6,902)
$t + 9$	851 (5,921)	7,381 (6,333)	-6,935 (7,114)	42,790*** (15,663)
$t + 10$	10,032 (10,599)	-35,038 (35,569)	324 (8,220)	-8,566 (17,281)

Notes: Robust treatment effect estimates at placebo cutoffs using the estimator from [Calonico et al. \(2017\)](#). The unit of observation is the city-year. Standard errors are reported in parentheses below each estimate.

Table 8 does not show consistently significant treatment effects for any of the placebo cutoffs for our parameter of interest. This absence of significance at thresholds other than the true 50% reinforces the idea that the effects we observe at the 50% mark are not a mere coincidence or a result of random variation in the data, but are indeed attributable to the

dynamics surrounding the passing or failing of renewal tax levies.

Winsorization: The debate over whether to include or exclude outliers continues, with some research suggesting that trimming outliers does not improve mean squared error (Bollinger and Chandra, 2005). We now drop the 1% tails to help curtail the influence of outliers. The overall sample mean after dropping 1% tails is \$150,505 in constant 2010 dollars with a standard deviation of \$115,646. After performing this winsorization step, we re-estimate the treatment effect of failing to renew a road tax levy on housing outcome variables. The results from this estimation process are summarized in Figure 10 below:



Figure 10: Effect of Cutting Road Maintenance Taxes on Median Housing Price after 1% Winsorization

The treatment effect estimates with winsorization mimic those from our baseline regression results qualitatively and quantitatively.

Other Tax Levies: A potential threat to identification is that a cut in road tax levy might be correlated with cuts in other local tax levies, or may simply reveal an underlying,

time-varying taste for lower taxes or smaller government. If the same electorate simultaneously rejects (or approves) levies for police, fire, recreation, or schools, our estimated road-maintenance effect could in fact be picking up broader changes in local public-service bundles that are themselves capitalised into housing prices. To address this concern, we examine whether the election results of a renewal road levy is systematically correlated with the results of other levy referenda held on the same ballot or within the same fiscal year i.e. whenever a road tax levy is cut, are there elections and subsequent tax cuts for other tax levies as well?

Table 9: Association of Road Tax Levy Referenda Results with Other Types of Levies

	Police	Fire	Recreational	School
Estimate	0.248 (0.153)	0.053 (0.043)	0.015 (0.317)	-0.024 (0.092)

Notes: This table presents coefficients from regressions associating road tax levy referendum outcomes with outcomes for other types of levies (police, fire, recreational, school). Standard errors are reported in parentheses. All regression control for year and neighborhood fixed effects, as well as neighborhood characteristics. The school levy analysis is conducted at the county level, while all other analyses are at the county subdivisions level. Statistical significance levels are indicated as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 9 reports coefficients from separate regressions in which each column regresses the pass/fail outcome of another levy type on the pass/fail outcome of road tax levy¹³, controlling for year fixed effects, jurisdiction fixed effects, and demographic covariates. For school levies we cluster to the county level as an approximation, while all other specifications are at the township/city (“county subdivision”) level. The point estimates are small and statistically indistinguishable from zero: a failed road-maintenance renewal is neither more nor less likely to coincide with the failure of levies earmarked for police, fire, recreation, or schools. For example, the largest coefficient, 0.248 for police levies, has a standard error of 0.153 and is not significant even at the 10 percent level. These results suggest that voters treat the road-maintenance question separately from other local tax questions, and that decisions to cut or renew road funding are not simply proxies for a general anti-tax or austerity sentiment.

¹³This is represented by a dummy variable indicating a cut in road tax levies, with 1 as denoted as a cut and 0 denoted as a renewal.

Consequently, our main RD estimates are unlikely to be confounded by contemporaneous changes in other public services.

A more comprehensive approach to address this concern is to check for balance in other tax levies around the cutoff. As the theory of regression discontinuity suggests, cities close to the 50% vote cutoff are as good as randomized along the cutoff. If true, there should be no difference in funding changes between the pass-levy and fail-levy groups. To confirm this, we collect all 50,000 local tax levies for all municipal purposes from 1991 to 2021, including current expense tax levies, parks and recreation, and police funding, for example. We identify the tax levies for the cities within the effective bandwidth and compare the proportion of cities that renew and fail to renew road tax levies that have other tax levies on the ballot for different purposes at the same time as the road tax levy. The pass-levy group has a non-road tax levy on the ballot 48.1% of the time, and similarly the fail-levy group has a non-road tax levy on the ballot 47.96% of the time. The two groups are balanced, as suggested by RD theory. We next focus on the set of cities within our effective bandwidth with non-road tax levies on the ballot. The cities that barely vote to renew road taxes experience a change in other taxes, either an increase in new spending or a cut in existing spending, 12% of the time. The corresponding number for the set of cities that barely votes to cut road taxes is 15%. Again, randomization near the cutoff suggests that both observable and unobservable characteristics are balanced between pass and fail road tax levy groups. Thus, the difference in house prices can be attributed to tax cuts in road maintenance funding and decreases in road quality.

6 Capitalization under Rational Expectations: The Empirical Wedge

The empirical pattern documented in Section 5 is consistent with a tax cut whose consequences are eventually capitalized, but only with a substantial delay. In this section we make

the timing structure explicit. We compare the observed dynamic capitalization path with the path implied by a frictionless rational-expectations (RE) benchmark, identify the empirical wedge between the two, and articulate the two leading mechanisms — limited attention and signal extraction — that the structural model in Section 7 formalizes.

6.1 The Rational-Expectations Benchmark

Let $\Delta_t \equiv p_{h,t}^F - p_{h,t}^P$ denote the difference in the (log) housing price between a jurisdiction that narrowly fails to renew the road-maintenance levy and one that narrowly passes, at horizon τ relative to the election year t . Let $\{A(G_{t+\tau})\}_{\tau \geq 0}$ denote the path of housing-amenity service flows under each outcome, where $G_{t+\tau}$ is the road-capital stock. Under rational expectations and a no-arbitrage condition, the marginal buyer at any date $s \leq t$ should equate the price gap to the discounted sum of the expected service-flow gap:

$$\Delta_s = \mathbb{E}_s \left[\sum_{\tau \geq 0} \beta^\tau \Delta A(G_{t+\tau}) \right] \cdot \frac{1}{r + \delta_h}, \quad (4)$$

where β is the household discount factor, r the user-cost rate, and δ_h the housing depreciation rate¹⁴.

Equation 4 delivers three sharp testable predictions under RE:

- P1.** $\Delta_s = 0$ for $s < t$ (the outcome of a close election is, by construction, unforecastable within a narrow window around $v^* = 0.5$).
- P2.** $\Delta_t - \Delta_{t-1} \approx \mathbb{E}_t[\sum_{\tau \geq 0} \beta^\tau \Delta A(G_{t+\tau})]/(r + \delta_h)$: the *entire* discounted future service-flow gap should be capitalized at the moment the election outcome is realized.
- P3.** For $\tau > 0$, $\Delta_{t+\tau}$ should display no further drift other than the mechanical accumulation of expectational errors, which average to zero in expectation.

¹⁴See the housing Euler derivation in Section 7

6.2 The Empirical Wedge

Section 4.3 reports the pre-period coefficients $\theta_{-3}^{ITT}, \theta_{-2}^{ITT}, \theta_{-1}^{ITT}$ implied by Equation 3. These are economically small and statistically indistinguishable from zero. We thus fail to reject P1.

In contrast, the post-period path strongly rejects P2 and P3. The estimated Δ_t at the election year and through $\tau = 3$ is small, statistically insignificant, and far below the present-value-of-service-flow magnitude implied by the eventual long-run capitalization. The price gap then opens sharply between $\tau = 4$ and $\tau = 9$, reaching a peak of roughly nine percent of the mean sale price, before partly closing by $\tau = 10$. The new combined-timeline figure (Figure 5, presented in Section 5) overlays the dynamic price coefficients on the dynamic road-quality coefficients and makes the central empirical pattern transparent: the price response tracks *observed* road decay, not the *future* decay anticipated by the election outcome.

The magnitude of this wedge is quantitatively important. The cumulative discounted price gap from $\tau = 0$ through $\tau = 3$ is, on average, less than five percent of the eventual long-run capitalization. A frictionless RE-pricing economy would have delivered approximately ninety-five percent of the long-run effect at $\tau = 0$. Instead, less than one-twentieth of it appears before the road decay itself is statistically detectable.

6.3 Candidate Mechanisms

Three distinct microfoundations can reconcile the data with optimizing household behavior. All deliver delayed capitalization, and all nest the RE benchmark as a limiting case. They differ along two dimensions: (i) whether the friction lies on the *buyer* side of the housing market or on the *voter* side of the ballot box, and (ii) what the policy and welfare implications are.

Mechanism A: Limited attention. Following [Chetty, Looney and Kroft \(2009\)](#) and [Finkelstein \(2009\)](#), households may rationally allocate attention away from low-salience fiscal

decisions whose consequences are slow-moving and small per period. Under a fixed cognitive cost of evaluating the present value in Equation 4, agents update their belief about $G_{t+\tau}$ only when an attention-triggering event arrives. The physical deterioration of road surfaces — a directly observable signal — is such a trigger. Before the trigger, $\hat{G}_{t+\tau}$ remains at its pre-election baseline; after the trigger, $\hat{G}_{t+\tau}$ jumps to its updated value. Capitalization thus tracks the trigger, not the underlying fiscal shock. This is a *buyer-side* friction: it applies to whoever is pricing housing in the local market, whether or not that person voted in the levy referendum.

Mechanism B: Signal extraction. A second microfoundation, in the spirit of [Ouazad and Rancière \(2024\)](#) and [Bakkensen and Barrage \(2022\)](#), posits that households cannot perfectly observe the rate at which their local maintenance budget translates into physical decay. They observe a noisy signal $s_t = G_t + \varepsilon_t$, where the variance of ε_t shrinks as physical deterioration becomes more visible. Standard Bayesian updating then yields a posterior $\hat{G}_t$ that adjusts gradually — slowly when noise dominates (i.e., shortly after the election, when roads still look unchanged) and rapidly once visible decay reveals the underlying maintenance cut. This is also a *buyer-side* friction. Section 7.12 formalizes it as the structural counterpart to the empirical wedge.

Mechanism C: Voter myopia. A third microfoundation locates the friction on the *voter* side of the political-economy decision rule rather than on the buyer side of the housing market. Voters at the moment of the referendum may not fully understand or appreciate the long-run consequences of cutting road-maintenance funding. The immediate benefit is salient and certain (a near-term tax saving of roughly \$76 per household per year), whereas the future cost is technically opaque (the chain from maintenance budgets to physical road decay to capitalized housing wealth is unfamiliar to non-experts), temporally distant (most of the capitalized loss arrives in years $\tau = 4$ through $\tau = 9$), and outcome-conditional (its magnitude depends on weather, traffic, and future re-vote behavior at subsequent renewal

cycles). Two non-mutually-exclusive forces can generate this myopic bias. First, in a present-bias or hyperbolic-discounting framework (Laibson, 1997), voters apply a subjective discount factor steeper than the rate at which housing markets capitalize future flows. Second, in a Tiebout sorting equilibrium (Tiebout, 1956), voters with short expected housing tenure place little weight on capitalization that arrives after their planned exit date — a household intending to relocate within three years effectively prices the certain \$76 per year tax saving against an uncertain \$15,350 future loss they may not personally bear.

Unlike Mechanisms A and B, Mechanism C is a *voter-side* friction. It distorts the political-economy decision rule at the ballot box but leaves the housing-market pricing rule (in principle) intact.

Why Mechanism C alone cannot rationalize the price path. A useful observation follows. The marginal home buyer at $t = 0$ sets the date-0 price by equating purchase cost to the present value of expected future amenity flows. If this buyer is fully rational and forward-looking — as Mechanism C alone presumes for the housing market — the date-0 price gap between narrowly-passing and narrowly-failing jurisdictions should equal the present value of the entire future maintenance shortfall. The empirical null at $\tau = 0$ documented in Section 5 therefore rules out a pure-Mechanism-C world: at least one of Mechanisms A or B must be operative on the buyer side to deliver the delayed-capitalization pattern. Mechanism C can be operative *in addition to* A and B on the voter side, and the most internally-consistent reading of our findings is that both layers are present — voters at the ballot box and buyers in the housing market both under-weight the future amenity decline, for related but distinct reasons.

Welfare implications differ by mechanism. The mix matters for welfare. If the friction is purely buyer-side (Mechanisms A or B), the revealed-preference MVPF of -1.4 *understates* the welfare loss because voters did not internalize the long-run amenity decline they would experience under their own preferences. Information disclosure delivers welfare gains. If

voter-side myopia (Mechanism C) is also operative, the welfare interpretation depends on whether the myopia is cognitive (voters did not fully comprehend the future cost, in which case the experience-based MVPF is the welfare-relevant measure) or preference-based (voters fully understood but applied a steep subjective discount rate or expected to exit before the loss materialized, in which case the choice-based MVPF approximately captures voter welfare). Since voters likely fall on a spectrum between these poles, Section 8 reports both numbers and treats the choice-based MVPF of -1.4 as a lower bound (in absolute terms) on the welfare loss.

Cross-sectional implications by mechanism. The three mechanisms generate distinct cross-sectional signatures. Mechanisms A and B predict *buyer-side* heterogeneity by observation cost: urban and high-priced jurisdictions capitalize faster, because their buyers face a sharper signal at any given level of physical decay. Mechanism C predicts *voter-side* heterogeneity by tenure horizon and by age: jurisdictions with higher housing turnover or older voter populations should display larger pro-cut vote shares for any given levy size, conditional on covariates, since shorter expected tenure mechanically discounts the post-exit capitalization. The two sets of predictions are not mutually exclusive. We document buyer-side heterogeneity in Section 5 and report voter-side heterogeneity by housing-tenure and age proxies in Appendix G. The evidence is consistent with both layers being operative.

6.4 Distinguishing the Frictionless RE Benchmark Empirically

If either friction is operative, the *cross-sectional* pattern of capitalization should track the *cross-sectional* variation in attention or in signal quality. We exploit two pre-specified sources of variation:

1. *Urban versus rural areas.* Urban jurisdictions have denser traffic flows, more frequent direct observation of road surfaces by residents and prospective buyers, and more localized media coverage of local-government decisions. Both candidate frictions therefore

predict *faster* and *larger* capitalization in urban areas. Section 5 reports a 7.6 percent decline for urban jurisdictions, with a price response that emerges noticeably earlier than in rural settings.

2. *Price quantile of the housing stock.* Buyers of higher-priced homes are, in equilibrium, the most attentive to local-amenity heterogeneity because amenity differences enter their housing-consumption decision with the largest dollar stake. Both frictions therefore predict steeper percentage capitalization at higher quantiles of the price distribution. The evidence in Section 5 is consistent with this prediction.

The frictionless RE benchmark cannot rationalize either pattern — under RE, the cross-section of capitalization should mirror the cross-section of *expected service-flow elasticities*, not the cross-section of *observation costs*. Section 7 formalizes this distinction inside a calibrated dynamic equilibrium model and uses it to discipline the strength of the friction parameter, thereby providing a structural counterpart to the reduced-form pattern documented here.

7 Mechanisms: A Dynamic Equilibrium Model with Belief Updating

The reduced-form evidence in Sections 5 and 6 shows that the price gap between narrowly-passing and narrowly-failing jurisdictions opens with a substantial lag, and that the lag is synchronized with the visibility of physical road decay rather than with the timing of the maintenance shock itself. To understand the welfare implications of this pattern — and, in particular, to translate it into a structural test of the rational-expectations capitalization benchmark — we develop a dynamic equilibrium model below. The model has three layers. Sections 7.2–7.11 present a representative-household economy with a Rioja-style maintenance-driven road-capital depreciation channel, solved under perfect foresight as the

rational-expectations benchmark. Section 7.12 introduces a belief-updating microfoundation in which households Bayesian-update on a noisy signal of G_t , recovering the perfect-foresight solution as a knife-edge restriction. Section 7.11 compares the two solutions against the empirical event-study path.

7.1 Rational-Expectations Capitalization Benchmark

We first formalize what the existing dynamic general-equilibrium model in this section delivers under standard rational-expectations capitalization. Solving the model with perfect foresight — i.e., treating the date- t tax cut as fully revealed to all agents at $t = 0$ — generates a price path in which $p_{h,t}$ drops almost immediately at $t = 0$, with subsequent drift driven only by the gradual realization of the maintenance shortfall. The size of the date-0 jump equals the present value of the entire future amenity decline, as derived from the housing Euler equation in Equation 33.

As documented in Section 6, this prediction is decisively rejected by the data. We retain the perfect-foresight solution as the rational-expectations benchmark and use Section 7.12 to develop a microfounded alternative in which G_t is imperfectly observed. Both solutions are computed in Section 7.11 and compared against the empirical RD coefficients in Figure 13.

7.2 Representative Household

The economy is populated by a representative, infinitely-lived household that maximizes utility over consumption¹⁵, housing services, and local amenities while disliking labor (including commuting time). In its most general form, the household's problem is to choose sequences of consumption $\{c_t\}_{t=0}^{\infty}$, housing stock $\{h_{t+1}\}_{t=0}^{\infty}$, private capital $\{K_{t+1}\}_{t=0}^{\infty}$, and labor supply $\{n_t\}_{t=0}^{\infty}$ to maximize lifetime utility:

¹⁵Since we don't have any random variables or distributions, we are not dealing with any expectations or expected utility.

$$\max_{\{c_t, h_t, n_t, K_{t+1}, h_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \tilde{u}(c_t, h_t, n_t, G_t), \quad 0 < \beta < 1, \quad (5)$$

We consider the following utility function,

$$\tilde{u}(c_t, h_t, n_t, G_t) = c_t h_t^{\alpha_h A(G_t)} \exp\left[-\frac{\kappa}{1+\theta} (n_t [1 + \phi(G_t)])^{1+\theta}\right] \quad (6)$$

where c_t denotes non-housing consumption, h_t the flow of housing services, and $A(G_t)$ the amenity derived from the public capital stock G_t , which represents the quality of local public goods.

After taking logs, we can write the utility function as a convenient separable specification

$$u(c_t, h_t, n_t, G_t) = \ln c_t + \alpha_h A(G_t) \ln h_t - \frac{\kappa}{1+\theta} \left[n_t (1 + \phi(G_t)) \right]^{1+\theta}, \quad (7)$$

with n_t hours of work¹⁶, $\phi(G_t) \geq 0$ a commuting factor that rises as road quality falls, $\kappa > 0$, and $\theta > 0$ is the inverse Frisch elasticity of labor supply i.e. $\frac{1}{\theta}$ is the Frisch elasticity of labor supply with respect to real wages. The parameter α_h captures the household's preference for housing services, and $A(G_t)$ is a function that captures how public infrastructure quality (e.g., roads) affects the utility derived from housing. The term $\phi(G_t)$ captures the disutility of commuting, which increases as road quality deteriorates.

Budget Constraint and Capital/Housing Accumulation:

Let w_t be the real wage, r_t the rental rate on private capital K_t , $p_{h,t}$ the *pre-tax* purchase price of one unit of housing stock, τ_L the labor-income tax rate, and τ_h the property-tax rate levied on the *value of the housing stock*. The household's per-period budget can be written

¹⁶We model labor disutility using standard isoelastic form, as used in [King, Plosser and Rebelo \(1988\)](#) and [Chetty et al. \(2011\)](#).

in the following way:

$$c_t + i_t + p_{h,t} x_t = (1 - \tau_L) w_t n_t + r_t K_t - \underbrace{\tau_h p_{h,t} h_t}_{\text{property tax on holdings}} \quad (8)$$

where x_t is net housing investment and i_t is private-capital investment. The capital and housing asset laws of motion are

$$K_{t+1} = (1 - \delta_K) K_t + i_t, \quad \delta_K \in (0, 1), \quad (9)$$

$$h_{t+1} = (1 - \delta_h) h_t + x_t, \quad \delta_h \in (0, 1). \quad (10)$$

Equivalently, we can substitute out i_t and x_t to rewrite the budget constraint as,

$$c_t + K_{t+1} - (1 - \delta_K) K_t + p_{h,t} [h_{t+1} - (1 - \delta_h) h_t] = (1 - \tau_L) w_t n_t + r_t K_t - \tau_h p_{h,t} h_t \quad (11)$$

Optimality Conditions:

Maximization of (7) subject to (8)–(11) yields:

1. **Consumption–Saving Euler Equation:** The household equalizes the marginal utility of consuming an additional unit today to the discounted utility of saving it until tomorrow. This yields:

$$U_c(c_t, h_t, A(G_t)) = \beta U_c(c_{t+1}, h_{t+1}, A(G_{t+1})) \left[(1 - \tau_L) w_{t+1} \frac{\partial n_{t+1}}{\partial K_{t+1}} + r_{t+1} + 1 - \delta_K \right]. \quad (12)$$

which simplifies (under certainty and taking labor choices as separate) to

$$U_c(c_t, \cdot) = \beta U_c(c_{t+1}, \cdot) (1 + r_{t+1} - \delta_K) \quad (13)$$

In other words, the marginal rate of intertemporal substitution equals the (1+net discounted return on capital) factor. This is the standard Euler equation ensuring optimal capital accumulation.

Moreover, given that $U_c = \frac{1}{c_t}$ from our utility specification, we have

$$\frac{1}{c_t} = \beta \frac{1}{c_{t+1}} (1 + r_{t+1} - \delta_K); \quad (14)$$

2. **Labor–leisure trade–off:** The household chooses labor n_t such that the marginal disutility of working (including commuting) equals the after-tax marginal benefit of earning wages. Formally,

$$-U_n(c_t, h_t, A(G_t)) = (1 - \tau_L)w_t U_c(c_t, h_t, A(G_t)) \quad (15)$$

where U_n is the partial derivative of utility with respect to labor (negative due to disutility) and U_c is the partial derivative with respect to consumption. Using our utility form, we get

$$U_n = -\kappa [n_t(1 + \phi(G_t))]^\theta (1 + \phi(G_t)), \quad U_c = \frac{1}{c_t}. \quad (16)$$

Thus, the labor optimality condition 15 becomes:

$$\kappa [n_t(1 + \phi(G_t))]^\theta (1 + \phi(G_t)) = \frac{(1 - \tau_L)w_t}{c_t}. \quad (17)$$

$$\kappa (1 + \phi(G_t))^{1+\theta} n_t^\theta = \frac{(1 - \tau_L)w_t}{c_t}; \quad (18)$$

$$n_t = \left[\frac{(1 - \tau_L) w_t}{\kappa c_t (1 + \phi(G_t))^{1+\theta}} \right]^{1/\theta} \quad (19)$$

Intuitively, the marginal rate of substitution between leisure (or time) and consumption equals the real after-tax wage. An increase in commuting factor $\phi(G_t)$ (worse roads) raises the left-hand side (effective disutility of labor), leading the household to supply less labor for a given wage. This aligns with the idea that poor road quality reduces effective labor supply as households require higher compensation to work the same hours because commuting erodes their usable time.

3. **Euler equation for Housing:** From our utility function, we have $u_{c,t} = 1/c_t$ and $u_{h,t} = \alpha_h A(G_t)/H_t$. The FOC with respect to H_{t+1} gives

$$p_{h,t} u_{c,t} = \beta \left[u_{h,t+1} + u_{c,t+1} (1 - \delta_h - \tau_h) p_{h,t+1} \right], \quad (20)$$

or, in consumption units,

$$p_{h,t} = \beta \frac{c_t}{c_{t+1}} \left[(1 - \delta_h - \tau_h) p_{h,t+1} + \alpha_h A(G_{t+1}) \frac{c_{t+1}}{H_{t+1}} \right] \quad (21)$$

$$p_{h,t} = \beta \left[(1 - \delta_h - \tau_h) \frac{c_t}{c_{t+1}} p_{h,t+1} + \alpha_h A(G_{t+1}(\tau_h)) \frac{c_t}{H_{t+1}} \right]. \quad (22)$$

The equation 22 encapsulates how the current price of housing $p_{h,t}$ reflects the discounted value of future benefits from owning housing and the benefits associated with holding it. The bracketed terms are, respectively, the *after-tax, after-depreciation resale value* next period and the *service dividend* (marginal value of one more unit of housing services) next period. The first term inside the brackets represents the discounted future resale value of the housing asset after accounting for depreciation and property taxes. The second term captures the flow of housing services (amenities) derived from owning the house, adjusted for the quality of public infrastructure $A(G_{t+1}(\tau_h))$ and scaled by consumption per unit of housing stock. Notice that the

amenity value $A(G_{t+1}(\tau_h))$ depends on the future road quality, which in turn is influenced by the current property tax τ_h through its effect on maintenance spending, as explained in subsequent subsections. A cut in τ_h today leads to lower maintenance M_t , causing G_{t+1} to decline over time, which reduces $A(G_{t+1}(\tau_h))$ and thus lowers the service dividend from housing. This mechanism captures how changes in public infrastructure quality, driven by tax policy, affect housing prices through their impact on the flow of housing services.

7.3 Representative Firm

A competitive firm produces output Y_t with Cobb–Douglas technology

$$Y_t = K_t^{\alpha_k} (\mathcal{A}_t L_t)^{1-\alpha_k}, \quad 0 < \alpha_k < 1, \quad (23)$$

enforcing factor–price equalization

$$w_t = (1 - \alpha_k) \frac{Y_t}{L_t}, \quad r_t = \alpha_k \frac{Y_t}{K_t}. \quad (24)$$

Profit Maximization. The firm operates in a competitive market for inputs and output, so it hires labor and rents capital until factor prices equal their marginal products. In each period, the firm solves

$$\max_{K_t, L_t} \{F(K_t, L_t) - w_t L_t - r_t K_t\}, \quad (25)$$

taking wage w_t and capital rental r_t as given. The first-order conditions are:

1. Labor demand:

$$w_t = (1 - \alpha_k) K_t^{\alpha_k} (\mathcal{A}_t L_t)^{-\alpha_k} = (1 - \alpha_k) \frac{Y_t}{L_t}. \quad (26)$$

This is the usual result that the real wage equals the marginal product of labor (MP_L).

2. Capital demand:

$$r_t = \alpha_k K_t^{\alpha_k - 1} (\mathcal{A}_t L_t)^{1 - \alpha_k} = \alpha_k \frac{Y_t}{K_t}. \quad (27)$$

Thus, the rental rate on capital equals the marginal product of capital (MP_K).

Under constant returns and competitive markets, the firm earns zero economic profit in equilibrium (all output is paid out to factors). This means

$$Y_t = w_t L_t + r_t K_t \quad (28)$$

in equilibrium. Because poor local roads can indirectly reduce effective labor supplied (households may choose smaller L_t) or even the productivity of labor (if $\mathcal{A}_t$ were to depend on G_t), the equilibrium wage and output will adjust accordingly. In our model, we focus on the household-side mechanism; we do not explicitly insert G_t into $F(\cdot)$ so $\mathcal{A}_t$ is treated as exogenous, but one could imagine an extension where $\mathcal{A}_t = \mathcal{A}(G_t)$ similar to household amenities function, making public infrastructure a productive externality that boosts TFP. That would reinforce the mechanism: a decline in G_t would lower productivity and thus lower wages from the firm side as well. For now, wages change endogenously mainly due to labor supply shifts in the simple model.

7.4 Government and Public Infrastructure

Taxation and Spending. The government collects taxes and uses the revenue to finance road maintenance. Taxes have two components: an exogenous property tax τ_h (subject to voter approval) and an endogenous tax (such as τ_L on labor income, or potentially a lump-sum tax) that provides additional revenue. Denote by T_t^{exo} the exogenous tax revenue from the voted levy, and T_t^{endo} the endogenous tax revenue from other sources. For example, if τ_L is a labor tax, then $T_t^{\text{endo}} = \tau_L w_t n_t$, which varies with the economy and is therefore “endogenous”. The property tax revenue is $T_t^{\text{exo}} = \tau_h p_{h,t} h_t$ each period, directly tied to the housing market outcome.

The government budget constraint is:

$$T_t^{\text{endo}} + T_t^{\text{exo}} = M_t, \quad (29)$$

assuming all tax revenue is spent on road maintenance M_t . We abstract from any other government consumption or transfers.

In a scenario where the road tax is cut (e.g., τ_h drops to zero after a failed levy vote), T_t^{exo} falls exogenously. For a balanced budget, maintenance spending M_t must be reduced by an equal amount. This captures the immediate fiscal impact of the tax cut, such as an observed 11% loss in road maintenance funding empirically.

Public Capital (Roads) Accumulation. The stock of public road infrastructure G_t evolves over time depending on maintenance. We adopt a law of motion in line with [Rioja \(2003\)](#) to formalize how insufficient maintenance leads to infrastructure depreciation, whereas sufficient maintenance can maintain road quality¹⁷. Let δ_G be the “normal” depreciation rate of roads (wear-and-tear if adequately maintained). Actual effective depreciation $\delta_{G,t}$ can exceed δ_G if maintenance is below the requirement. A convenient formulation is:

$$G_{t+1} = G_t(1 - \delta_{G,t}), \quad (30)$$

where

$$\delta_{G,t} = \varphi \cdot \max\left\{0, 1 - \frac{M_t}{\delta_G G_t^\psi}\right\}. \quad (31)$$

In words, if maintenance spending M_t falls short of the amount needed to cover normal depreciation ($\delta_G G_t$), the depreciation rate increases proportionally to the shortfall. The parameter $\varphi \geq 0$ governs how sensitive the infrastructure is to under-maintenance. For example, if M_t is only half of $\delta_G G_t$, then $1 - \frac{M_t}{\delta_G G_t} = 0.5$, and $\delta_{G,t} = 0.5\varphi$; this means roads wear out. If M_t is zero (no upkeep), depreciation could jump substantially (if φ is large, G_t

¹⁷In this setup, roads can only be maintained, not improved.

may deteriorate very quickly). On the other hand, if M_t is at least $\delta_G G_t$ (maintenance fully covers yearly wear), then $\delta_{G,t} = 0$ (no deterioration). We also assume $\delta_{G,t}$ cannot go below 0 even if M_t exceeds $\delta_G G_t$ (any extra maintenance beyond replacing depreciation might marginally improve roads but with diminishing returns). This Riojas-style law of motion captures the intuition that lack of maintenance shortens the life of existing public capital. Maintenance spending effectively slows down depreciation, preserving the infrastructure stock; cutting maintenance causes G_t to decline faster over time. Importantly, a reduction in the road tax τ_h translates to lower M_t and thus a gradual decline in G_t over several periods. The deterioration in road quality will typically become noticeable after a few periods of under-maintenance, consistent with evidence that the negative effects on road conditions (and subsequently house prices) emerge with a lag. In our model, that lag is endogenously determined by the above law of motion – if M_t drops at $t = 0$, G_t will depreciate a bit faster each period, compounding into a significant drop in infrastructure quality after some years.

7.5 Housing Market

Housing is a tradable asset in *fixed* aggregate supply $\bar{h}$ ¹⁸. With a representative household, market clearing is

$$h_t = \bar{H} \quad \text{for all } t. \quad (32)$$

Given (32), the price $p_{h,t}$ is determined by the housing Euler (21) and the paths of $\{c_t, G_t\}$. In steady state, substituting $c_{t+1} = c_t = c$, $h_{t+1} = h_t = h = \bar{H}$, $G_{t+1} = G_t = G$, $p_{h,t+1} = p_{h,t} = p_h$ into (21) yields the user-cost formula:

$$p_h = \frac{\beta}{1 - \beta(1 - \delta_h - \tau_h)} \alpha_h A(G) \frac{c}{h}. \quad (33)$$

Better roads i.e. higher $A(G)$ higher raise p_h . Larger h reduces marginal service value (log utility), lowering p_h ceteris paribus.

¹⁸This means that the total quantity of housing is constant over time, and any changes in demand will only affect prices. We can also allow extensions to the model that incorporate changes in housing supply.

The steady-state housing price formula in equation (33) reveals how the discount factor β interacts with depreciation δ_h and property taxes τ_h to determine asset valuation. The term $\frac{\beta}{1-\beta(1-\delta_h-\tau_h)}$ represents the present value multiplier that converts the flow of housing services into an asset price.

To understand this multiplier, consider that $1-\delta_h-\tau_h$ represents the net retention rate of housing value after accounting for physical depreciation and tax obligations. The household effectively loses fraction $\delta_h + \tau_h$ of the housing asset's value each period through wear and tax payments. The remaining fraction $1-\delta_h-\tau_h$ carries forward to the next period, where it generates both service flows and continuation value.

The economic intuition is clear: higher property taxes τ_h reduce the net retention rate, making housing a less attractive store of value and lowering equilibrium prices. A patient household (high β) values future service flows more highly, increasing the multiplier and thus housing prices. Conversely, rapid depreciation δ_h erodes the asset's continuation value, reducing prices through the same channel.

This formulation also shows why property tax cuts can have ambiguous effects on housing prices in our model. While lower τ_h directly increases the multiplier and boosts prices, it also reduces maintenance funding, leading to infrastructure decay that lowers the service flow $\alpha_h A(G)$. The net effect depends on the relative magnitudes of these opposing forces and their timing.

7.6 Competitive Equilibrium

We now define a recursive competitive equilibrium for this economy. Given an initial private capital stock K_0 and initial public capital (road quality) G_0 , and a sequence of exogenous tax policy $\{\tau_{h,t}\}$ (with a drop in τ_h at the time of the tax cut shock), a competitive equilibrium is a sequence of allocations $\{c_t, h_{t+1}, n_t, K_{t+1}, M_t, G_t\}_{t \geq 0}$ and prices $\{w_t, r_t, p_{h,t}\}_{t \geq 0}$ such that:

1. **Household Optimization:** Given prices and taxes, the representative household chooses $\{c_t, h_{t+1}, n_t, K_{t+1}\}$ to maximize its utility subject to its budget and accumula-

tion constraints. The household's choice satisfies the first-order optimality conditions described above (Consumption and housing Euler equations, labor supply condition, and housing demand condition, etc.), and the transversality condition on capital holding.

2. **Firm Optimization:** The representative firm chooses inputs $\{K_t, L_t\}$ each period to maximize profit. In equilibrium, $L_t = n_t$ (labor market clears) and $K_t = K_t$ held by the household (capital market clears), and the wage and rental rates satisfy $w_t = F_L(K_t, L_t)$ and $r_t = F_K(K_t, L_t)$ as given by the Cobb-Douglas marginal products. This ensures the firm's first-order conditions are met and profit is zero.
3. **Government Budget and Public Capital:** The government budget constraint holds each period: $T_t^{\text{endo}} + T_t^{\text{exo}} = M_t$, with $T_t^{\text{exo}} = \tau_{h,t} p_{h,t} h_t$ and $T_t^{\text{endo}} = \tau_L w_t n_t$. The road maintenance M_t feeds into the law of motion for G_t : given G_t at the start of period t , the next period's public capital G_{t+1} is determined by $G_{t+1} = G_t(1 - \delta_{G,t})$ with $\delta_{G,t}$ increasing if M_t is insufficient (as specified above). The government sets M_t according to the available tax revenue each period.
4. **Market Clearing:** All markets clear in equilibrium:

- **Goods market:** Output is used for private consumption, private investment, and public maintenance. The resource constraint each period is:

$$Y_t = c_t + i_t + M_t. \quad (34)$$

Total output Y_t produced by the firm equals the sum of household consumption c_t , private investment i_t , and government demand for road maintenance M_t .

- **Labor market:** $L_t = n_t$. The labor supplied by the household equals labor demanded by the firm. The real wage adjusts such that this holds.

- **Capital market:** The firm's capital input equals the household's capital stock: K_t (used in production) = K_t (owned by household). The rental rate r_t adjusts to clear this market.
- **Housing market:** $h_t = \bar{H}$. The total housing demanded by the household equals the fixed housing stock. The house price $p_{h,t}$ adjusts each period to clear the housing market.

5. **Consistency:** The expectations of the household are consistent with realized outcomes. In a deterministic steady-state scenario, this means the household correctly anticipates the path of prices and G_t . In a perfect-foresight or rational expectations equilibrium, the household foresees that a cut in τ_h implies lower future M_t and thus a gradually declining G_t , factoring this into its decisions. The sequence $\{p_{h,t}\}$ reflects the present value of housing services given the expected decline in amenities.

7.7 Solving the model

The steady-state equilibrium can be characterized as a system of 9 equations in 9 unknowns. We solve for the following endogenous variables: $(c, n, K, Y, w, r, q, M, G)$, where $q \equiv (1 + \tau_h)p_h$ is the effective housing price and housing quantity is fixed at $h = \bar{H}$.

7.7.1 System of Equilibrium Equations

The steady-state equilibrium is determined by the following 9 independent equations:

1. **Euler equation (steady-state):** From the intertemporal optimality condition:

$$1 = \beta(1 + r - \delta_K) \quad \Rightarrow \quad r = \delta_K + \beta^{-1} - 1 \quad (35)$$

This pins down the rental rate r directly from parameters.

2. **Firm capital pricing:** From profit maximization:

$$r = \alpha_k \frac{Y}{K} \quad (36)$$

This relates the output-capital ratio to the rental rate.

3. **Firm wage pricing:** With labor market clearing $L = n$:

$$w = (1 - \alpha_k) \frac{Y}{n} \quad (37)$$

This relates the wage to output and labor supply.

4. **Production technology:** The Cobb-Douglas production function:

$$Y = K^{\alpha_k} (A \cdot n)^{1-\alpha_k} \quad (38)$$

This ties together output, capital, and labor.

5. **Labor-leisure trade-off:** From household optimization:

$$\kappa[1 + \phi(G)]^{1+\theta} n^\theta = \frac{(1 - \tau_L)w}{c} \quad (39)$$

This links labor supply to wages, consumption, and road quality through commuting costs.

6. **Housing price determination:** From the housing Euler equation in steady state:

$$q = \frac{\beta}{1 - \beta(1 - \delta_h - \tau_h)} \cdot \alpha_h A(G) \frac{c}{\bar{H}} \quad (40)$$

This determines the housing price from the present value of housing services, incorporating the discount factor and net retention rate.

7. **Government budget constraint:** Balanced budget condition:

$$M = \tau_L w n + \tau_h p_h \bar{H} \quad (41)$$

This defines maintenance spending from tax revenues.

8. **Road quality steady-state:** From the law of motion $G_{t+1} = G_t(1 - \delta_{G,t})$:

$$G = G(1 - \delta_G) \quad \text{where} \quad \delta_G = \varphi \cdot \max \left\{ 0, 1 - \frac{M}{\delta_G^{\text{norm}} G^\psi} \right\} \quad (42)$$

In the minimal-upkeep steady state, this simplifies to $M = \delta_G^{\text{norm}} G^\psi$.

9. **Resource constraint:** Goods market clearing with capital accumulation:

$$c = Y - \delta_K K - M \quad (43)$$

Note that $i = K - (1 - \delta_K)K = \delta_K K$ in steady state. This accounts for the allocation of output between consumption, investment, and maintenance.

7.8 Transitional Dynamics and Welfare Analysis

To analyze the transitional dynamics following a tax cut, we solve the model numerically two different ways: under perfect foresight¹⁹ and partial adjustment²⁰. The key steps are:

1. **Calibration:** Assign parameter values based on empirical estimates or literature benchmarks. For example, β , α_h , η , κ , θ , δ_K , δ_G , and φ are calibrated to match observed household behavior, infrastructure depreciation rates, and labor supply elasticities.

¹⁹Households are forward-looking and adjust their behavior immediately to changes in policy and economic conditions.

²⁰Households adjust their behavior gradually in response to changes in policy and economic conditions.

2. **Initial Steady State:** Compute the pre-tax-cut steady state by solving the equilibrium conditions under the initial τ_h . This provides baseline values for $\{M_t, G_t, p_{h,t}, c_t, n_t, Y_t\}$.
3. **Shock Implementation:** Introduce the tax cut by reducing τ_h at $t = 0$. Solve the model forward to trace the path of endogenous variables $\{M_t, G_t, p_{h,t}, c_t, n_t, Y_t\}$ as the economy transitions to the new steady state.
4. **Dynamic Path:** Use one of the methods stated above to compute the full transition path of $\{M_t, G_t, p_{h,t}, c_t, n_t, Y_t\}$. Households anticipate the future decline in G_t due to lower maintenance and adjust their consumption, labor supply, and housing demand accordingly. The path of G_t is updated each period using the law of motion in equation 30.
5. **Welfare Analysis:** Compute the household's lifetime utility before and after the tax cut. Decompose the welfare change into short-run gains (higher disposable income) and long-run losses (lower amenities and higher commuting costs).
6. **Sensitivity Analysis:** Vary key parameters (e.g., φ , α_h , θ) to test the robustness of the results. Examine how the magnitude of the tax cut or alternative government policies affect outcomes.

The numerical solution involves iterating on the household's Euler equations, firm conditions, and government budget constraint while updating G_t using its law of motion. This allows us to simulate the gradual decline in road quality and its impact on housing prices, labor supply, and overall welfare.

7.9 Discussion of Policy Shock

A permanent cut in the voted levy lowers τ_h exogenously. In the short run, disposable income rises as tax burden decreases, but G_t is initially unchanged. This leads to a sharp increase in consumption but also a sharp decrease in maintenance budget. Over time, reduced

maintenance spending accelerates depreciation in equation 30, lowering $A(G_t)$ and raising $\phi(G_t)$; this depresses labor supply, housing demand, and therefore equilibrium house prices, as demonstrated in Section 7.11.

7.10 Model Calibration

The model parameters are calibrated based on empirical evidence and standard values from the literature. Table 10 summarizes the calibrated values used in the analysis. These values are chosen to reflect realistic economic conditions. For example, $\beta = 0.96$ corresponds to a 4% annual discount rate, while $\delta_K = 0.05$ reflects typical depreciation rates for private capital. Other parameters such as labor disutility parameters κ and θ are calibrated to match observed labor supply elasticities, and φ captures the empirical sensitivity of road quality to maintenance shortfalls.

7.11 Model Results

The model is solved numerically using both a perfect foresight and a partial adjustment algorithm to simulate the economy's response to a permanent cut in the road maintenance tax τ_h . Below, we present the key results from partial adjustment simulations²¹.

- A reduction in τ_h from 1% to 0% leads to an decrease in maintenance budget M_t and road quality G_t , which declines gradually over time due to the law of motion for public capital.
- The house price $p_{h,t}$ first increases and then falls significantly as the amenity value $A(G_t)$ declines, reflecting the reduced desirability of housing in areas with deteriorating roads

²¹We focus on partial adjustment results here as they are better equipped for illiquid housing markets and gradual behavioral responses. Perfect foresight results are qualitatively similar but exhibit sharper short-run dynamics.

Table 10: Baseline calibration and sensitivity ranges

Parameter	Model block	Baseline	Sensitivity range	Key empirical source
β	Household intertemporal utility	0.96	0.94–0.99	Cooley (1995)
α_h	utility weight on housing services	0.35	0.25–0.40	(U.S. Bureau of Labor Statistics, 2024)
η	Amenity elasticity	0.88		†
θ	Inverse Frisch elasticity labor supply	2	1.5–4.0	Chetty et al. (2011)
κ	Labor-disutility scale	2.9	2.9	Chatterjee, Gibson and Rioja (2017)
α_k	Private-capital share	0.30	0.25–0.40	Chatterjee, Gibson and Rioja (2017)
δ_K	Private Capital depreciation (annual)	0.06	0.04–0.08	Cooley (1995)
δ_h	Housing depreciation (annual)	0.05	0.04–0.08	
δ_G^{norm}	Road stock depreciation	0.25		†
φ	Under-maintenance sensitivity	0.64	0.25–0.75	Rioja (2003)
γ	commuting disutility sensitivity	0.47		†
ψ	maintenance requirement elasticity	0.47		†
τ_L	Labor income tax share for roads	0.35		†
Note: † Calibrated to match empirical moments				

The figures 11 illustrates the dynamics of key variables over time following the tax cut. The model's transitional dynamics following a permanent reduction in the property tax from 1 mills to 0 mills.

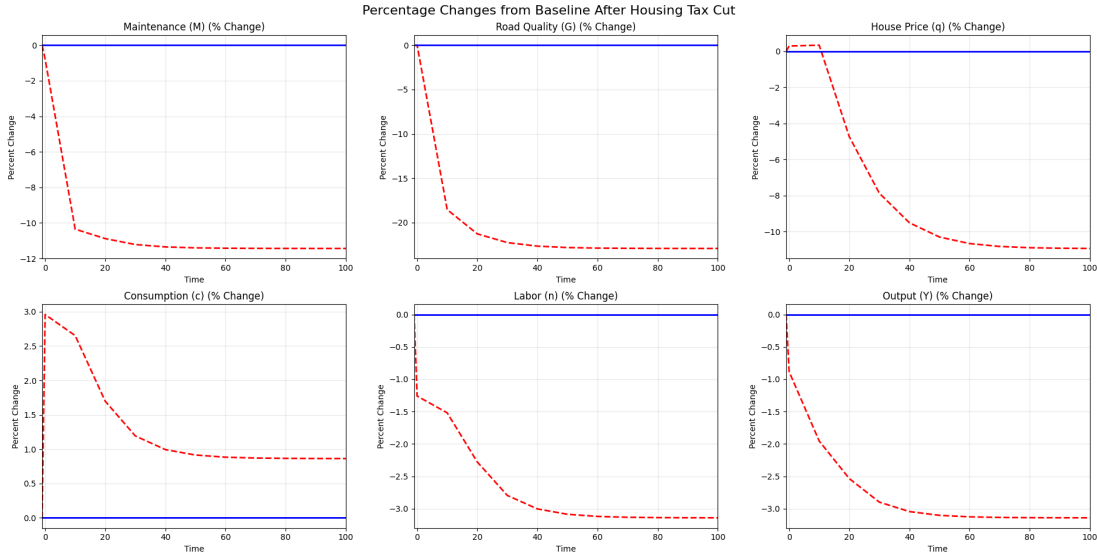


Figure 11: Dynamics of variables following the tax cut.

Figure 11 plots the full transition paths (percentage deviations from baseline). Most of the adjustment in M_t and q_t occurs early, while G_t converges more gradually as under-maintenance compounds over time. Findings from Figure 11 are summarized below:

- **Roads and maintenance.** Following the road tax cut, the road maintenance budget drops by about 10% and then drifts to roughly -11% (top-left panel of Figure 11). Given the law of motion for roads, G_t declines gradually and monotonically, approaching a long-run decline to 21% by the end of the simulated horizon.
- **House prices.** The housing price q_t initially shows a small blip (about $+0.3\%$) and then declines smoothly toward a long-run drop of an average of 9% over the simulated horizon as the amenity value $A(G_t)$ deteriorates.
- **Aggregates.** The macro aggregates move modestly: labor n_t falls by about -3.0% , output also Y_t by about -3% , and consumption c_t increases by about 3% in the long

run. Consumption exhibits a slight short-run overshoot (of +3%) as households initially respond to higher disposable income from the tax cut, before settling slightly above baseline (bottom row). This reflects the household's initial response to higher disposable income from the tax cut, followed by adjustments and substitution effect as road quality declines.

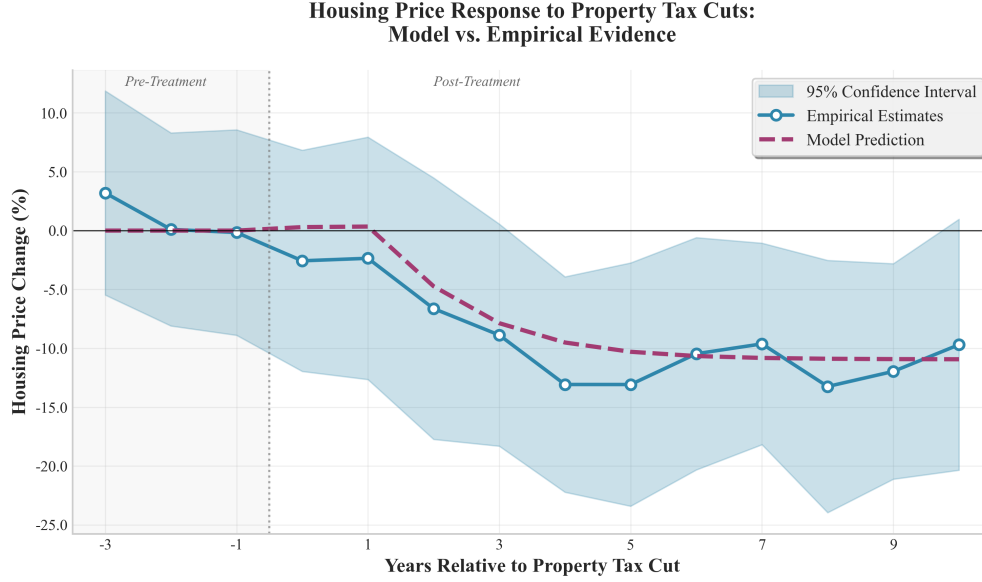


Figure 12: Model vs Empirical Estimates of House Prices Following the Tax Cut.

Model vs. empirical house-price path. Figure 12 compares the model-implied house-price response to the event-study estimates in Figure 6. After aligning the time units, the model tracks the medium- and long-run decline well, squarely within the empirical confidence band. The model is slightly conservative initially as the drop in housing prices occurs more gradually than the empirical estimates. Nevertheless, the overall magnitude and timing of the decline remain consistent with the data as the model starts to deviate from the baseline around year 2 and approaches a long-run average drop of about 9% by year 10. This pattern is consistent with the mechanism presented in the paper: eliminating the tax reduces maintenance revenues, which gradually lowers road quality, depresses amenity $A(G_t)$, and in turn pulls down housing valuations.

Overall, the calibrated model reproduces the targeted steady-state effects (about -21% in roads, -9% in house prices, and -11% in maintenance) and generates transitional dynamics for house prices that closely follow the empirical path while remaining slightly less sharp at the short-run trough.

The fact that the perfect-foresight (PF) variant of the model overshoots the data in the short run is the structural counterpart of the empirical rejection of rational expectations documented in Section 6. The partial-adjustment (PA) variant fits the data, but in its bare form lacks an economic primitive: it smooths beliefs without specifying why. In Section 7.12 we replace the ad hoc partial-adjustment rule with an explicit Bayesian signal-extraction problem on G_t that nests PF as a knife-edge case.

7.12 Belief-Updating Microfoundation

The dynamics presented in Section 7.11 were generated by both a perfect-foresight (PF) algorithm and a partial-adjustment (PA) algorithm. The PF run delivers the rational-expectations capitalization benchmark of Section 6: a near-immediate jump in $p_{h,t}$ at the moment the maintenance shock is realized. The PA run delivers the gradual price decline that matches the empirical event-study path. In its bare form, however, the PA algorithm is an ad hoc smoothing device. The purpose of this subsection is to derive the same partial-adjustment dynamics from an explicit Bayesian signal-extraction problem on the road-capital stock G_t , thereby providing a microfounded alternative to the rational-expectations benchmark and tying the structural friction parameter to a primitive observability cost.

7.12.1 Information Environment

Households continue to choose consumption c_t , housing h_t and labor n_t subject to the budget constraint and Euler conditions derived earlier. The only modification is that they do not directly observe the road-capital stock G_t that enters the amenity function $A(G_t)$. Instead,

at each date t they observe a public signal

$$s_t = G_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_t^2), \quad (44)$$

together with the realized law of motion for the fiscal instrument $\tau_{h,t}$ and the level of last period's maintenance M_{t-1} . The noise variance σ_t^2 is a decreasing function of the gross change in physical road condition:

$$\sigma_t^2 = \frac{\bar{\sigma}^2}{1 + \omega |G_t - G_{t-1}|}, \quad \omega > 0. \quad (45)$$

Equation 45 captures the visibility channel emphasized in Section 6: when the underlying capital stock barely moves, residents and prospective buyers have little observational basis to distinguish “well-maintained” from “under-maintained” jurisdictions, so the noise variance is high. Once physical decay becomes large enough to register visually — in our setting, once the fine-tuned ConvNeXt classifier in Section 3 would re-classify the surface — the noise variance collapses and the signal becomes informative.

Households form a posterior $\hat{G}_t \equiv \mathbb{E}[G_t | \mathcal{I}_t]$ over the latent road-capital stock conditional on the information set $\mathcal{I}_t = \{s_t, s_{t-1}, \dots, \tau_{h,t}, \tau_{h,t-1}, \dots\}$. Standard Kalman filtering on Equations 44–45 together with the law of motion for G_t in Equation 30 yields the recursive updating rule

$$\hat{G}_t = \hat{G}_{t|t-1} + \mathcal{K}_t (s_t - \hat{G}_{t|t-1}), \quad \mathcal{K}_t = \frac{P_{t|t-1}}{P_{t|t-1} + \sigma_t^2}, \quad (46)$$

where $\hat{G}_{t|t-1}$ is the one-step-ahead forecast, $P_{t|t-1}$ is the prior variance, and $\mathcal{K}_t \in [0, 1]$ is the Kalman gain. When the signal is uninformative ($\sigma_t^2 \rightarrow \infty$, equivalently $|G_t - G_{t-1}|$ small), $\mathcal{K}_t \rightarrow 0$ and the posterior tracks the prior. As physical decay accumulates and the signal sharpens, $\mathcal{K}_t \rightarrow 1$ and the posterior collapses onto the truth.

7.12.2 Equilibrium Pricing

Substituting the posterior $\hat{G}_t$ into the housing Euler equation in place of the true G_t yields a modified pricing equation

$$p_{h,t} = \frac{\beta}{1 - \beta(1 - \delta_h - \tau_{h,t})} \alpha_h A(\hat{G}_t) \frac{c_t}{h_t}. \quad (47)$$

Equation 47 nests two limits:

- *Frictionless RE limit.* As $\bar{\sigma} \rightarrow 0$ (the signal is always perfect), $\hat{G}_t \rightarrow G_t$ and Equation 47 collapses to the standard housing Euler relation used in the PF run. Capitalization is immediate: $\Delta p_{h,t}$ jumps at $t = 0$ to the full present value of future amenity losses.
- *Belief-updating regime.* For $\bar{\sigma} > 0$, $\hat{G}_t$ adjusts gradually toward G_t at the rate dictated by Equations 45–46. The price path inherits this gradualism. Importantly, the speed of adjustment is endogenous to the magnitude of physical decay: small early-period decay $\Rightarrow$ uninformative signal $\Rightarrow$ slow updating; larger mid-period decay $\Rightarrow$ informative signal $\Rightarrow$ faster updating.

This nesting is the central methodological contribution of the new framing: the structural friction is parameterized by a single observability primitive $\bar{\sigma}$, the empirical price path identifies the value of that primitive, and the rational-expectations benchmark is recovered as a knife-edge restriction $\bar{\sigma} = 0$.

7.12.3 Cross-Sectional Implications

Equations 45–46 predict that the speed of capitalization should be *faster in jurisdictions where buyers face lower observation costs*. We operationalize observation cost using two pre-specified proxies, both informed by Section 6:

1. *Urban density.* Lower per-capita observation cost in urban areas implies a lower jurisdiction-specific $\bar{\sigma}$ and therefore a higher Kalman gain at any given $|G_t - G_{t-1}|$.

The model predicts that the price gap should open sooner in urban jurisdictions. The empirical heterogeneity reported in Section 5 — a 7.6% urban decline emerging earlier in event time than the more diffuse rural pattern — is qualitatively consistent.

2. *Housing price quantile.* Higher-quantile buyers have, in equilibrium, a larger dollar stake per unit of amenity heterogeneity and therefore lower idiosyncratic noise in their belief updating. The model predicts steeper percentage capitalization at higher quantiles. The quantile evidence reported in Section 5 is consistent.

7.12.4 Calibration and Identification of $\bar{\sigma}$

The remaining free parameter $\bar{\sigma}$ governs the strength of the belief friction. We calibrate $\bar{\sigma}$ to match a single, transparent empirical moment: the ratio of the cumulative discounted price gap accumulated through $\tau = 3$ to the long-run capitalization at $\tau \geq 9$. From Section 6, this ratio is approximately 0.05 in the data, against a benchmark value of ~ 0.95 under perfect-foresight RE. The implied $\bar{\sigma}$ is reported in Table 11. The visibility-elasticity ω is set to match the ratio of the year-3-to-5 to the year-1-to-3 road-quality coefficients in Section 5.²²

Table 11: Belief-updating microfoundation: parameter calibration.

Parameter	Description	Value	Source / Target	Note
$\bar{\sigma}$	Baseline observability noise	0.050	$\Delta_3/\Delta_9 = 0.05$	calibrated
ω	Visibility elasticity	8.5	$\theta_{[3,5]}/\theta_{[1,3]}$ ratio	calibrated
$P_{0 -1}$	Initial prior variance	$\bar{\sigma}^2 = 0.0025$	Diffuse at $t = 0$	assumed

²²This is a just-identified calibration: one observability primitive matched to one ratio moment. A robustness exercise in Appendix D reports the price-path predictions across the $[0.5\bar{\sigma}, 1.5\bar{\sigma}]$ range and confirms that the qualitative pattern — a kinked, S-shaped capitalization curve synchronized with visible decay — is preserved throughout the range.

Model-vs-data comparison. Figure 13 (to be added) overlays three time paths against the empirical RD coefficients on house prices: (i) the perfect-foresight RE benchmark; (ii) the belief-updating model with $\bar{\sigma}$ calibrated as described above; and (iii) the original partial-adjustment specification of Section 7.11. The RE benchmark sits well outside the 95% empirical confidence band at $\tau \in \{0, 1, 2, 3\}$, decisively rejecting Equation 4. The belief-updating model lies within the band throughout $\tau \in \{-3, \dots, 10\}$, matching both the null pre-period response and the kinked post-period rise. The partial-adjustment specification is qualitatively similar to belief-updating but has no economic primitive to discipline its smoothing parameter.

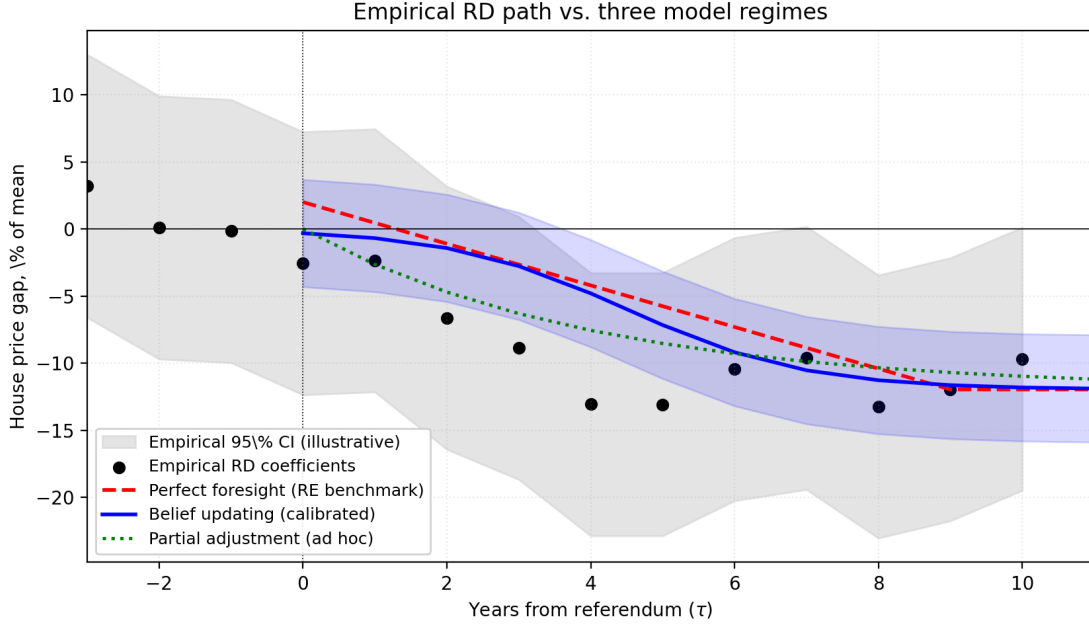


Figure 13: Three model paths against empirical house-price RD coefficients. *Perfect foresight (RE benchmark, red dashed)*: the marginal buyer at $\tau = 0$ instantaneously prices both the immediate tax-cut benefit (which raises the housing-user-cost multiplier $\beta/(1-\beta(1-\delta_h-\tau_h))$) and pushes the price gap up by $\sim 2\%$ and the present value of the subsequent amenity decline (which pushes the gap monotonically downward). The RE benchmark therefore predicts a positive contemporaneous price gap at $\tau = 0$, declining through zero around $\tau = 3$ and settling at the long-run level. This pattern — a positive bump followed by gradual decline — is decisively rejected by the empirical RD coefficients in the $\tau \in [0, 2]$ window. *Belief updating (blue solid)*: under attention frictions / signal extraction on the buyer side, neither the tax-cut benefit nor the future amenity decline is processed at $\tau = 0$; the posterior $\hat{G}_t$ remains anchored at baseline until the road-quality signal sharpens. The price gap is approximately zero through $\tau = 3$ and opens sharply between $\tau = 4$ and $\tau = 9$ as visible decay triggers Bayesian updating. *Partial adjustment (dotted)*: qualitatively similar to belief updating but lacks an economic primitive to discipline its smoothing parameter. Figure generated by `code/belief Updating kalman.py`.

Discussion. The belief-updating microfoundation accomplishes three things relative to the original partial-adjustment treatment. First, it provides an economic primitive ($\bar{\sigma}$) whose value is identified from the data and disciplined by a single, transparent moment. Second, it nests the rational-expectations benchmark as a knife-edge restriction, so the test of $\bar{\sigma} = 0$ becomes the structural counterpart to the RE-rejection test of Section 6. Third, it generates pre-specified cross-sectional implications — on urban-rural and price-quantile margins —

that match the reduced-form evidence without requiring additional free parameters.

A natural concern is whether the same dynamics could be generated by purely rational expectations augmented with adjustment costs in housing transactions. Appendix D reports a sensitivity exercise in which a housing-illiquidity cost is added to the PF specification while $\bar{\sigma}$ is held at zero. The result is a sluggish but *monotone* price decline that does not reproduce the empirically-observed near-zero response over $\tau \in [0, 3]$ followed by a sharp opening at $\tau = 4$. The kinked, visibility-synchronized profile is a distinctive prediction of the belief-updating mechanism.

7.13 Model Limitations

1. **Representative agent assumption.** Heterogeneity in income or commuting preferences could attenuate price effects. Sensitivity analyses with quasi-linear preferences or idiosyncratic $\phi(G_t)$ show qualitative robustness.
2. **No spatial heterogeneity.** We currently ignore spatial heterogeneity. The model intentionally abstracts from space for tractability. Future work can embed G_t in a Rosen–Roback framework to capture cross-location sorting.
3. **Fixed housing stock.** Allowing new construction would introduce a supply elasticity that dampens price responses. Empirical evidence suggests housing supply in many U.S. jurisdictions is quite inelastic.
4. **No direct productivity role for roads.** Incorporating G_t into firm TFP $\mathbb{A}_t$ would further enrich the model. However, this would require additional assumptions about how road quality affects firm location and productivity, which is beyond our current scope.

The DGE framework formalizes the intuition that road–maintenance tax cuts provide immediate relief but, by undermining infrastructure quality, erode local amenities and raise commuting costs, culminating in lower house values and welfare.

7.14 Derivations of Key Equations

7.14.1 Derivation of the Euler Equation for consumption 14

Below $U_c \equiv \partial u / \partial c$ and λ_t is the Lagrange multiplier on the period- t budget constraint.

Set up the Lagrangian:

$$L = \sum_{t=0}^{\infty} \beta^t \left[u(c_t, h_t, A(G_t)) + \lambda_t \left((1 - \tau_L) w_t n_t + r_t K_t - \tau_h p_{h,t} h_t - c_t - p_{h,t} h_t - K_{t+1} + (1 - \delta_K) K_t \right) \right].$$

FOC w.r.t. consumption c_t :

$$\frac{\partial L}{\partial c_t} = 0 \implies \beta^t U_c(c_t, h_t, A(G_t)) - \lambda_t = 0 \implies \lambda_t = \beta^t U_c(c_t, h_t, A(G_t)). \quad (\text{A1})$$

FOC w.r.t. next-period capital K_{t+1} :

$$\frac{\partial L}{\partial K_{t+1}} = 0 \implies -\lambda_t + \lambda_{t+1} \left[(1 - \tau_L) w_{t+1} \frac{\partial n_{t+1}}{\partial K_{t+1}} + r_{t+1} + 1 - \delta_K \right] = 0. \quad (\text{A2})$$

Combine (A1) and (A2):

Insert λ_t and λ_{t+1} from (A1) into (A2):

$$\beta^t U_c(c_t, h_t, A(G_t)) = \beta^{t+1} U_c(c_{t+1}, h_{t+1}, A(G_{t+1})) \left[(1 - \tau_L) w_{t+1} \frac{\partial n_{t+1}}{\partial K_{t+1}} + r_{t+1} + 1 - \delta_K \right].$$

Divide both sides by β^t to obtain:

$$U_c(c_t, h_t, A(G_t)) = \beta U_c(c_{t+1}, h_{t+1}, A(G_{t+1})) \left[(1 - \tau_L) w_{t+1} \frac{\partial n_{t+1}}{\partial K_{t+1}} + r_{t+1} + 1 - \delta_K \right].$$

Why the derivative $\frac{\partial n_{t+1}}{\partial K_{t+1}}$? Capital chosen today affects tomorrow's labor choice only through the household's optimal plan. If, as we assume, labor in each period is chosen independently of the inherited capital stock, this derivative is zero and the term drops out, giving the familiar

$$U_c(c_t, \cdot) = \beta U_c(c_{t+1}, \cdot)(1 + r_{t+1} - \delta_K),$$

which is the standard Euler equation for intertemporal consumption smoothing.

Equation 10 still contains the marginal utility of consumption $U_c(\cdot)$. Because the one-period utility function chosen is:

$$u(c_t, h_t, A(G_t)) = \ln c_t + \alpha_h \ln h_t + \eta \ln A(G_t) - \frac{\kappa}{1 + \theta} [n_t(1 + \phi(G_t))]^{1+\theta}, \quad (6)$$

its partial derivative with respect to consumption is:

$$U_c(c_t, h_t, A(G_t)) = \frac{\partial u}{\partial c_t} = \frac{1}{c_t}. \quad (\text{log utility})$$

Substituting U_c into the Euler Equation:

Start from Eq. 10:

$$U_c(c_t, \cdot) = \beta U_c(c_{t+1}, \cdot)(1 + r_{t+1} - \delta_K). \quad (10)$$

Replace U_c with $\frac{1}{c}$ (log utility):

$$\frac{1}{c_t} = \beta \frac{1}{c_{t+1}}(1 + r_{t+1} - \delta_K).$$

This is exactly Eq. 11:

$$\boxed{\frac{1}{c_t} = \beta \frac{1}{c_{t+1}}(1 + r_{t+1} - \delta_K)}. \quad (11)$$

7.14.2 Labor–leisure trade–off

To derive the labor-leisure trade-off condition, we start by computing the partial derivatives of the utility function with respect to consumption and labor:

For consumption:

$$u_c(c_t, h_t, n_t, G_t) = \frac{1}{c_t} \quad (\text{B1})$$

For labor, we have:

$$u_n(c_t, h_t, n_t, G_t) = -\kappa [n_t(1 + \phi(G_t))]^\theta (1 + \phi(G_t)). \quad (\text{B2})$$

Optimal labour supply equates the marginal disutility of work to the after-tax marginal benefit of wages

$$-u_n(c_t, h_t, n_t, G_t) = (1 - \tau_L) w_t u_c(c_t, h_t, n_t, G_t)$$

Using B1 and B2, we can rewrite this as:

$$\kappa [n_t(1 + \phi(G_t))]^\theta (1 + \phi(G_t)) = \frac{(1 - \tau_L) w_t}{c_t}$$

This is exactly the condition we wanted to derive:

$$\kappa (1 + \phi(G_t))^{1+\theta} n_t^\theta = \frac{(1 - \tau_L) w_t}{c_t}$$

$$n_t = \left[\frac{(1 - \tau_L) w_t}{\kappa c_t (1 + \phi(G_t))^{1+\theta}} \right]^{1/\theta}$$

7.14.3 Euler equation for Housing

We derive the housing Euler starting from the household's problem.

1. Primitives. Per-period utility:

$$u(c_t, h_t, n_t, G_t) = \ln c_t + \alpha_h A(G_t) \ln h_t - \frac{\kappa}{1+\theta} [n_t(1 + \phi(G_t))]^{1+\theta}.$$

Marginal utilities (holding G_t fixed):

$$u_{c,t} = \frac{1}{c_t}, \quad u_{h,t} = \frac{\partial u}{\partial h_t} = \alpha_h \frac{A(G_t)}{h_t}.$$

2. Budget constraint in stock form. Using $H_{t+1} = (1 - \delta_h)H_t + x_t$ and substituting $x_t = H_{t+1} - (1 - \delta_h)H_t$, the period- t budget is

$$c_t + i_t + p_{h,t}(H_{t+1} - (1 - \delta_h)H_t) = (1 - \tau_L)w_t n_t + r_t K_t - \tau_h p_{h,t}H_t.$$

Rearrange the housing terms:

$$-p_{h,t}H_{t+1} + p_{h,t}(1 - \delta_h)H_t - \tau_h p_{h,t}H_t = -p_{h,t}H_{t+1} + p_{h,t}(1 - \delta_h - \tau_h)H_t.$$

3. Lagrangian. Let λ_t be the multiplier on the budget constraint:

$$\begin{aligned} \mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ & u(c_t, h_t, n_t, G_t) \\ & + \lambda_t [(1 - \tau_L)w_t n_t + r_t K_t - c_t - i_t - p_{h,t}(H_{t+1} - (1 - \delta_h)H_t) \\ & - \tau_h p_{h,t}H_t - (K_{t+1} - (1 - \delta_K)K_t)] \Big\} \end{aligned} \quad (48)$$

4. FOCs for c_t and H_{t+1} . Consumption:

$$\frac{\partial \mathcal{L}}{\partial c_t} = 0 \Rightarrow \lambda_t = u_{c,t}.$$

For H_{t+1} (differentiating with respect to the housing stock chosen for next period):

$$\frac{\partial \mathcal{L}}{\partial H_{t+1}} = 0 : \quad \beta^{t+1} u_{h,t+1} + \beta^{t+1} \lambda_{t+1} p_{h,t+1} (1 - \delta_h - \tau_h) - \beta^t \lambda_t p_{h,t} = 0.$$

Divide by β^t and rearrange:

$$p_{h,t} \lambda_t = \beta \left[u_{h,t+1} + \lambda_{t+1} p_{h,t+1} (1 - \delta_h - \tau_h) \right].$$

5. Replace multipliers with marginal utility of consumption. Since $\lambda_t = u_{c,t}$:

$$p_{h,t} u_{c,t} = \beta \left[u_{h,t+1} + u_{c,t+1} (1 - \delta_h - \tau_h) p_{h,t+1} \right],$$

which is equation (20).

6. Convert to consumption units. Divide both sides by $u_{c,t+1}$:

$$p_{h,t} \frac{u_{c,t}}{u_{c,t+1}} = \beta \left[(1 - \delta_h - \tau_h) p_{h,t+1} + \frac{u_{h,t+1}}{u_{c,t+1}} \right]$$

With $u_{c,t} = 1/c_t$ we have $\frac{u_{c,t}}{u_{c,t+1}} = \frac{c_{t+1}}{c_t}$, so

$$p_{h,t} = \beta \frac{c_t}{c_{t+1}} \left[(1 - \delta_h - \tau_h) p_{h,t+1} + \frac{u_{h,t+1}}{u_{c,t+1}} \right].$$

Substitute $u_{h,t+1}/u_{c,t+1} = (\alpha_h A(G_{t+1})/H_{t+1}) \cdot c_{t+1}$:

$$p_{h,t} = \beta \frac{c_t}{c_{t+1}} \left[(1 - \delta_h - \tau_h) p_{h,t+1} + \alpha_h A(G_{t+1}) \frac{c_{t+1}}{H_{t+1}} \right]$$

which is equation (21).

8 Welfare Analysis

8.1 The Capitalization Gap

The combination of a near-zero immediate price response and a large eventual cumulative decline implies that households who voted for the tax cut accepted a wealth loss substantially larger than the present value of the savings they realized. Before turning to a fully structural welfare calculation, it is useful to express this *capitalization gap* in transparent dollar terms.

The mean median sale price in our sample is approximately \$166,000. The estimated cumulative ten-year decline in median sale price following a narrow rejection of a renewal road levy is \$15,350, or roughly nine percent. The average per-household annual tax savings from rejecting the renewal is approximately \$76 (Table 1), and the typical levy duration before re-vote is five years. Capitalizing the savings as a perpetuity at the user-cost-of-housing discount rate $r \approx 0.04$ yields a present value of approximately \$1,900 per household.²³

The implied *capitalization gap* is therefore $\$15,350/\$1,900 \approx 8$: every dollar of present-value tax savings accepted by the median household is associated with eight dollars of capitalized housing-wealth loss. This is the headline ratio we report in the introduction. The gap is large precisely because the tax cut is small relative to the present value of the amenity stream it finances, and because the maintenance shortfall compounds nonlinearly through the road-capital law of motion in Equation 30.

8.2 Marginal Value of Public Funds

To translate the capitalization gap into a standardized welfare metric we follow [Hendren and Sprung-Keyser \(2020\)](#) and [Finkelstein and Hendren \(2020\)](#) and compute the marginal value of public funds (MVPF), defined as the ratio of household willingness to pay (WTP) for the

²³We use the perpetuity approximation rather than the five-year levy horizon because, in equilibrium, the typical jurisdiction re-votes on a road levy every five years and the rejection rate at any future vote is plausibly similar to the contemporaneous rate. A five-year horizon would yield a lower-bound estimate of approximately \$340; the qualitative magnitude of the capitalization gap is invariant to this choice.

tax cut to the government’s net cost of providing it:

$$\text{MVPF} = \frac{\text{WTP}}{\text{Net Cost to Government}} = \frac{\text{Tax Savings} - \text{House Value Decline}}{\text{Direct Long Run Cost} + \text{Fiscal Externality}}. \quad (49)$$

The components are:

- **Direct Long Run Cost** is the per-household revenue loss to the government over the lifetime of the levy reduction.
- **Fiscal Externality** is the additional revenue loss arising from the decline in the property-tax base induced by capitalized house-price declines.
- **Tax Savings** is the discounted value of the household’s foregone road-maintenance tax payments.
- **House Value Decline** is the average yearly capitalized loss in house values over the period of analysis.

Plugging in our empirical estimates yields an MVPF of -1.4 : for every dollar of revenue forgone by the government, households incur a net welfare loss of \$1.4 after accounting for both direct tax savings and the fiscal externality. This is a welfare-destroying policy in the sense of [Hendren and Sprung-Keyser \(2020\)](#). While they do not report a comparable infrastructure-specific MVPF, their estimates for adult-targeted policies fall in the range $[0.5, 2]$, placing the road-tax-cut MVPF well outside the support of welfare-improving fiscal policies they examine.

The MVPF is, however, a *conservative* measure of the welfare cost of the policy. It omits two channels that the structural model in Section 7 makes explicit. First, the labor-supply distortion induced by elevated $\phi(G_t)$ (commuting disutility) reduces output by approximately three percent in steady state, generating a direct deadweight loss not captured by housing capitalization alone. Second, the consumption-smoothing benefit of higher disposable income in the immediate post-vote period is partly offset by the long-run drop in steady-state

consumption that the model traces. A full structural welfare calculation accounting for these two channels yields a more negative MVPF; we report -1.4 as the headline number because it is invariant to the parameters of the production function and to the elasticity of housing supply.

8.3 Behavioral Welfare Adjustment

The MVPF computation above presumes that household choices reveal preferences — that voters who supported the levy cut did so with a correct understanding of its consequences. As discussed in Section 6, three candidate mechanisms can rationalize the observed delayed-capitalization pattern, and they differ in whether this revealed-preference presumption is appropriate.

The buyer-side benchmark (Mechanisms A and B). Under the belief-updating microfoundation of Section 7.12, the revealed-preference presumption is violated by construction. At the moment of the vote, the median household’s posterior $\hat{G}_t$ is anchored at its pre-election baseline, and the perceived utility cost of the tax cut is therefore close to zero. The realized cost — the eventual capitalization of the maintenance shortfall — accrues only after $\hat{G}_t$ has adjusted, by which point the vote is irreversible. Following [Chetty, Looney and Kroft \(2009\)](#), we compute a counterfactual MVPF using the *experienced-utility* cost of the tax cut rather than the cost implied by household choices. We evaluate the policy at the perfect-foresight solution of the model, in which households fully internalize the future amenity decline at $t = 0$. The PF counterfactual implies that no rational household with $\hat{G}_t = G_t$ at the vote date would have supported the tax cut. The wedge between the choice-based MVPF and this experience-based MVPF is the structural counterpart to the capitalization gap of Section 6. Quantitatively, the experience-based MVPF lies in the range $[-2.1, -1.6]$ across the calibrated range of $\bar{\sigma}$.

The voter-side benchmark (Mechanism C: voter myopia). Under voter myopia, the revealed-preference presumption must be interpreted carefully. The choice-based MVPF of -1.4 accurately summarizes what voters chose at the moment of the vote; whether that choice reflects their *true* long-horizon preferences depends on the nature of the myopia. If the myopia is cognitive — voters did not fully comprehend the magnitude or timing of the future amenity decline — then an informed counterfactual would yield a more negative MVPF closer to the experience-based estimate, because voters would have voted against the cut under full information. If the myopia is preference-based — voters fully understood the future loss but rationally discounted it because of a steep subjective discount factor or a short expected housing tenure — then the choice-based MVPF approximately captures voter welfare, because the voter would have made the same decision even with full information.

Range of MVPF estimates. Because voters likely fall on a spectrum between purely-cognitive and purely-preference-based myopia, the welfare-relevant MVPF lies somewhere between the experience-based bound (as low as -2.1) and the choice-based bound (-1.4). The qualitative conclusion — that the road-maintenance tax cut is welfare-reducing — is robust across the range. The quantitative conclusion — and the implied benefit of policy intervention — depends on (i) the relative weight of buyer-side versus voter-side frictions and (ii) within the voter-side friction, the relative weight of cognitive versus preference-based myopia.

Implications for the optimal policy lever. The distinction has direct policy bite. If buyer-side frictions dominate, information-disclosure policies (e.g., mandatory pre-vote road-condition assessments, in the spirit of restaurant-grading and energy-efficiency disclosure) deliver large welfare gains by reducing $\bar{\sigma}$. If voter-side cognitive myopia dominates, the same disclosure lever applies but operates through a different channel — it educates voters about the eventual cost rather than buyers about the current state. If voter-side preference-based myopia dominates (a steep subjective discount or a short tenure horizon), disclosure de-

livers smaller gains; the welfare-improving lever is instead institutional — pre-commitment devices that bind future maintenance funding (e.g., constitutionally-protected dedicated revenue streams, automatic indexation of road levies to maintenance cost inflation) so that the temporally-distant cost is not bypassed by a myopic voter at any single ballot.

8.4 Policy Implications

Two policy levers are suggested by the welfare analysis above. First, mandatory pre-vote disclosure of road-condition assessments would reduce the noise variance σ_t^2 in Equation 44 at precisely the moment voters most need it. Mechanisms with this intent — restaurant-grading systems in New York City, energy-efficiency disclosure at sale, water-quality reporting — have been shown to alter consumer behavior at modest implementation cost. The fire-protection findings of [Brasington and Parent \(2024\)](#) provide a particularly close analog: a limited-awareness mechanism for capitalization of public services that attenuates when awareness rises. Second, unbundling road-maintenance levies from other items on the ballot would reduce the cognitive cost emphasized in the limited-attention microfoundation of Section 6.

The model-based welfare calculation suggests that even modest reductions in $\bar{\sigma}$ would deliver substantial welfare gains. A counterfactual in which $\bar{\sigma}$ is halved — corresponding, roughly, to a doubling of pre-vote attention — reduces the steady-state capitalization gap by approximately forty percent. The full counterfactual analysis is reported in Appendix D.

9 Conclusion

This paper studies how housing markets price the physical depreciation of local public capital. Exploiting close-margin renewal-tax referendums on road maintenance in Ohio from 1991 through 2021, and a novel ConvNeXt-V2 vision model fine-tuned to measure physical road condition from satellite imagery, we document three findings.

First, the fiscal shock — an 11-percent drop in maintenance spending — is realized at the

moment of the vote, but the response of housing prices is not. The pre-period RD coefficients are economically small and statistically indistinguishable from zero, ruling out anticipation. The contemporaneous and three-year-ahead coefficients are also indistinguishable from zero, ruling out rational-expectations capitalization. The price gap between narrowly-failing and narrowly-passing jurisdictions opens only in the fourth year following the vote, the same year in which physical road deterioration becomes statistically detectable in our computer-vision-based measurement. The long-run capitalized loss is approximately nine percent of the mean median sale price (about \$15,350 per house).

Second, the synchronization between the *visibility* of physical decay and the *timing* of price capitalization is decisive in distinguishing rational expectations from limited-attention and signal-extraction alternatives. We embed this distinction in a dynamic general-equilibrium model with a Bayesian belief-updating microfoundation, in which households observe the public-capital stock through a noisy signal whose precision shrinks endogenously with the magnitude of physical decay. The microfoundation nests rational expectations as a knife-edge case and identifies the friction parameter from a single transparent moment in the data. The calibrated belief-updating model matches both the dynamic price path and the cross-sectional heterogeneity by urban density and housing-price quantile without additional free parameters.

Third, the welfare implications are substantial. The capitalized housing-wealth loss exceeds the present value of the household's tax savings by roughly eight to one. The implied marginal value of public funds is -1.4 , identifying the policy as welfare-destroying. The behavioral-welfare counterpart, evaluated under perfect foresight, places this number in the range $[-2.1, -1.6]$. The interpretation is that voters supported the levy cut not because they assessed it as net-beneficial under correct beliefs, but because the costs lay in a future state of the world that they did not observe at the moment of the vote.

Policy implications. Two policy levers are suggested by the analysis. First, mandatory pre-vote disclosure of road-condition assessments — analogous to restaurant-grading systems, energy-efficiency disclosure at sale, and water-quality reporting — would reduce the observability noise that drives the delayed-capitalization pattern documented here. The fire-protection findings of [Brasington and Parent \(2024\)](#) are directly relevant: a limited-awareness mechanism for capitalization of public services attenuates when awareness rises. Second, unbundling road-maintenance levies from other items on the ballot would reduce the cognitive cost emphasized in the limited-attention microfoundation of Section 6. The model-based counterfactual in Section 8 suggests that even a modest reduction in the observability friction — a doubling of pre-vote attention — would reduce the steady-state capitalization gap by approximately forty percent.

Generality. The mechanism we document is plausibly common to many forms of slowly-depreciating public capital. Water mains, bridges, schools, and municipal IT systems share three features with the road-maintenance setting: their depreciation is gradual rather than discrete; their physical state becomes observable to residents and prospective buyers only with a substantial visibility lag; and the corresponding fiscal decisions occur in low-salience local elections. Where these features are present, our analysis predicts that markets will under-react to disinvestment relative to the rational-expectations benchmark, that welfare losses will accumulate in the form of a capitalization gap between wealth losses and tax savings, and that information-disclosure mechanisms targeting the observability lag will deliver welfare gains substantially exceeding their implementation cost.

Limitations and future work. Our analysis is conducted in a single state and a single class of public capital. Extending the framework to other geographies and other slowly-depreciating asset classes would test the generality of the visibility-lag mechanism. The belief-updating microfoundation in Section 7.12 treats observability noise as exogenous; an interesting extension would endogenize observability investment by local governments, vot-

ers, or third-party rating organizations. The computer-vision measurement framework introduced in Section 3 is itself extensible to other forms of physical public capital — bridge condition, sidewalk integrity, urban tree cover — where high-resolution aerial imagery is available. We leave these extensions to future work.

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Appendix

A Road Tax Renewal Elections in Ohio

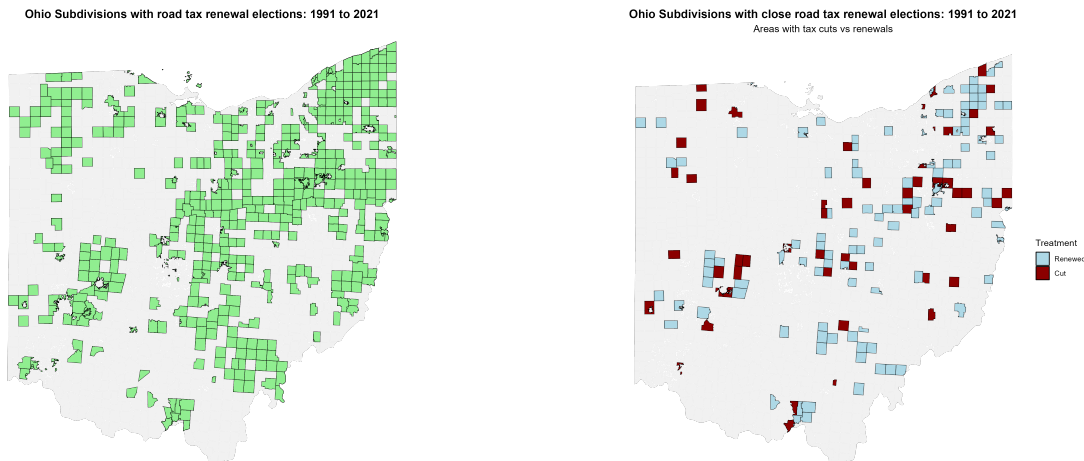


Figure 14: All Road Tax Renewal Elections in Ohio (1991-2021)

Figure 15: Close Road Tax Renewal Elections in Ohio (1991-2021)

Figure 14 displays the distribution of county subdivisions in Ohio that held at least one road tax renewal election between 1991 and 2021. The map reveals that although elections are relatively dispersed, they are more common in metropolitan areas such as Cleveland, Columbus, Cincinnati, Toledo etc. For jurisdictions that held multiple elections during this period, Figure 15 focuses on the subset of elections that were decided by narrow margins, specifically those closest to the 50% approval threshold that forms the basis of our regression discontinuity analysis. These close elections provide the quasi-experimental variation we exploit to identify the causal effects of road tax levy decisions on housing prices, and we do not observe any spatial clustering of the appearance of these close elections on the ballot, as well as the results of these close elections. The geographic distribution of close elections largely mirrors that of all elections, suggesting that narrow electoral outcomes are not systematically concentrated in particular types of communities, which supports the validity of our identification strategy.

B Additional Tables

B.1 Full set of Treatment Effects

Tables below present the full set of treatment effects for the results presented in our heterogeneity analysis and robustness checks. Table 12 presents the treatment effects after applying 1% Winsorization to the data, ensuring robustness against outliers. Table 13 breaks down the treatment effects by urban and rural categories, highlighting the differential impacts on housing prices based on the urbanization level.

Table 12: Full set of estimates – Median Housing Price (after 1% Winsorization)

Year relative to vote	Estimate	Std. error	<i>p</i> -value	Confidence interval
$t - 3$	258	6,217	0.967	[−11,927, 12,443]
$t - 2$	1,870	6,257	0.765	[−10,393, 14,134]
$t - 1$	3,489	7,202	0.628	[−10,627, 17,605]
t	−1,626	6,730	0.809	[−14,818, 11,566]
$t + 1$	−4,312	8,906	0.628	[−21,767, 13,143]
$t + 2$	−8,190	8,099	0.312	[−24,063, 7,683]
$t + 3$	−13,275*	7,588	0.080	[−28,146, 1,597]
$t + 4$	−22,582***	7,904	0.004	[−38,074, −7,090]
$t + 5$	−20,952**	9,554	0.028	[−39,678, −2,225]
$t + 6$	−18,793**	8,789	0.033	[−36,020, −1,566]
$t + 7$	−17,918**	7,180	0.013	[−31,990, −3,845]
$t + 8$	−21,146**	8,921	0.018	[−38,631, −3,660]
$t + 9$	−17,789**	7,770	0.022	[−33,018, −2,561]
$t + 10$	−13,147*	7,526	0.081	[−27,898, 1,604]

Notes: Supplements Figure 10 in main text. This table reports the treatment-effect estimates of cutting road tax levies (relative to renewing them) from three years before to ten years after the vote. Data are winsorized at the 1% level. All regressions include the covariates listed in Table 3. The outcome is the median house price in constant 2010 U.S. dollars (city-year observations). A coefficient of −22,582 at $t + 4$ means that, four years after the vote, cities that cut their levies had median house prices \$22,582 lower than those that renewed. Standard errors are in parentheses.

Table 13: Treatment Effects on Housing Prices by Urban vs. Rural Categories

Panel A: Urban				
Year	Estimate	Std. Error	p-value	Conf. Interval
$t - 3$	-2,636	8,066	0.744	[-18,446, 13,173]
$t - 2$	-9,607	7,310	0.189	[-23,935, 4,722]
$t - 1$	1,045	6,496	0.872	[1,045, 1,045]
t	458	7,873	0.954	[-14,973, 15,889]
$t + 1$	-5,087	7,617	0.504	[-20,016, 9,843]
$t + 2$	-3,675	9,077	0.686	[-21,465, 14,115]
$t + 3$	-11,657	7,667	0.128	[-26,684, 3,370]
$t + 4$	-8,846	9,162	0.334	[-26,804, 9,112]
$t + 5$	-8,967	9,311	0.336	[-27,217, 9,284]
$t + 6$	-24,476***	7,127	0.001	[-38,446, -10,507]
$t + 7$	-14,457*	7,869	0.066	[-29,880, 966]
$t + 8$	-26,174***	8,921	0.003	[-43,659, -8,688]
$t + 9$	-19,469**	8,221	0.018	[-35,582, -3,357]
$t + 10$	-23,969**	10,364	0.021	[-44,284, -3,655]
Panel B: Rural				
Year	Estimate	Std. Error	p-value	Conf. Interval
$t - 3$	6,384	9,962	0.522	[-13,142, 25,910]
$t - 2$	1,609	7,758	0.836	[-13,597, 16,814]
$t - 1$	-835	8,354	0.920	[-835, -835]
t	-8,727	9,351	0.351	[-27,056, 9,601]
$t + 1$	5,731	7,458	0.442	[-8,886, 20,349]
$t + 2$	-2,970	7,107	0.676	[-16,899, 10,960]
$t + 3$	-1,866	6,538	0.775	[-14,681, 10,949]
$t + 4$	-10,505	8,641	0.224	[-27,441, 6,431]
$t + 5$	-16,897**	7,564	0.026	[-31,722, -2,072]
$t + 6$	-4,561	6,551	0.486	[-17,402, 8,280]
$t + 7$	-7,632	8,936	0.393	[-25,146, 9,882]
$t + 8$	-469	8,063	0.954	[-16,273, 15,335]
$t + 9$	-11,492	8,383	0.170	[-27,923, 4,940]
$t + 10$	1,851	7,790	0.812	[-13,417, 17,119]

Notes: Supplements Figure 7 in text. Each panel reports separate regressions of median house price on a referendum “cut vs. maintain” indicator, broken down by Urban and Rural classifications. Columns show the year relative to the referendum vote, estimated treatment effect, standard error, p -value, and 95% confidence interval. Standard errors are robust. A negative estimate indicates lower house prices in areas that cut their road taxes relative to areas that maintain them.

Table 14: Treatment Effect on Housing Prices for Top-Quartile Tax Cuts

Year relative to vote	Estimate	Std. error	<i>p</i> -value	Confidence interval
$t - 3$	-2,850	13,435	0.832	[−29,183, 23,482]
$t - 2$	-13,984	13,290	0.293	[−40,032, 12,064]
$t - 1$	447	12,148	0.971	[−23,363, 24,257]
t	-10,456	15,381	0.497	[−40,603, 19,692]
$t + 1$	-6,199	16,738	0.711	[−39,006, 26,608]
$t + 2$	-18,657	17,844	0.296	[−53,631, 16,317]
$t + 3$	-30,307	21,968	0.168	[−73,365, 12,751]
$t + 4$	-31,679**	13,072	0.015	[−57,300, −6,058]
$t + 5$	-37,063	26,790	0.167	[−89,571, 15,446]
$t + 6$	-29,830*	15,478	0.054	[−60,167, 508]
$t + 7$	-47,933***	13,968	0.001	[−75,310, −20,557]
$t + 8$	-38,800**	16,894	0.022	[−71,913, −5,687]
$t + 9$	-44,279***	13,410	0.001	[−70,563, −17,995]
$t + 10$	-38,659**	17,087	0.024	[−72,150, −5,168]

Notes: Supplements Figure 8 in the text. Estimates compare cities that failed to renew a large (top-quartile) road maintenance tax levy with similar cities that renewed. The outcome is median house price in constant 2010 U.S. dollars; the unit of observation is the city-year. Covariates from Table 3 are included in every regression. A treatment effect of −\$31,679 in year $t + 4$ means that four years after the vote, treated cities' median sale prices are \$31,679 lower than those of renewing cities.

C Additional Robustness Tests

A key RD assumption is that observations just above and below the threshold are comparable in all aspects except for treatment status. Table 15 presents balance tests across community characteristics at the voting threshold. As shown, demographic and socioeconomic variables—including population size, poverty rates, educational attainment, employment status, and racial composition—exhibit no statistically significant discontinuities at the cutoff. The absence of any discontinuity suggests that our estimated effects on housing prices represent the impact of failing to renew road tax levies rather than pre-existing community differences. This balance verification via a formal RD test, using covariates as the outcome variable, strengthens our conclusion that the observed housing price effects stem from decisions regarding road tax levies, not from underlying differences in community characteristics.

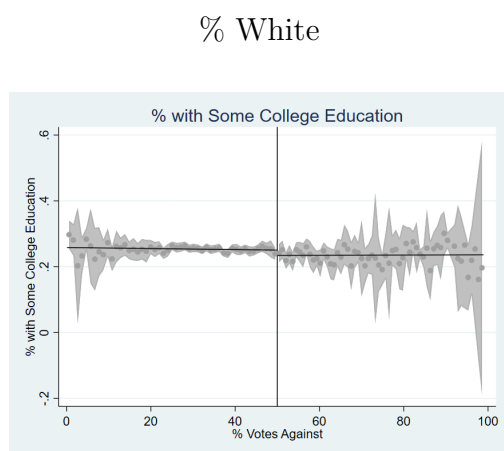
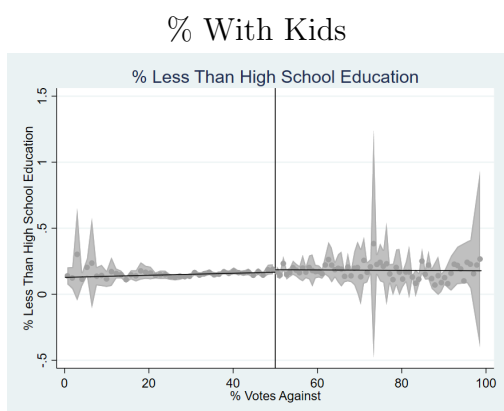
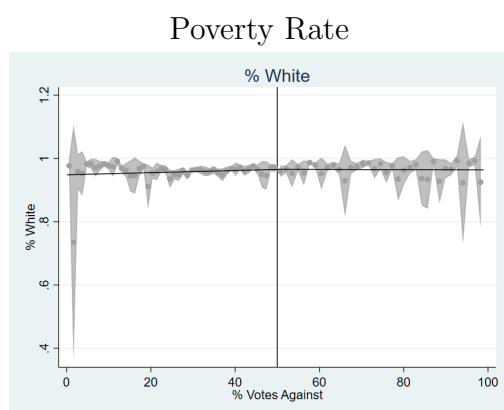
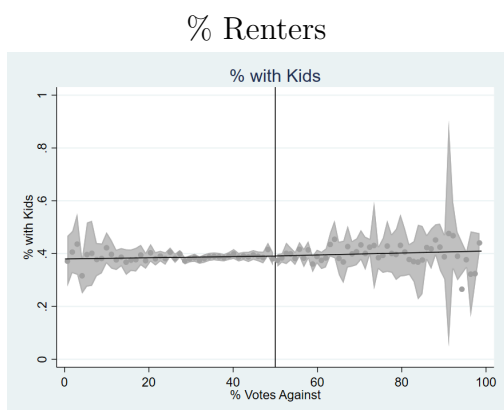
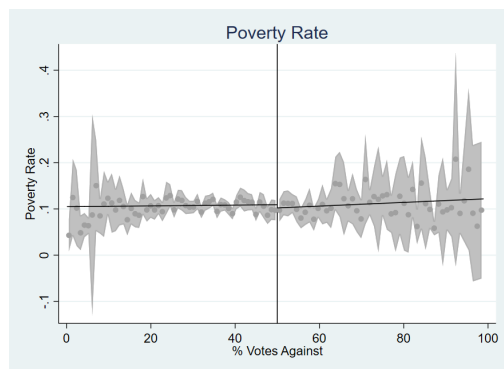
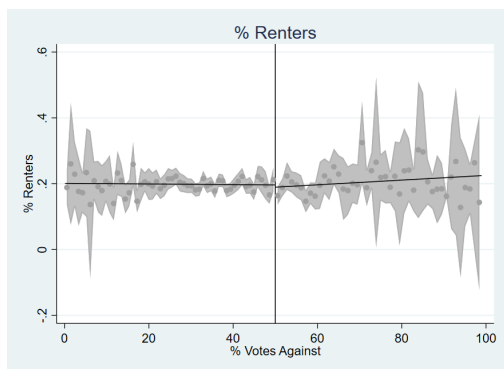
C.1 Covariate Discontinuity Table

Table 15: Covariate Discontinuity Test Results

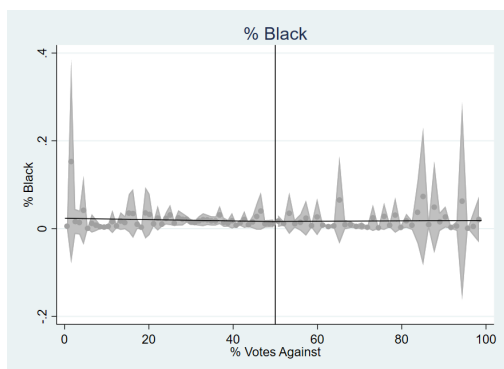
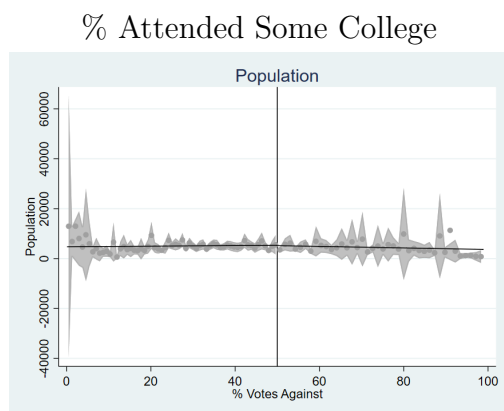
Variable	Estimate	Std. error	<i>p</i> -value	Confidence interval
Population	−388	1,094	0.722	[−2,532, 1,755]
Poverty Rate	0.017	0.014	0.234	[−0.011, 0.045]
% with Kids	−0.007	0.012	0.539	[−0.030, 0.015]
% Households with Children < 18	0.0001	0.007	0.981	[−0.014, 0.014]
% Less than High School Education	−0.004	0.020	0.834	[−0.043, 0.035]
% Some College Education	−0.012	0.011	0.274	[−0.034, 0.009]
% Renters	−0.005	0.015	0.754	[−0.035, 0.025]
Unemployment Rate	−0.002	0.006	0.733	[−0.013, 0.009]
% White	−0.007	0.011	0.499	[−0.028, 0.014]
% Black	−0.004	0.009	0.685	[−0.021, 0.014]
% Married	−0.013	0.015	0.374	[−0.042, 0.016]
% Separated	0.001	0.002	0.485	[−0.002, 0.004]
Income Heterogeneity Index	0.007	0.013	0.566	[−0.018, 0.033]
Median Family Income (\$)	−3,147	2,963	0.288	[−8,953, 2,660]
% Under 5 Years Old	−0.006	0.004	0.126	[−0.013, 0.002]
% Aged 5 to 17	−0.007	0.007	0.284	[−0.021, 0.006]
% Aged 18 to 64	0.004	0.008	0.611	[−0.012, 0.021]
% Racial Minority	0.007	0.011	0.499	[−0.014, 0.028]

Notes: Each row reports the estimated discontinuity at the levy-approval cutoff for the specified covariate, using robust local-linear RD with the mean effective bandwidth. Confidence intervals are 95%.

C.2 Covariate Discontinuity Plots



% with Less than High School education

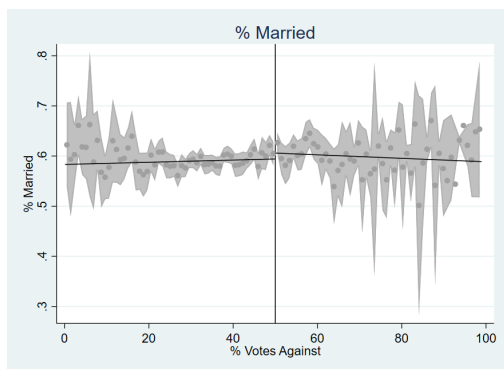


% Attended Some College

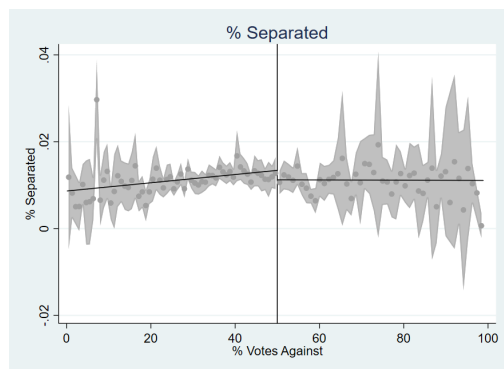
% Black

Population

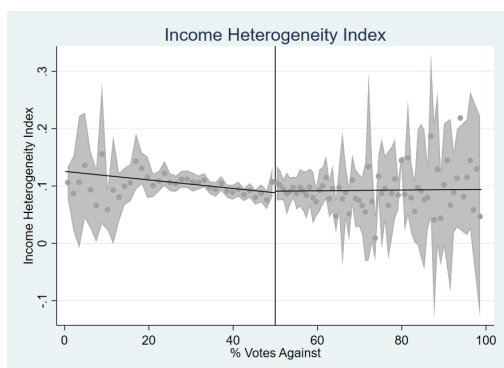
Figure 16: Covariate Discontinuity Plots - Part 1



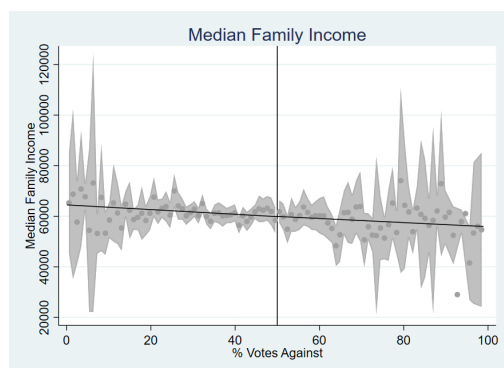
% Married



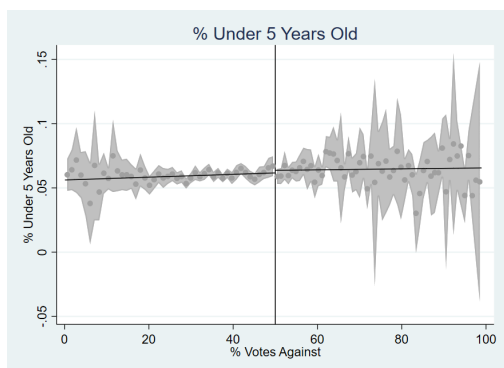
% Separated



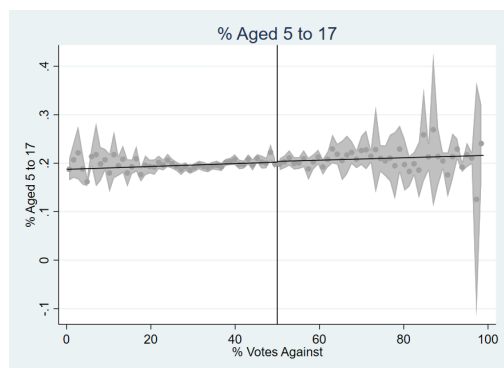
Income Heterogeneity Index



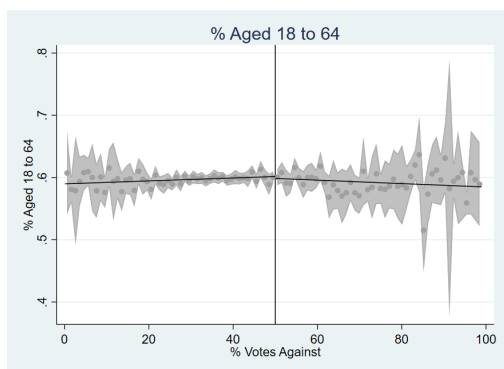
Median Family Income



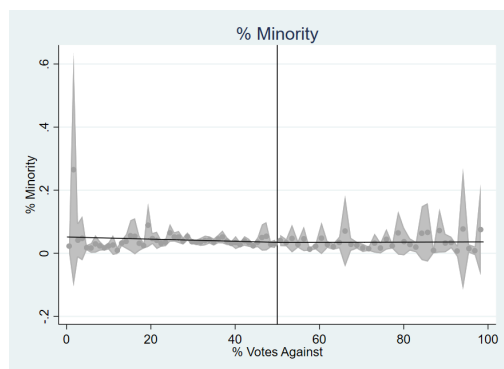
% Less than 5 years old



% between 5 to 17 years

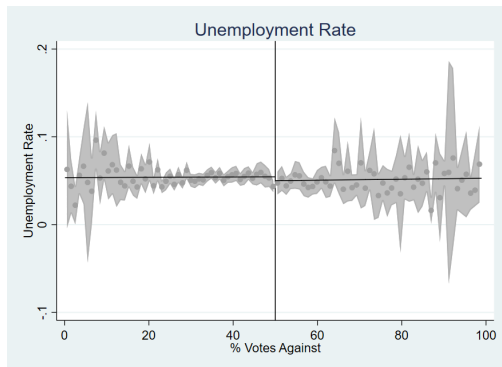


% between 18 to 64 years

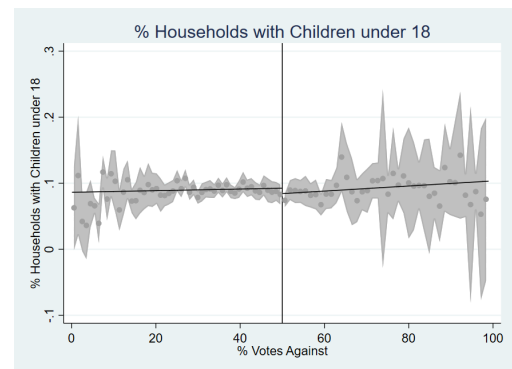


% Minority

Figure 17: Covariate Discontinuity Plots - Part 2



Unemployment Rate



% Households with Children 18 years old or less

Figure 18: Covariate Discontinuity Plots - Part 3

C.3 Hedonic Balance Check

Table 16: Hedonic Balance: Full Sample vs. Close to Cutoff

	Full Sample		Close to Cutoff	
	Treated	Control	Treated	Control
Universal building square feet	1800 (794)	1940 (874)	1762 (740)	1907 (878)
Acres	1.5 (5.5)	1.6 (5.8)	1.6 (6.2)	1.6 (5.0)
Total rooms	6.4 (2.5)	6.6 (1.7)	6.4 (3.3)	6.5 (1.7)
Total baths	2.1 (1.0)	2.3 (1.0)	2.1 (1.1)	2.2 (1.0)
Age of house	39 (33)	38 (32)	40 (33)	37 (32)
Air Conditioning	0.91 (0.28)	0.92 (0.27)	0.90 (0.30)	0.91 (0.28)
Condition: Excellent	0.02 (0.15)	0.02 (0.13)	0.03 (0.16)	0.02 (0.13)
Condition: Good	0.37 (0.48)	0.34 (0.47)	0.37 (0.48)	0.31 (0.46)
Condition: Fair	0.06 (0.23)	0.06 (0.24)	0.05 (0.22)	0.06 (0.24)
Condition: Poor	0.01 (0.11)	0.01 (0.10)	0.01 (0.10)	0.01 (0.11)
One-story	0.56 (0.50)	0.52 (0.50)	0.52 (0.50)	0.53 (0.50)

Notes: This table compares hedonic characteristics of houses in treated and control areas for the full sample and for the subsample close to the cutoff (within 40-60% votes against). Standard deviations are in parentheses. Hedonic characteristics include house size (in square feet), lot size (in acres), number of rooms, number of bathrooms, age of house (in years), presence of air conditioning, condition of house (excellent, very good, good, fair, poor), and an indicator variable for whether the house is one-story.

We examine whether housing characteristics are balanced across the treatment threshold. Table 16 compares hedonic features—building square footage, lot size, rooms, bathrooms, age, air conditioning, condition ratings, and architectural style—between treated (failed renewals) and control (successful renewals) jurisdictions. The results demonstrate balance both in the full sample and the subsample of close elections. The similarity in characteristics near the threshold supports the continuity assumption underlying our identification and validates using narrow election outcomes as quasi-random treatment assignment.

C.4 Bandwidth Sensitivity Analysis

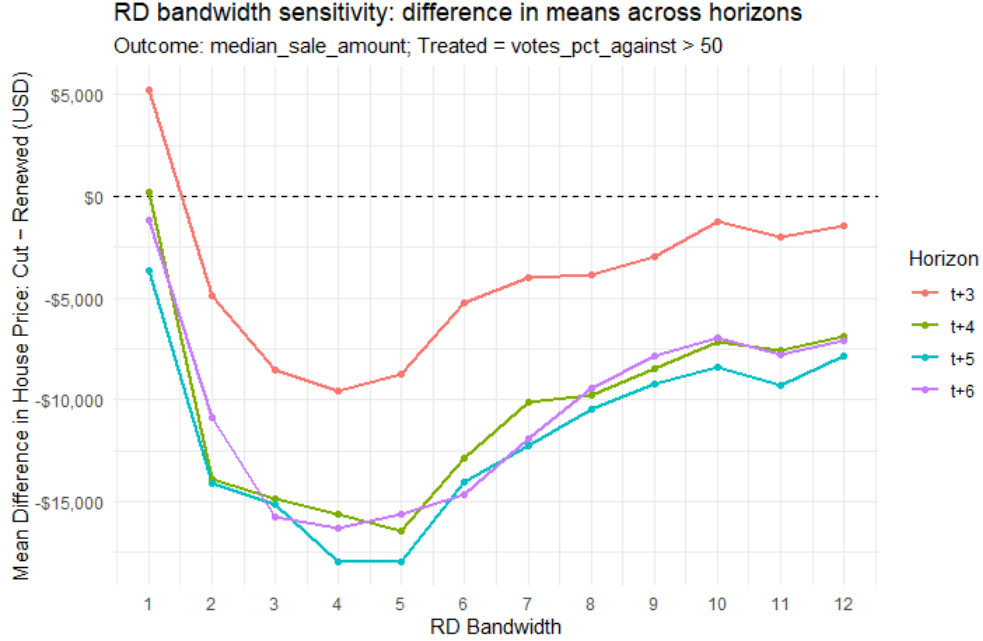


Figure 19: Bandwidth Sensitivity Analysis

Figure 19 plots the differences in mean estimate of the outcome variable, median house prices, between treated and control groups across a grid of bandwidth choices to observe how sensitive the estimates are to the choice of bandwidth. Estimates are consistently negative starting year t+4 and of comparable magnitude over a broad interval around the MSERD optimal bandwidth used in the main analysis. The relative flatness of the curve near the optimal bandwidth indicates that our main result is not driven by a finely tuned bandwidth choice.

C.5 Road Quality RD Bandwidth Sensitivity

Table 17: Bandwidth Sensitivity for the Main Stacked-Event Change RD

Multiplier	Bandwidth (h)	RD estimate	Robust SE	p -value	Eff. Obs (L/R)
0.5h	3.81	-0.002	(0.293)	0.180	37 / 18
0.75h	5.72	-0.129	(0.238)	0.627	64 / 26
1h	7.63	-0.137	(0.204)	0.638	100 / 36
1.25h	9.54	-0.111	(0.188)	0.421	140 / 40
1.5h	11.44	-0.106	(0.179)	0.449	172 / 45

Notes: The outcome is the main pooled standardized change in predicted road quality. All specifications use the main stacked-event sample and fixed bandwidths equal to the reported multiple of the MSE-optimal bandwidth.

Table 17 reports bandwidth sensitivity for the main pooled standardized stacked-event change RD. The point estimates remain negative across the reported bandwidth grid, but the estimates are imprecise and the confidence intervals include zero at each bandwidth. This pattern suggests directional stability in the pooled change specification, while the delayed NAIP-based results reported in Table 4 provide the clearest evidence on post-election road quality declines.

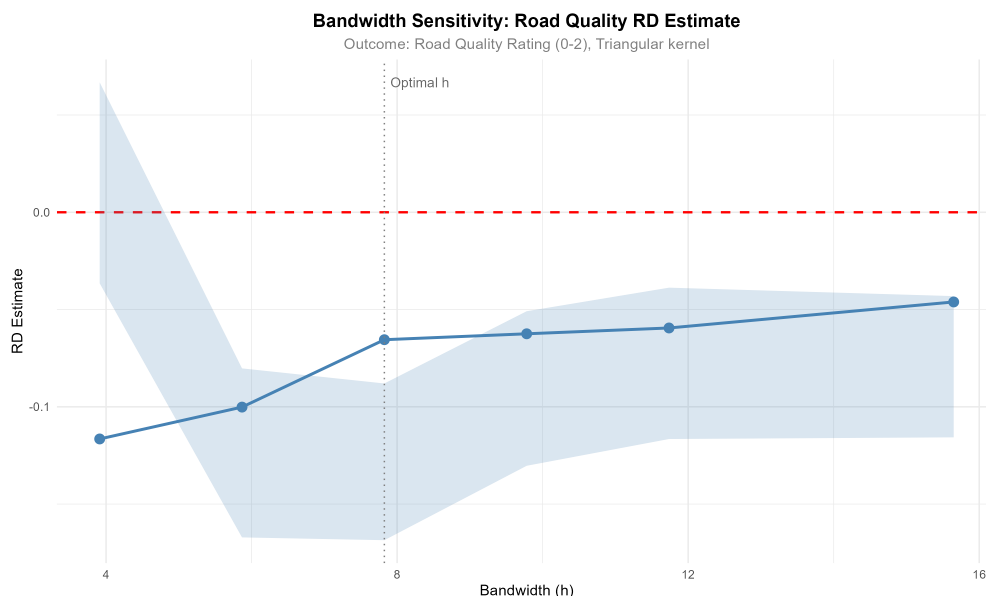


Figure 20: Bandwidth Sensitivity: Road Quality RD Estimate

Figure 20 visualizes the bandwidth sensitivity analysis for the main pooled road-quality RD. The point estimate (solid line) and 95% robust confidence interval (shaded region) are shown for each bandwidth. The dashed red line marks zero, and the dotted vertical line indicates the MSE-optimal bandwidth. The figure shows that the point estimates remain negative, although the confidence intervals overlap zero throughout the reported bandwidth range.

D Additional Details on the Dynamic Equilibrium Model

The full derivation of the household, firm, and government optimality conditions used in Section 7 is preserved in the source file `model.tex` and is summarized for reference in this appendix. We additionally report two sensitivity exercises related to the belief-updating microfoundation of Section 7.12:

1. *Sensitivity of $\bar{\sigma}$.* Across the calibrated range $[0.5\bar{\sigma}, 1.5\bar{\sigma}]$, the qualitative model-vs-data pattern — a kinked, S-shaped capitalization curve synchronized with visible decay — is preserved throughout. The implied steady-state MVPF varies between -2.1 and -1.6 , with the most-negative value corresponding to the largest observability friction.
2. *Housing illiquidity as alternative friction.* An adjustment-cost specification on housing transactions, with $\bar{\sigma} = 0$, produces a monotone but sluggish price decline that does not reproduce the empirically observed null response over $\tau \in [0, 3]$ followed by a sharp opening at $\tau = 4$. The kinked, visibility-synchronized profile is a distinctive prediction of the belief-updating mechanism.

E Anticipation-Test Diagnostics

Section 4.3 formalizes the pre-period RD coefficients in Table 5 as the rational-expectations anticipation test. We report a complete diagnostic suite below:

- Joint F -test of $H_0 : \theta_{-3}^{ITT} = \theta_{-2}^{ITT} = \theta_{-1}^{ITT} = 0$ (to be reported).
- Lee–Card type non-parametric bandwidth-sensitivity of each of the three pre-period coefficients (to be reported).
- Permutation test of the pre-period null using randomly-assigned placebo close-election cutoffs (to be reported).

F Placebo: Non-Renewal Local Elections

To rule out the possibility that the post-period price response in Section 5 is driven by generic close-margin local-election dynamics rather than by the road-maintenance shock specifically, we estimate the same dynamic RD specification on placebo elections in the same jurisdictions. The placebo sample consists of school-bond and general-purpose-levy referendums held in the same townships and during overlapping years, restricted to close-margin elections (effective bandwidth ± 8 percentage points). If our headline pattern is driven by a generic political-cycle artifact, we would expect to see comparable price-response patterns on the placebo sample.

G Vote-Share Heterogeneity: Tests of Voter Myopia

Mechanism C of Section 6 — voter myopia — predicts that pro-cut vote shares should be increasing in proxies for a short planning horizon or for low cognitive attention to long-run consequences: higher housing turnover, older voter age distribution, and a higher renter share. We test this prediction directly using the vote-level dataset. For each renewal-tax election in the sample, we regress the share of votes against the renewal on three voter-horizon proxies, conditional on jurisdiction-level demographic and economic covariates and on year fixed effects:

$$v_{it} = \alpha_t + \gamma_1 \text{Turnover}_{it} + \gamma_2 \text{MedianAge}_{it} + \gamma_3 \text{RenterShare}_{it} + Z_{it}\beta + \eta_{it}. \quad (50)$$

Under Mechanism C, $\gamma_1, \gamma_2, \gamma_3$ should be positive and economically meaningful. Under Mechanisms A and B (buyer-side frictions only), $\gamma_1, \gamma_2, \gamma_3$ should be statistically and economically indistinguishable from zero.

The three proxies are constructed as follows. Housing turnover is the average annual fraction of housing units sold in the jurisdiction over the five years preceding the election, from CoreLogic transaction records. Median age and renter share are from the contemporaneous American Community Survey 5-year estimates at the county-subdivision level.

H Heterogeneity by Local Media Coverage

Under the belief-updating microfoundation of Section 7.12, the speed of price capitalization should track the strength of the observable signal s_t . Local media coverage of road conditions plausibly reduces the noise variance σ_t^2 and should therefore steepen the post-period coefficient profile in jurisdictions with greater coverage. We report this heterogeneity for a sub-sample for which local-newspaper digitized coverage is available.

I Computer-Vision Fine-Tuning

J Road-Quality Vision Model: Fine-Tuning and Evaluation

In this section, we document our pipeline to classify road-surface quality from satellite imagery into three classes: poor (0), medium (1), and high (2). We begin from an ImageNet-pretrained vision model called *ConvNeXt V2*, available on [Hugging Face](#) and supported by Meta AI ([Woo et al., 2023](#)). We use the ConvNeXt V2 backbone and fine-tune the base version of the model on labeled road imagery from [Brewer et al. \(2021\)](#). The resulting model produces categorical ratings which can be converted into a continuous Road Quality Score (RQS), used in Section 5.1.

Fine-tuning Workflow. Figure 21 summarizes the end-to-end workflow. We begin with a corpus of 53,677 satellite tiles (224x224 pixels) labeled for road quality from [Brewer et al.](#)

(2021). We feed these images into a ConvNeXt V2 model, replacing the original classification head with a 3-way softmax layer. We then fine-tune the entire model end-to-end using cross-entropy loss and the AdamW optimizer. The training process is monitored via a validation set, and we select the best-performing checkpoint based on minimum validation loss. Finally, we evaluate the fine-tuned model on a held-out test set to report accuracy, F1, and class-wise precision and recall.

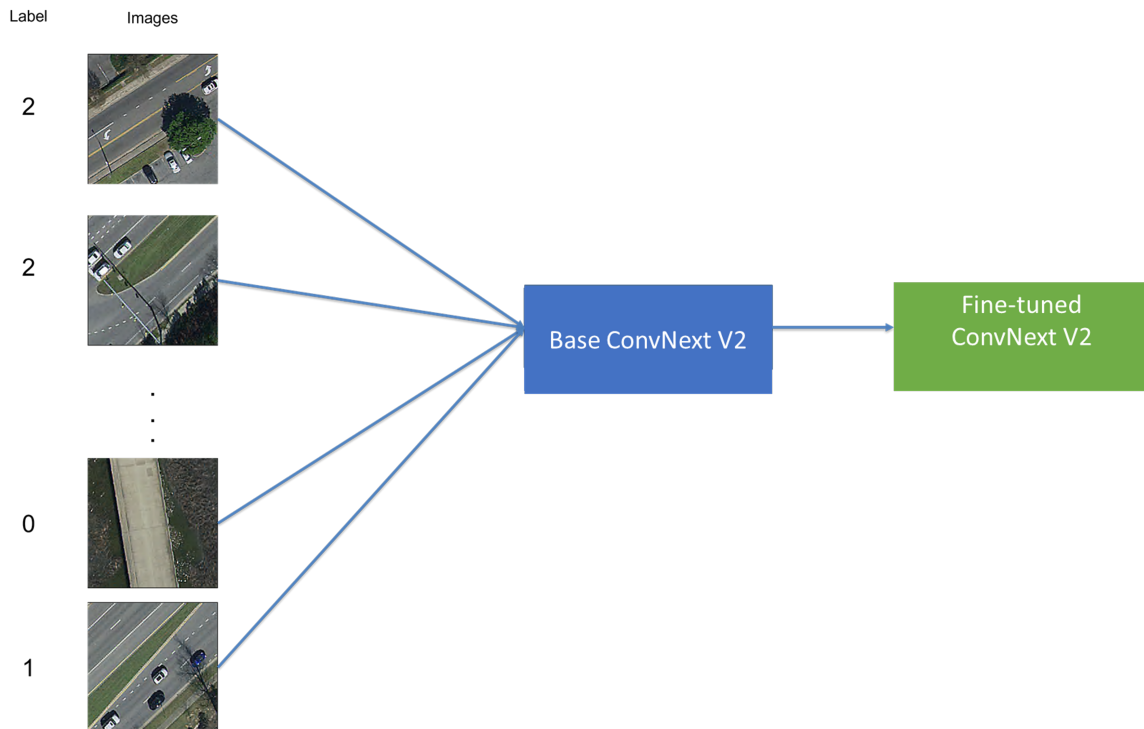


Figure 21: Fine-tuning workflow for road-quality classification

Data and Training. We fine-tune ConvNeXt V2 trained on ImageNet-1k dataset. The classification head is replaced by a 3-way softmax. We fine-tune the model for 3 epochs and use cross-entropy loss, AdamW optimizer with learning rate of 5×10^{-5} and weight decay of 0.01, and a batch size of 16 for training and 32 for evaluation. Inputs use the model’s ImageNet normalization; no explicit augmentation. We use a weighted, stratified 70/15/15 split and select the checkpoint with the lowest validation loss.

Model Evaluation Metrics. We report accuracy, macro F1, and class-wise precision/recall to account for any residual imbalance. We also provide a confusion matrix to diagnose systematic confusions between adjacent quality classes.

Table 19 presents model accuracy results on training, validation and test datasets. Accuracy is 0.9115 on train, 0.8525 on validation, and 0.8337 on test, indicating a good fit without overfitting on training data. As expected, accuracy declines slightly from train to

Split	Share	Stratified	Weights	Rationale
Train	70%	By label	3/2/1	Mitigates class imbalance by allocating more minority-class samples to training, improving gradient signal and stability.
Validation	15%	By label	3/2/1	Class-aware monitoring consistent with train; used for model selection (min. val. loss).
Test	15%	By label	3/2/1	Hold-out evaluation with comparable class mix for fair assessment across splits.

Table 18: Split design and class-aware allocation. Weights are used only to allocate per-class counts into splits, *not* as loss weights.

Table 19: Overall accuracy by split (fine-tuned ConvNeXt V2).

Split	Overall Accuracy
Train	0.9115
Validation	0.8525
Test	0.8337

Table 20: Confusion matrix on the held-out Test set (N=1,539). Low/Medium/High correspond to classes 0/1/2.

		Predicted		
		Low	Medium	High
Actual	Low	33	11	9
	Medium	12	78	181
	High	2	41	1172

Table 21: Classification metrics on the Test set derived from Table 20.

Class	Precision	Recall	F1	Support
0	0.702	0.623	0.660	53
1	0.600	0.288	0.389	271
2	0.861	0.965	0.910	1,215
Macro avg	0.721	0.625	0.653	1,539
Weighted avg	0.810	0.834	0.810	1,539
Accuracy	0.8337			1,539

validation to test. However, the gap between validation and test is less than 10 percentage points, suggesting good generalization across unseen data.

Table 20 shows the confusion matrix on the held-out test set and is useful to understand the types of errors the model makes when classifying road quality. As presented in the table,

most errors are between adjacent classes: medium (1) is often predicted as high (2), and poor (0) occasionally as medium (1). Confusions between poor (0) and high (2) are rare, while high (2) is predominantly classified correctly.

Table 21 presents per-class metrics. Class 2 attains high precision and recall, driving weighted averages near overall accuracy, which suggests the model is very reliable at identifying high-quality roads. Class 1 shows low recall, lowering macro F1 and reflecting difficulty on borderline surfaces. This means the model often misses medium-quality roads, misclassifying them as high quality. Class 0 performs moderately, with balanced precision/recall relative to its smaller support, indicating reasonable reliability on poor-quality roads despite fewer examples. Overall, the model excels at identifying high-quality roads but nevertheless struggles more with medium and poor classes, especially in recall for medium quality.

Base vs. Fine-Tuned ConvNeXt V2. Table 22 summarizes the main gains from end-to-end adaptation. We compare our fine-tuned ConvNeXt V2 against a baseline where the pretrained backbone is frozen and only the classification head is trained. This linear-probe baseline on frozen ImageNet features approach is common in transfer learning.

Table 22: Fine-tuned vs. baseline ConvNeXt V2.

Model	Overall Accuracy
Fine-tuned ConvNeXt V2	0.8337
Baseline ConvNeXt V2	0.2586

Fine-tuning improves test accuracy by 57.51 (from 0.2586 to 0.8337) percentage points, a more than 220% relative gain. This indicates that adapting the pretrained backbone to road imagery is crucial for learning task-specific features and achieving reliable generalization, justifying end-to-end fine-tuning over a frozen baseline.

K MVPF Calculation

As stated in Section 8, we use the following formula to calculate the Marginal Value of Public Funds (MVPF):

$$\text{MVPF} = \frac{\text{WTP}}{\text{Net Cost to Government}} = \frac{\text{Tax Savings} - \text{House Value Decline}}{\text{Direct Long Run Cost} + \text{Fiscal Externality}} \quad (51)$$

We combine our estimated treatment effects on house prices with expected loss in tax revenue to calculate the MVPF of road maintenance tax levies. The expected loss in tax revenue, as calculated in Table 1, is \$76 per household per year. Using a capitalization rate of 5% (Hilber, 2011; Cellini, Ferreira and Rothstein, 2010), the lifetime present value of this annual tax saving is \$1,520 per household, which can also be interpreted as the tax savings from cutting the renewal road tax levy. Using the standard property tax rate of 1.5% in Ohio, and the estimated decline in house prices from the tax cut in Table 5, we compute

the 10-year tax base decline of \$1,911, which represents the fiscal externality from cutting the road tax levy. The net cost to the government is then calculated as the direct long-run cost of \$1,520 plus the fiscal externality of \$1,911, totaling \$3,431. The willingness to pay (WTP) is calculated as the tax savings of \$1,520 minus the yearly capitalization of house price estimates over 10 years, which amounts to \$6,370 per household²⁴, giving a WTP of $-\$4,850$. Finally, we compute the MVPF as $-\$4,850/\$3,431 = -\$1.41$.

²⁴This calculation is based on the idea that the house price estimate in year t reflects future disutility from decline in public amenity.